

MightyXDDoS™ - DDoS Mitigator

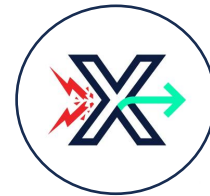
- High-Performance Security Bridge based on FD.io VPP and eBPF XDP
- L2 Mitigator for AI LLM Inference Server

MightyLink Lab.
(www.mightyconn.com)
Doc. Revision: 1.00

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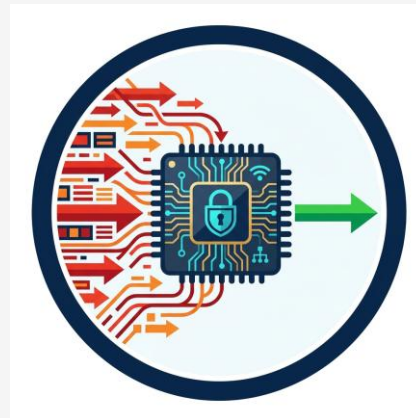
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2. Base Technology - **DPDK, FD.io VPP, eBPF XDP**
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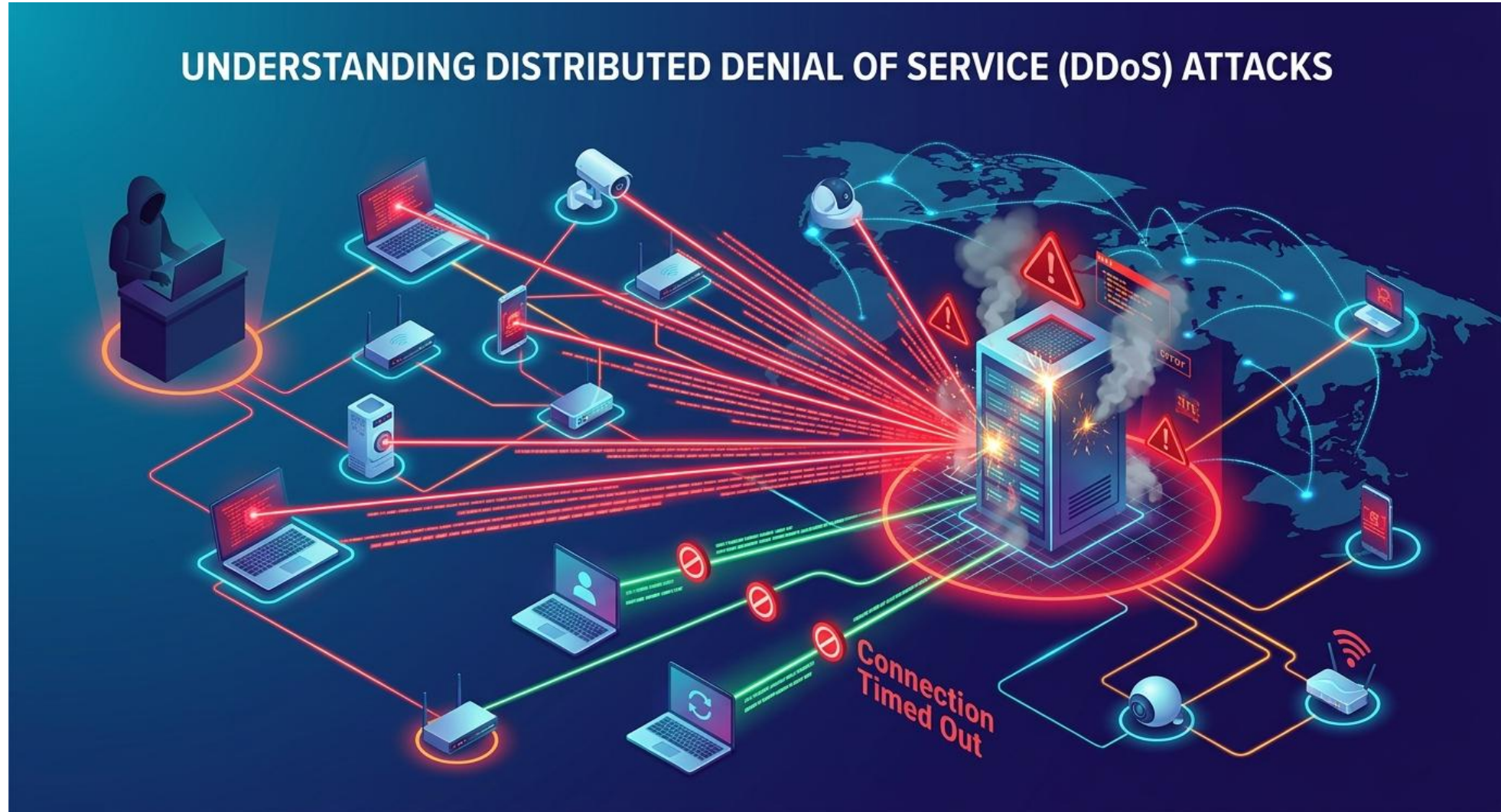


1. DDoS Attack & Mitigator

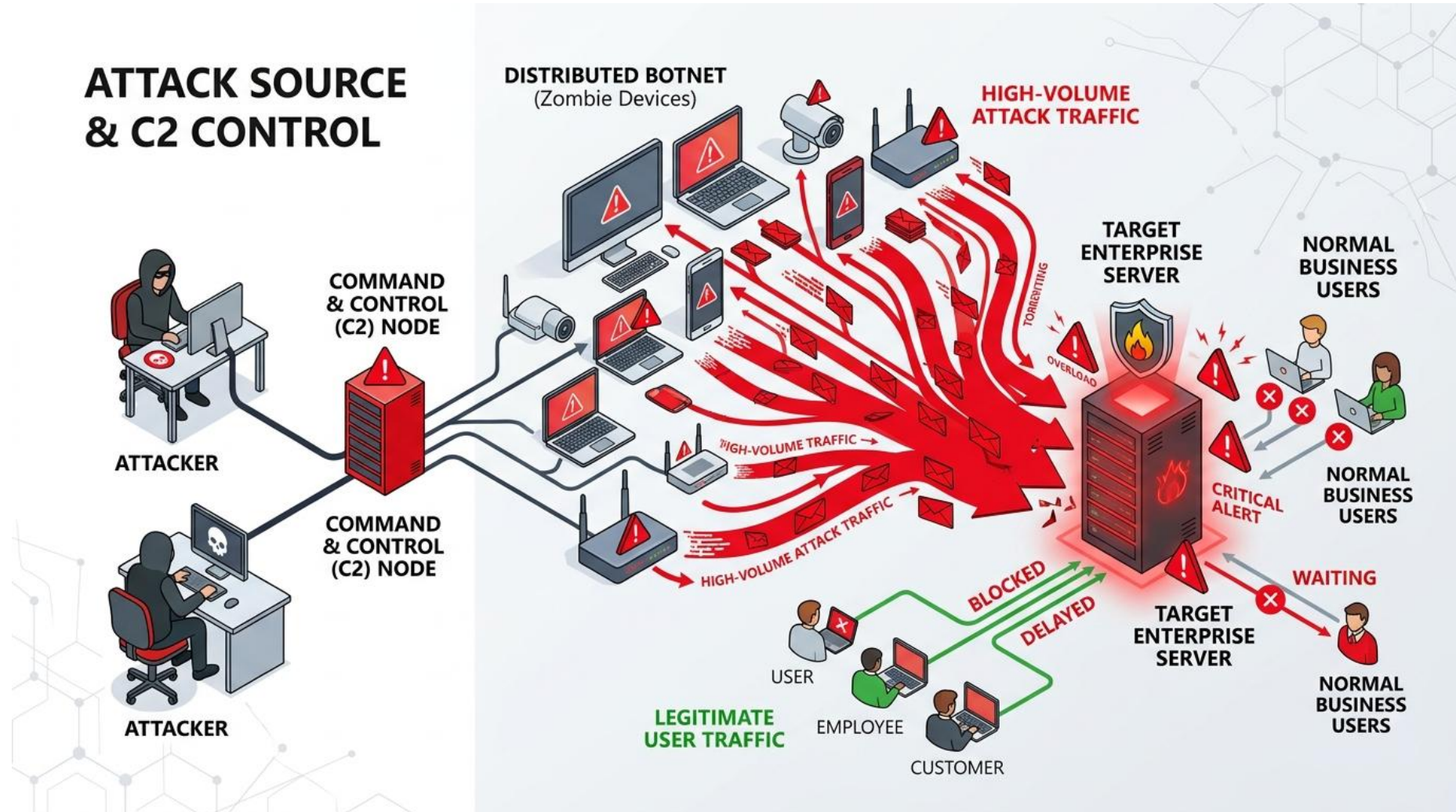
DDoS Attack and DDoS Mitigator



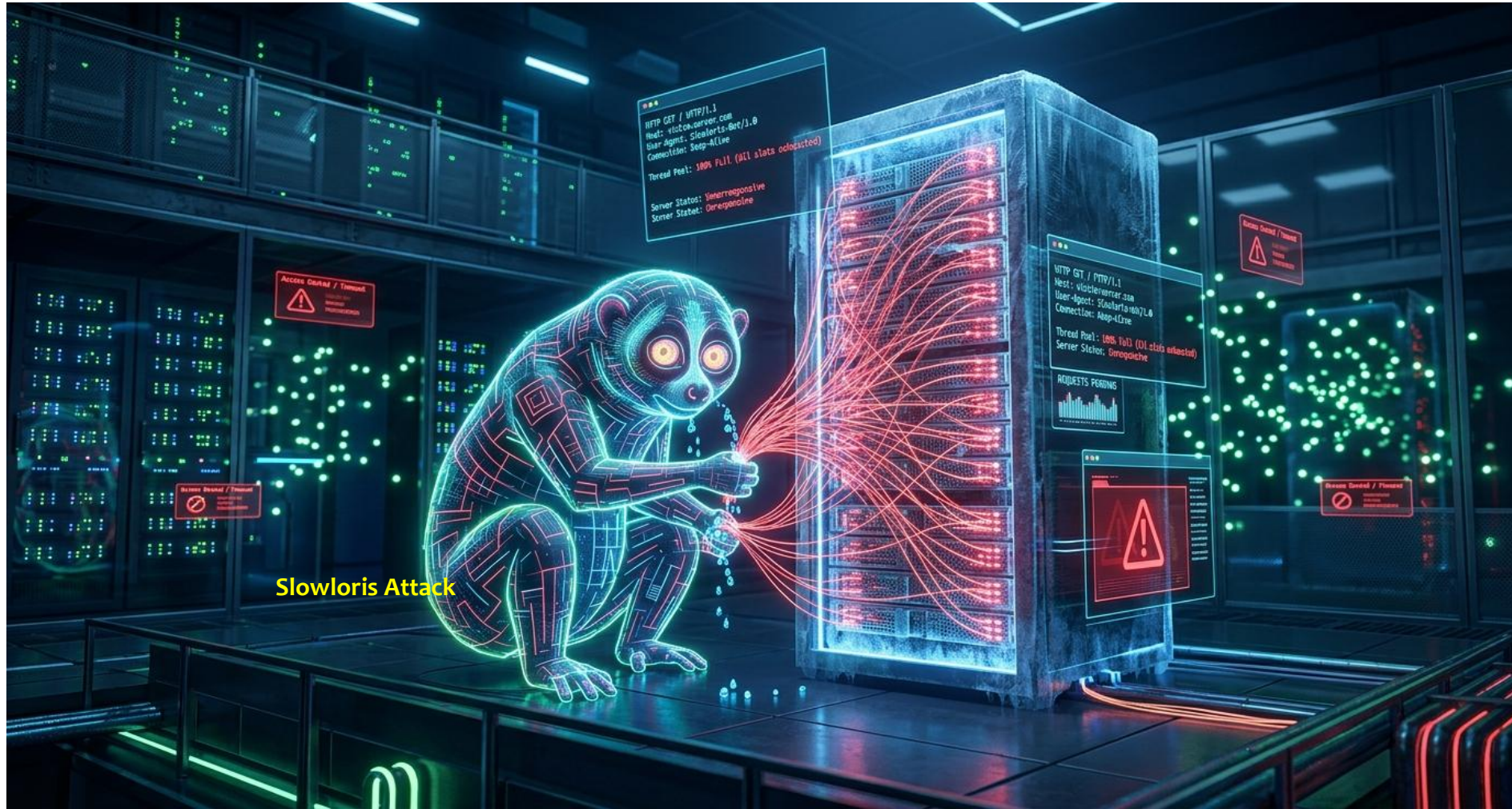
1. DDoS Attack & Mitigator(1) – Attack(1)



1. DDoS Attack & Mitigator(1) – Attack(2)



1. DDoS Attack & Mitigator(1) – Attack(3)



1. DDoS Attack & Mitigator(1) – Attack(4)

Types of DDoS Attacks

Volumetric attacks(L3/L4)

- Flooding the network with a large volume of data to consume available bandwidth..

Ex) UDP Flood, ICMP Flood(Ping Flood), DNS amplification attack

Protocol attacks(L3/L4)

- Exhausting server or firewall resources by exploiting vulnerabilities in communication protocols..

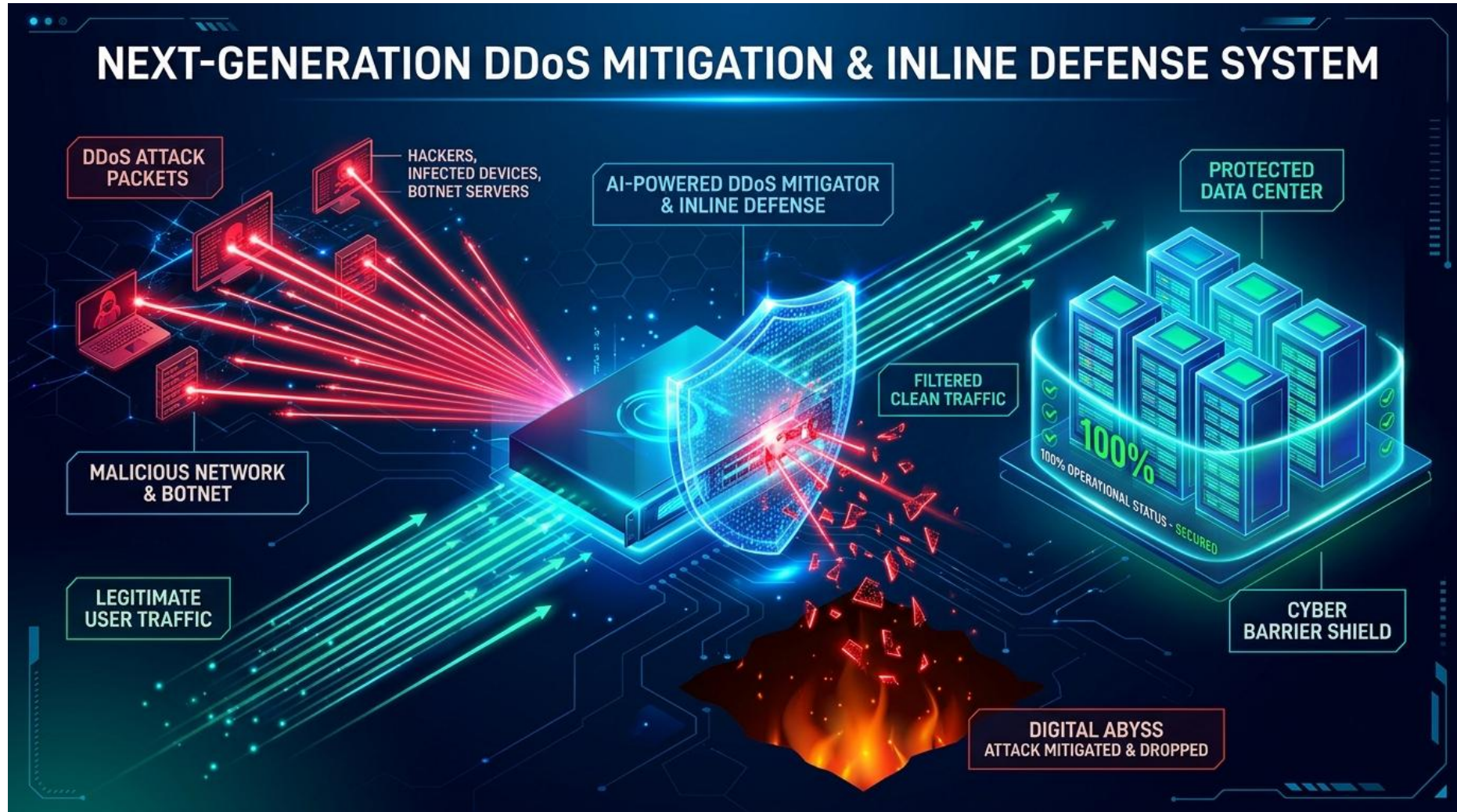
Ex) SYN Flood, Ping of Death, Smurf Attack, Fragmentation Packet Attack

Application layer attacks(L7)

- Mimics the behavior of legitimate users to target specific applications and evade detection..

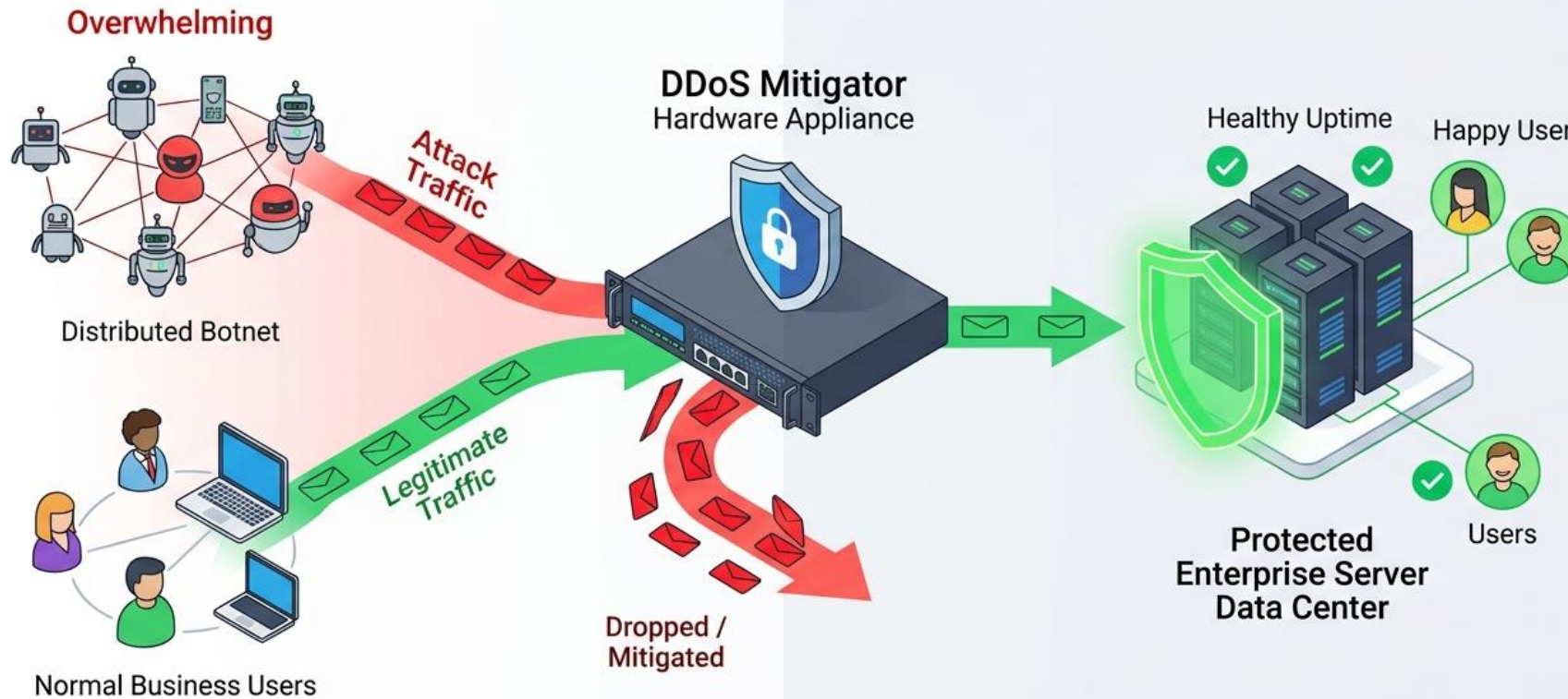
Ex) HTTP GET/POST Flood, Slowloris Attack, XML-RPC Flood, WordPress Administrator Login Flood

1. DDoS Attack & Mitigator(2) – Mitigator(1)

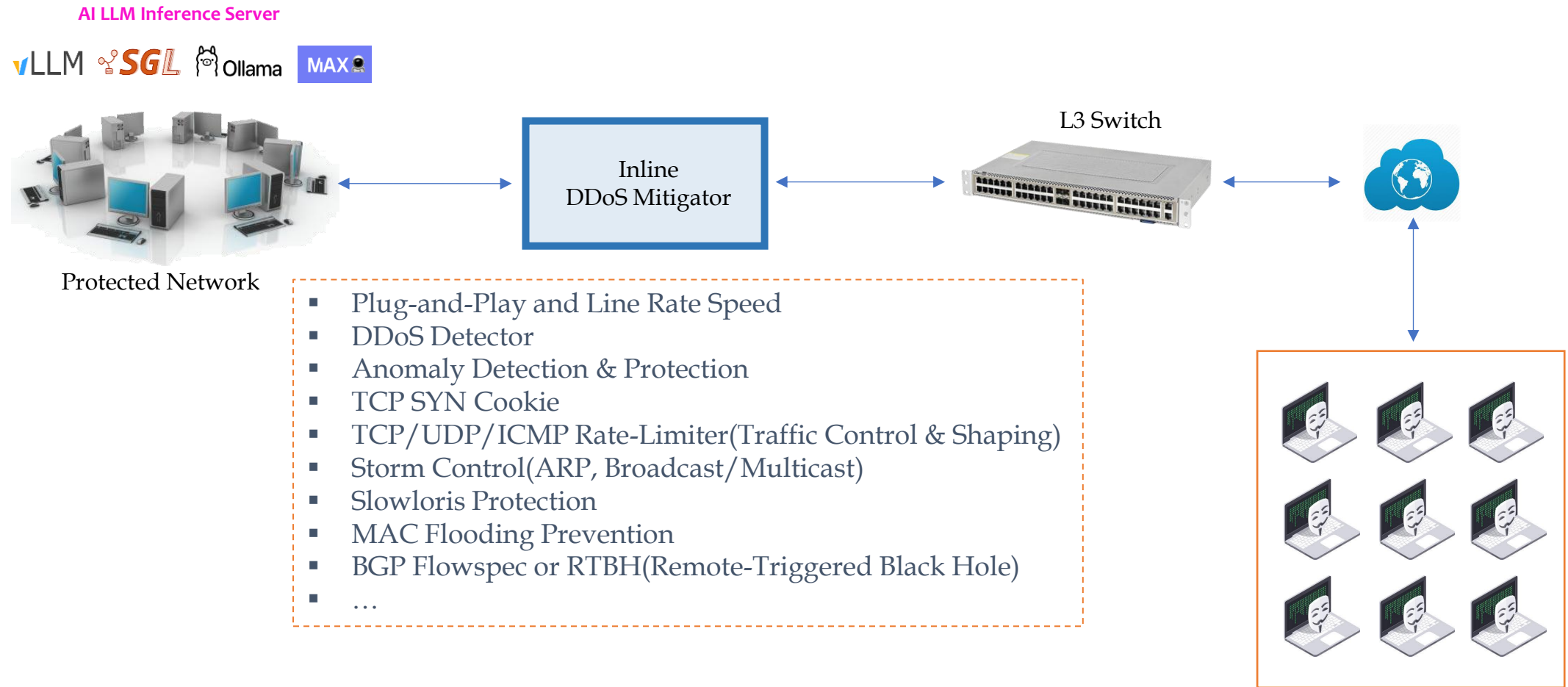


1. DDoS Attack & Mitigator(2) – Mitigator(2)

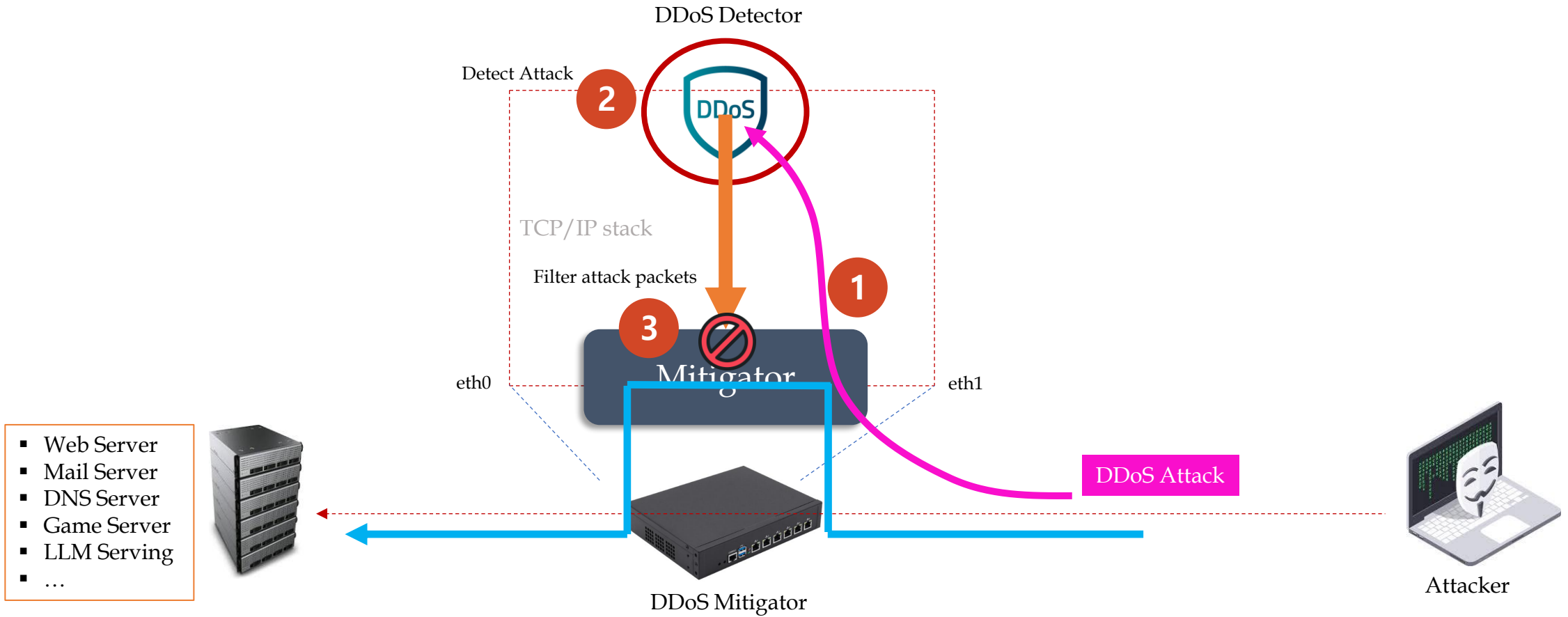
DDoS Mitigator and Inline Traffic Scrubber



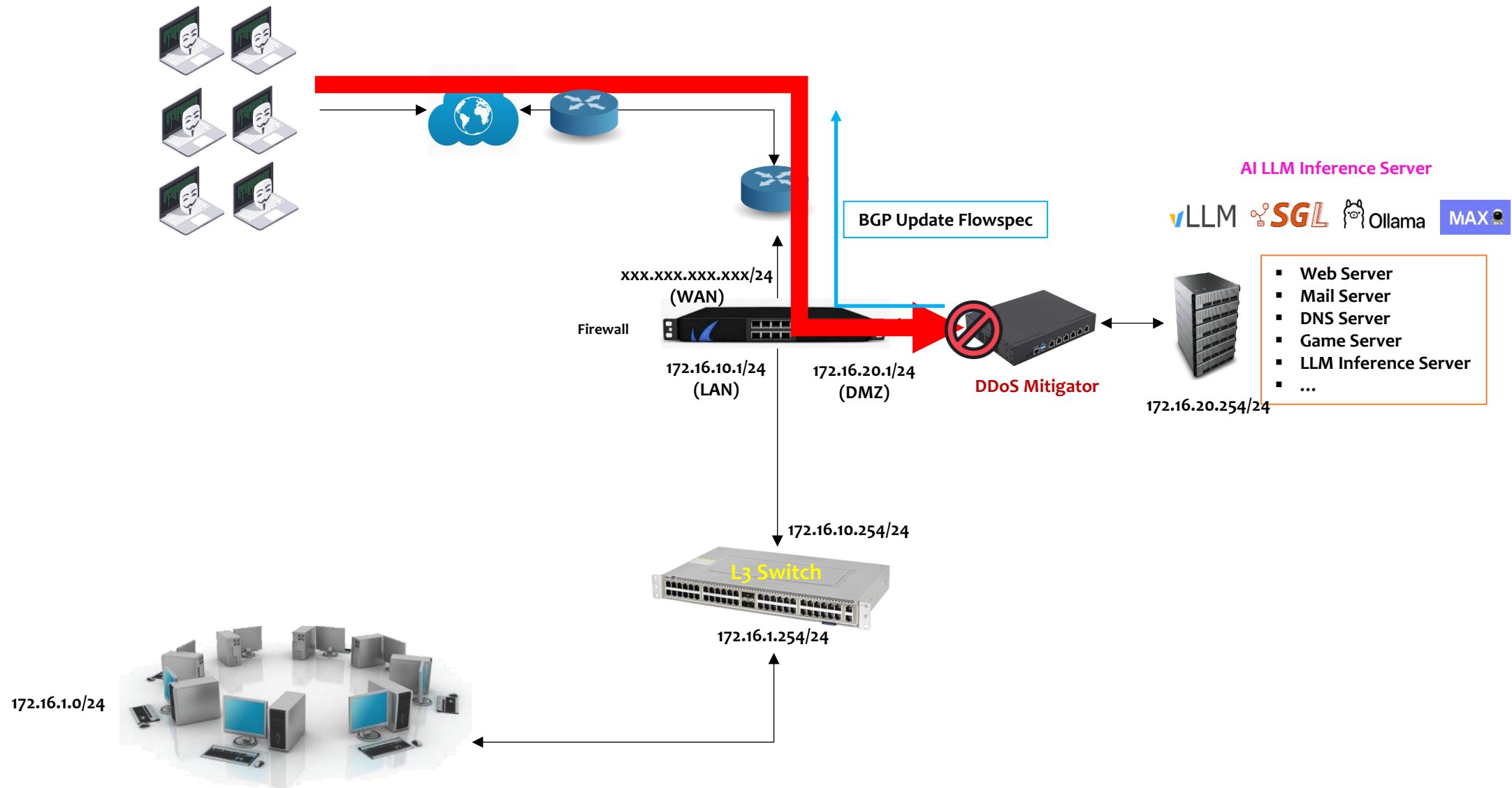
1. DDoS Attack & Mitigator(2) – Mitigator(3)



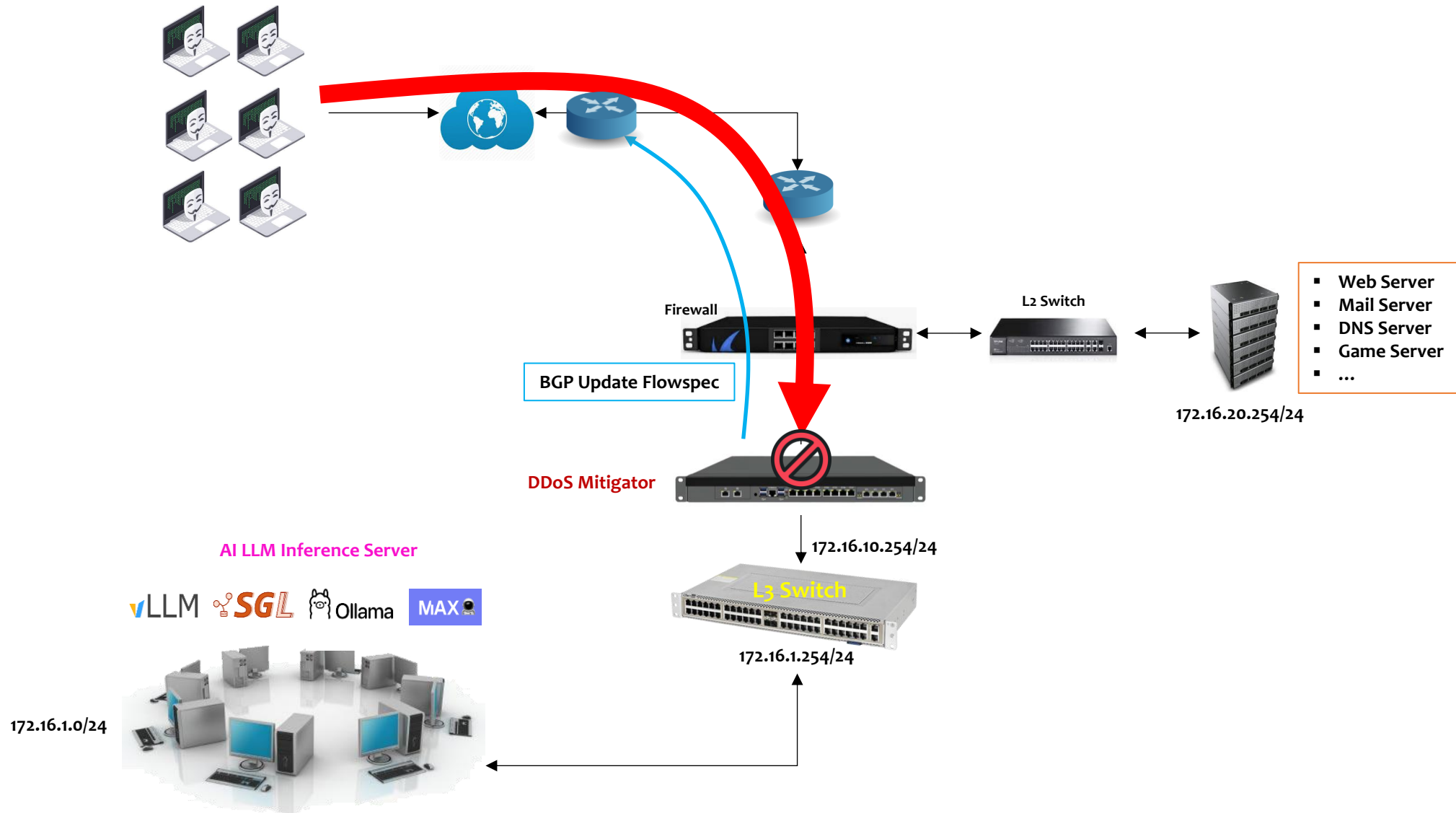
1. DDoS Attack & Mitigator(2) – Mitigator(4)



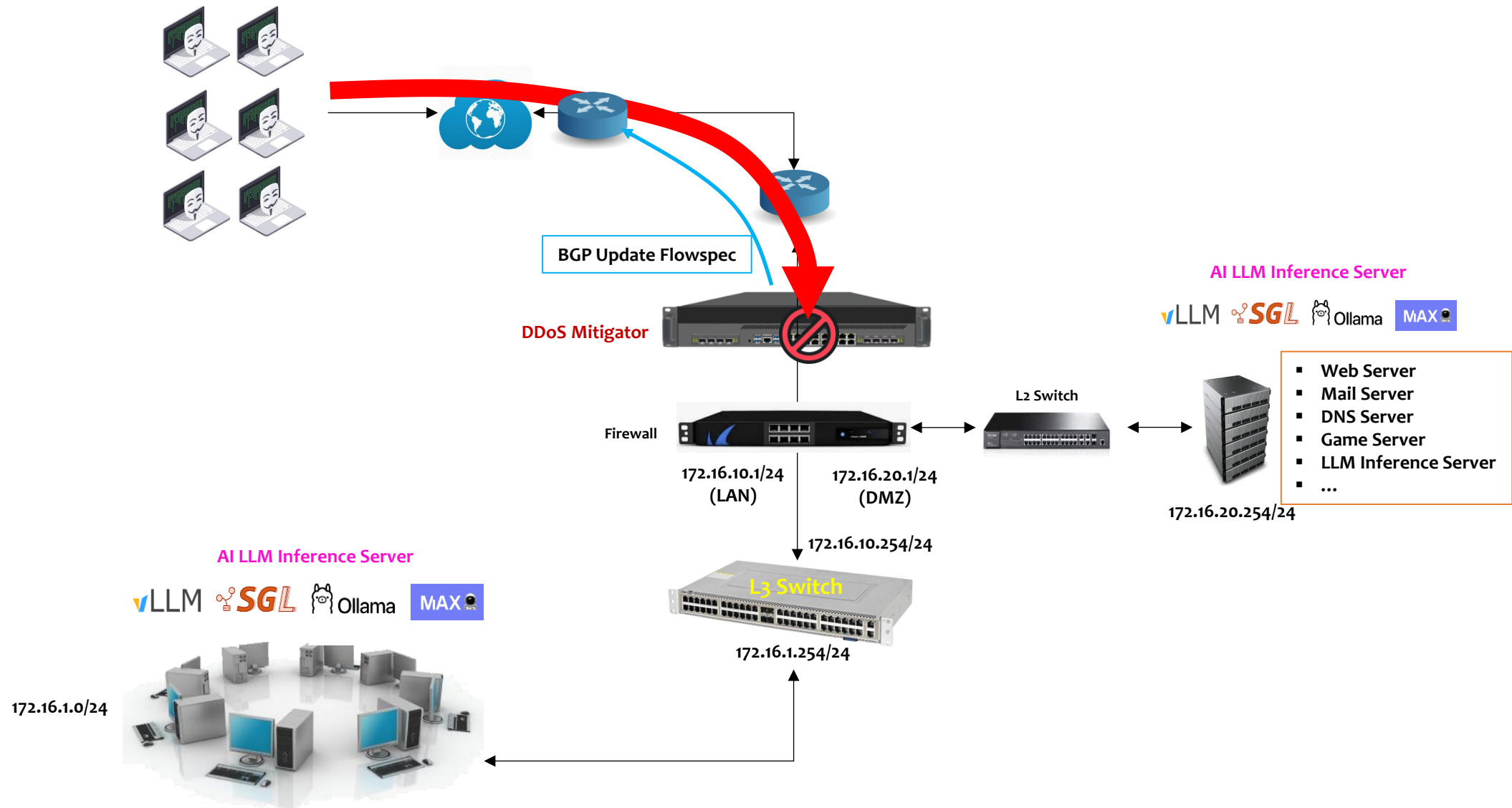
1. DDoS Attack & Mitigator(3) – Network 구성(1)



1. DDoS Attack & Mitigator(3) – Network 구성(2)



1. DDoS Attack & Mitigator(3) – Network 구성(3)



2. Base Technology

DPDK, FD.io VPP and XDP

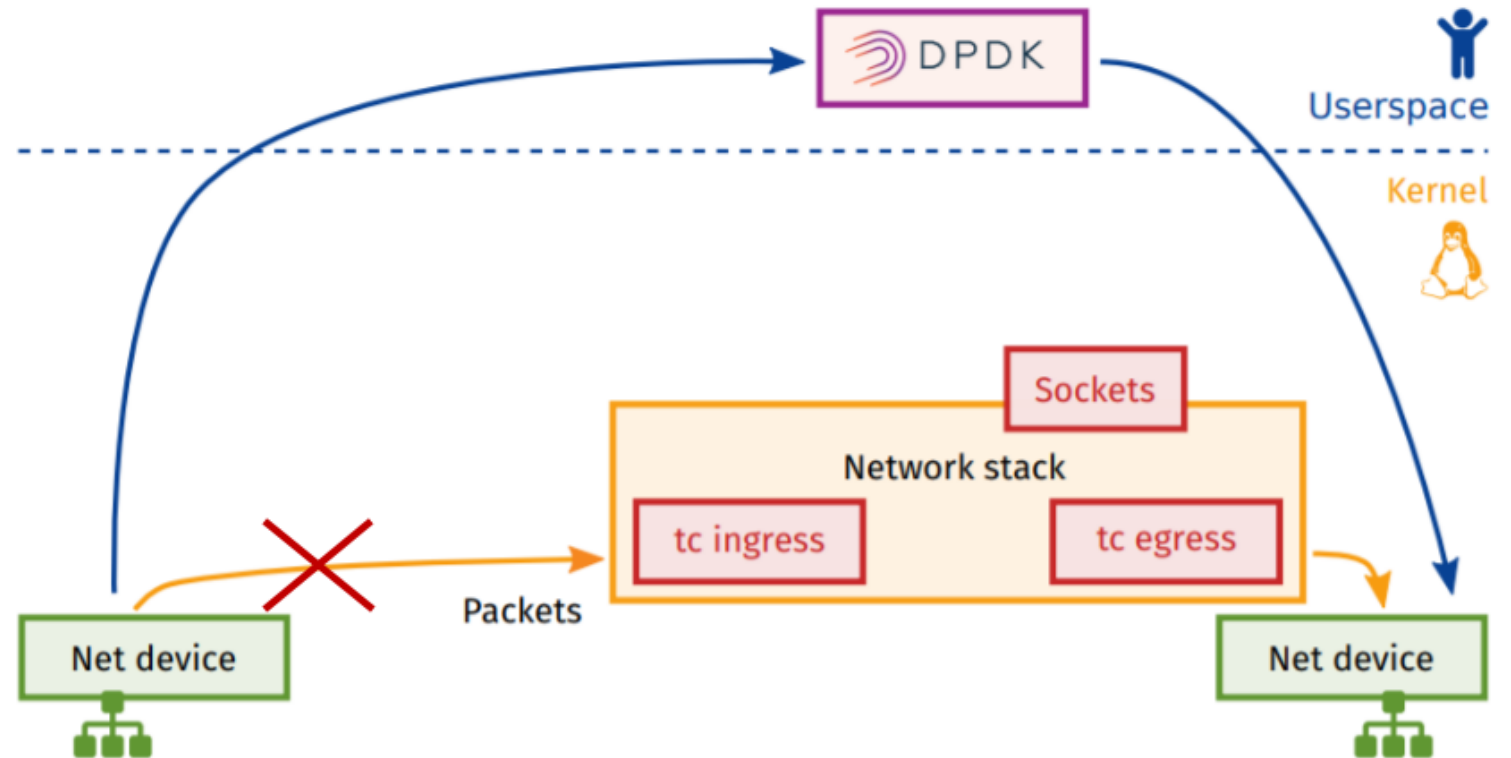
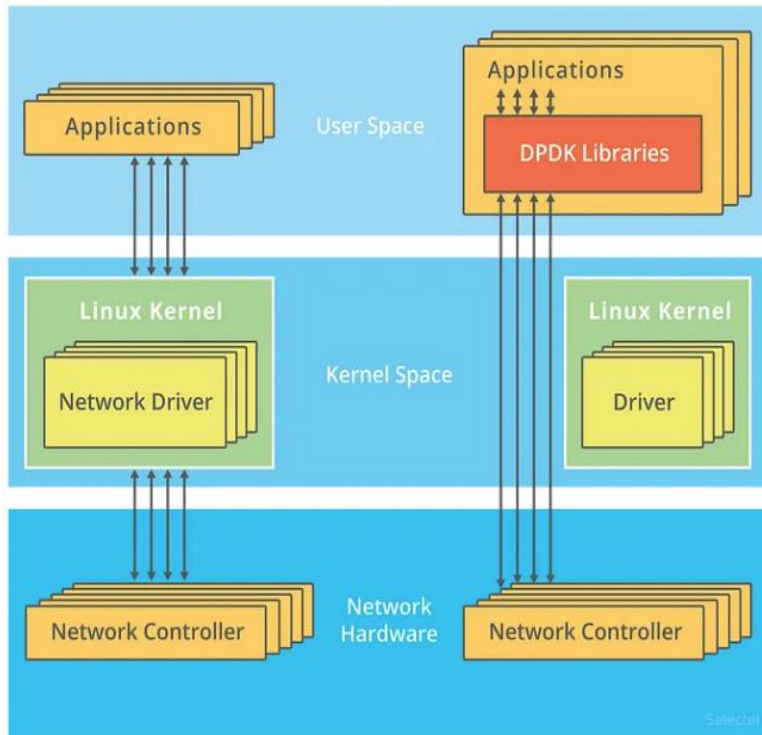
2.1 DPDK

Data Plane Development Kit



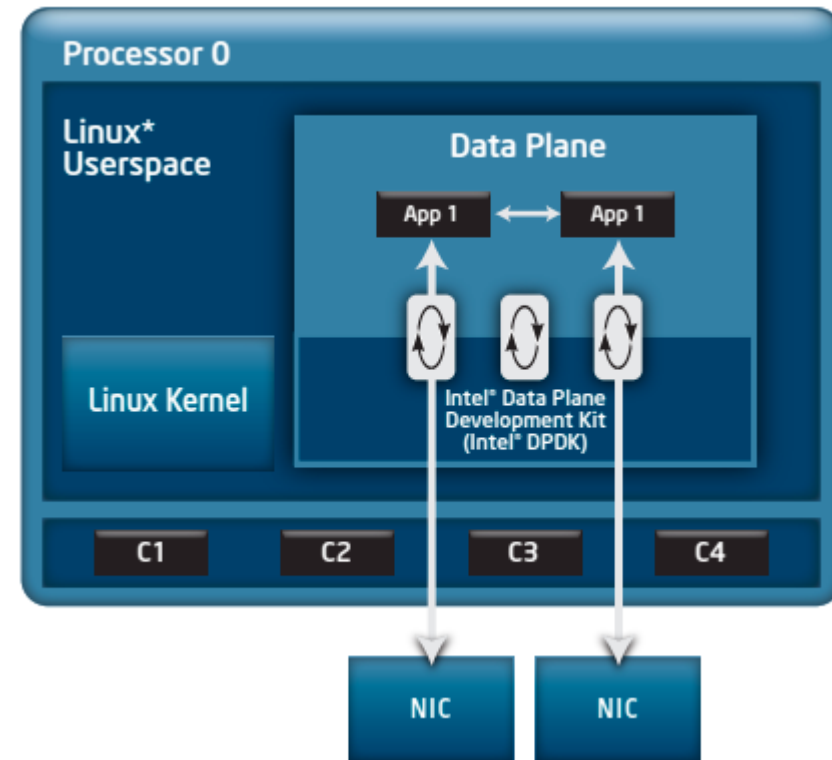
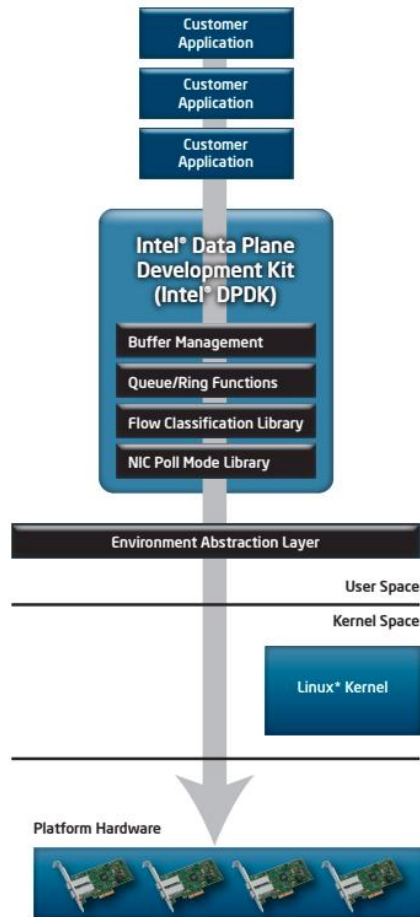
<https://www.dpdk.org/>

2.1 DPDK(1)



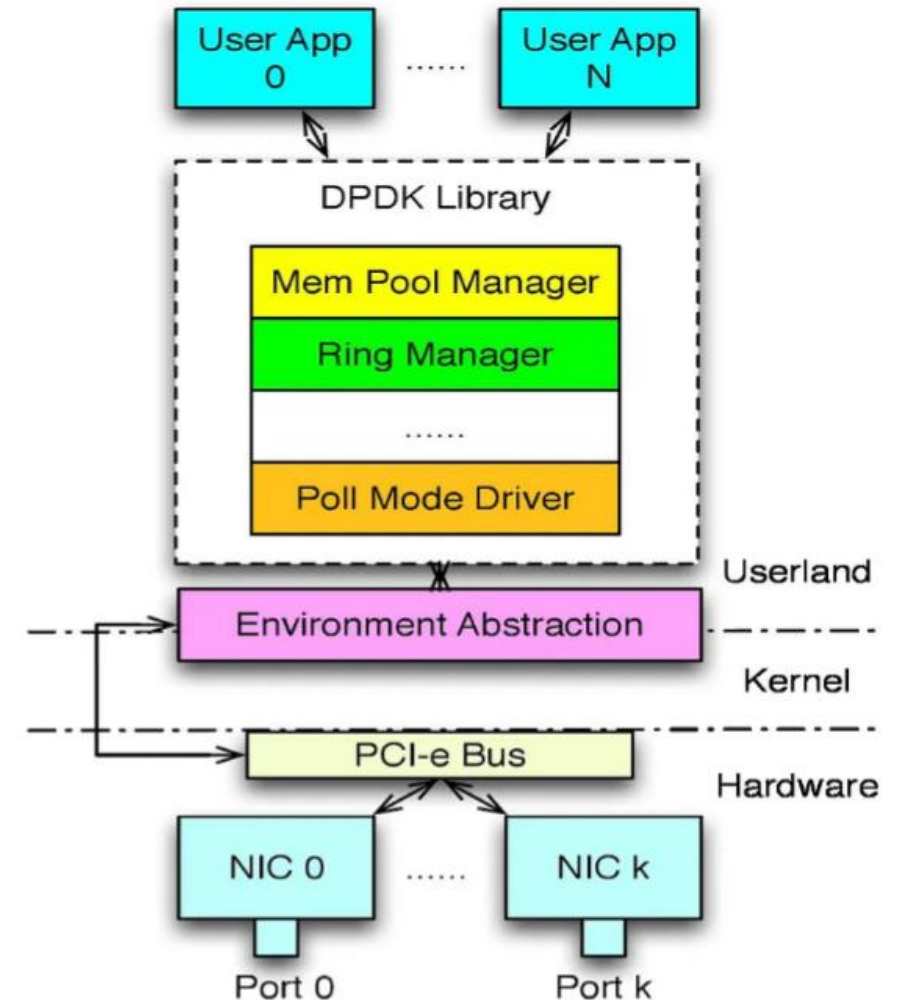
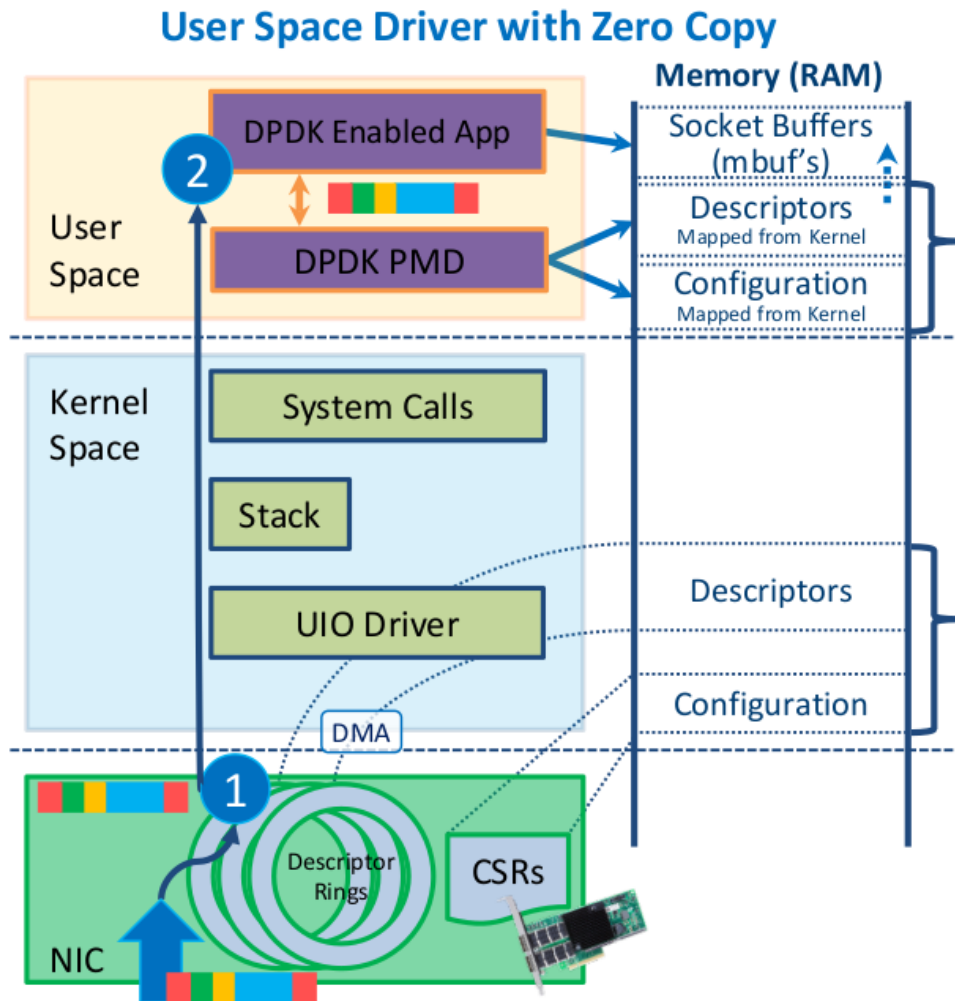
Ref) https://media.frnog.org/FRnOG_28/FRnOG_28-3.pdf

2.1 DPDK(2)

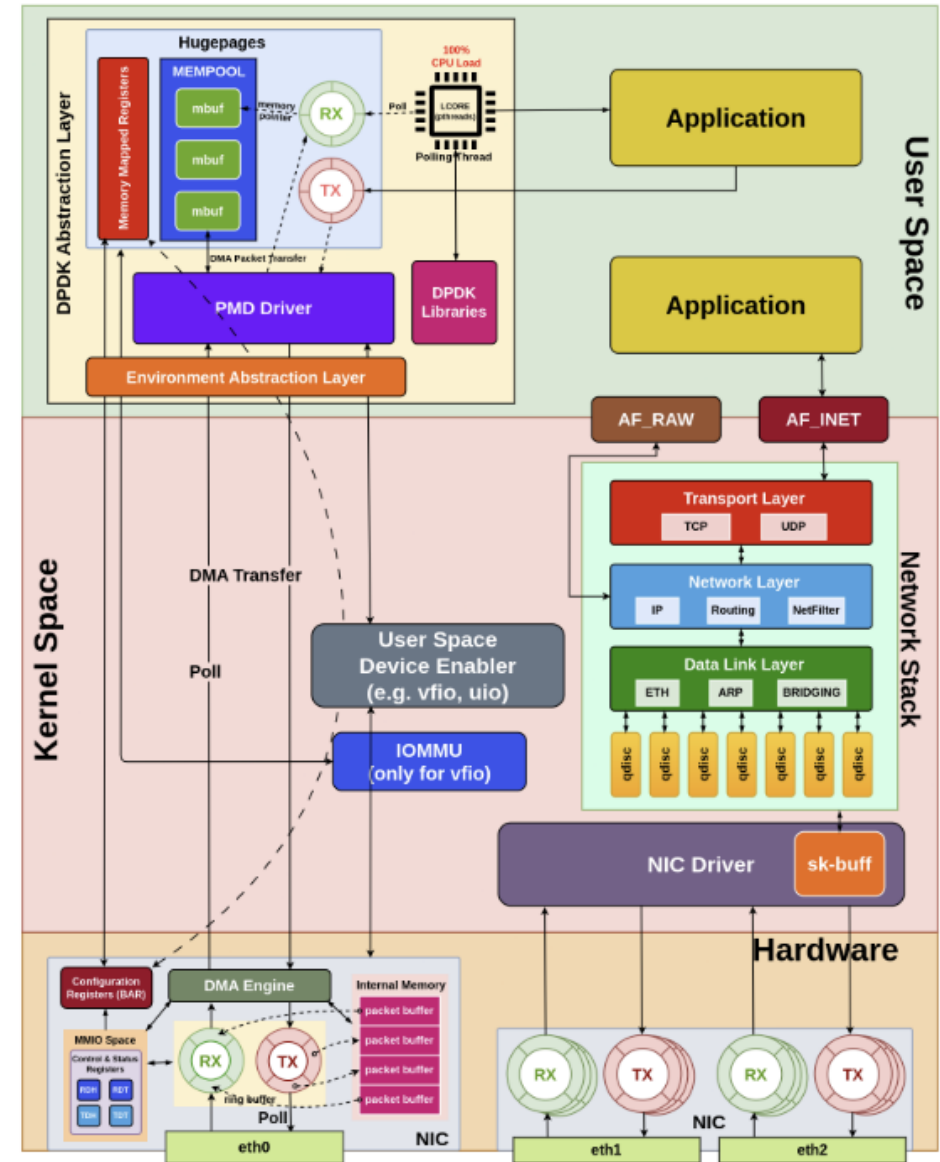
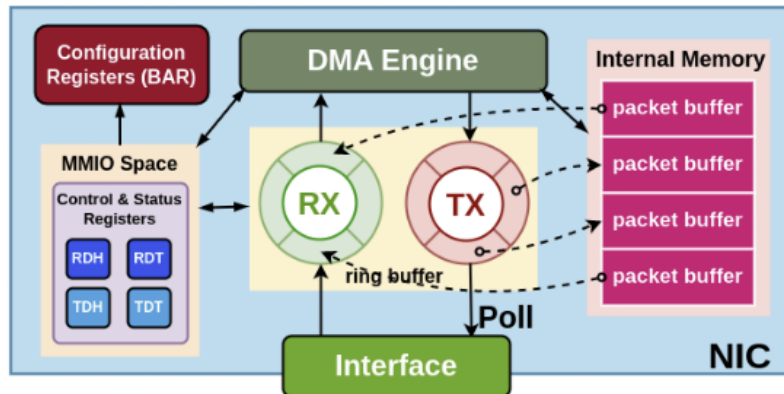
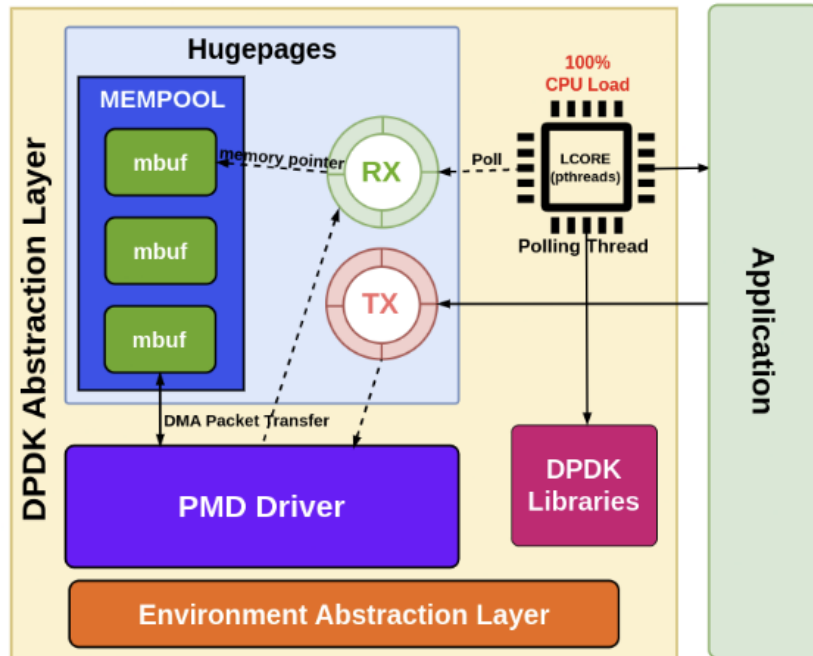


Ref) Impressive Packet Processing Performance Enables Greater Workload Consolidation, Intel

2.1 DPDK(3)



2.1 DPDK(4)



Ref) <https://docs.ngkore.com/blogs/dpdk/dpdk.html#hardware-requirements>

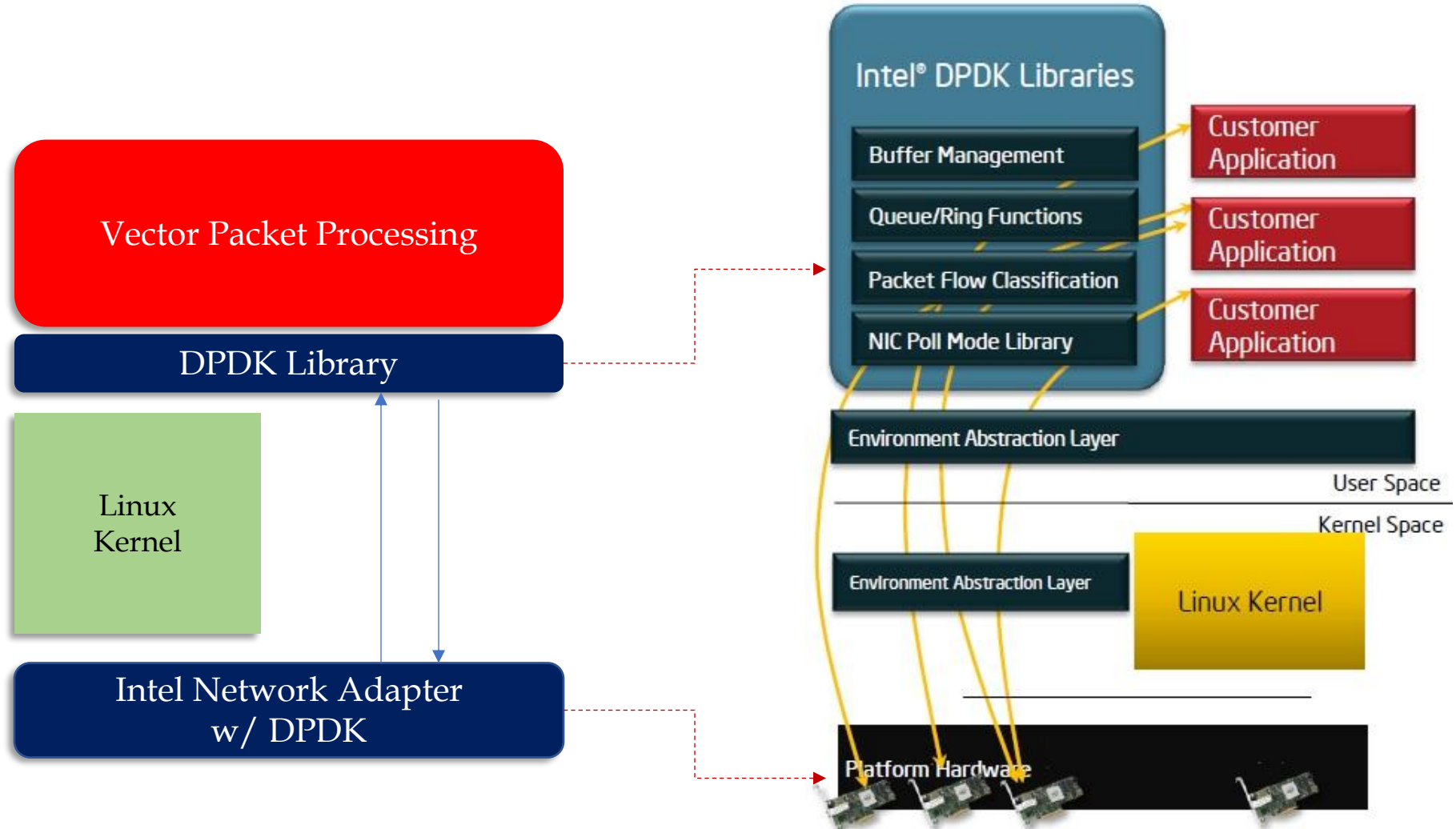
2.2 FD.io VPP

Vector Packet Processing

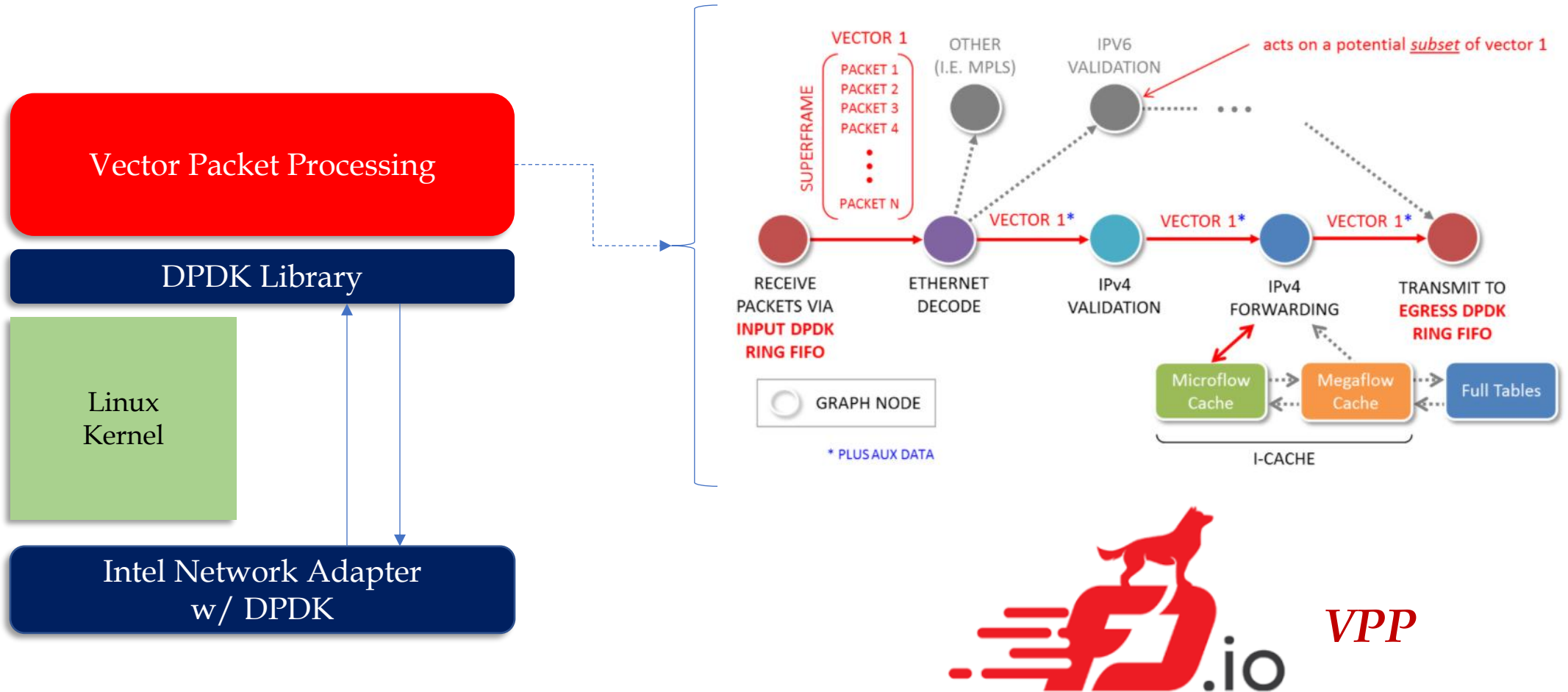


<https://fd.io/>

2.2 FD.io VPP(1)

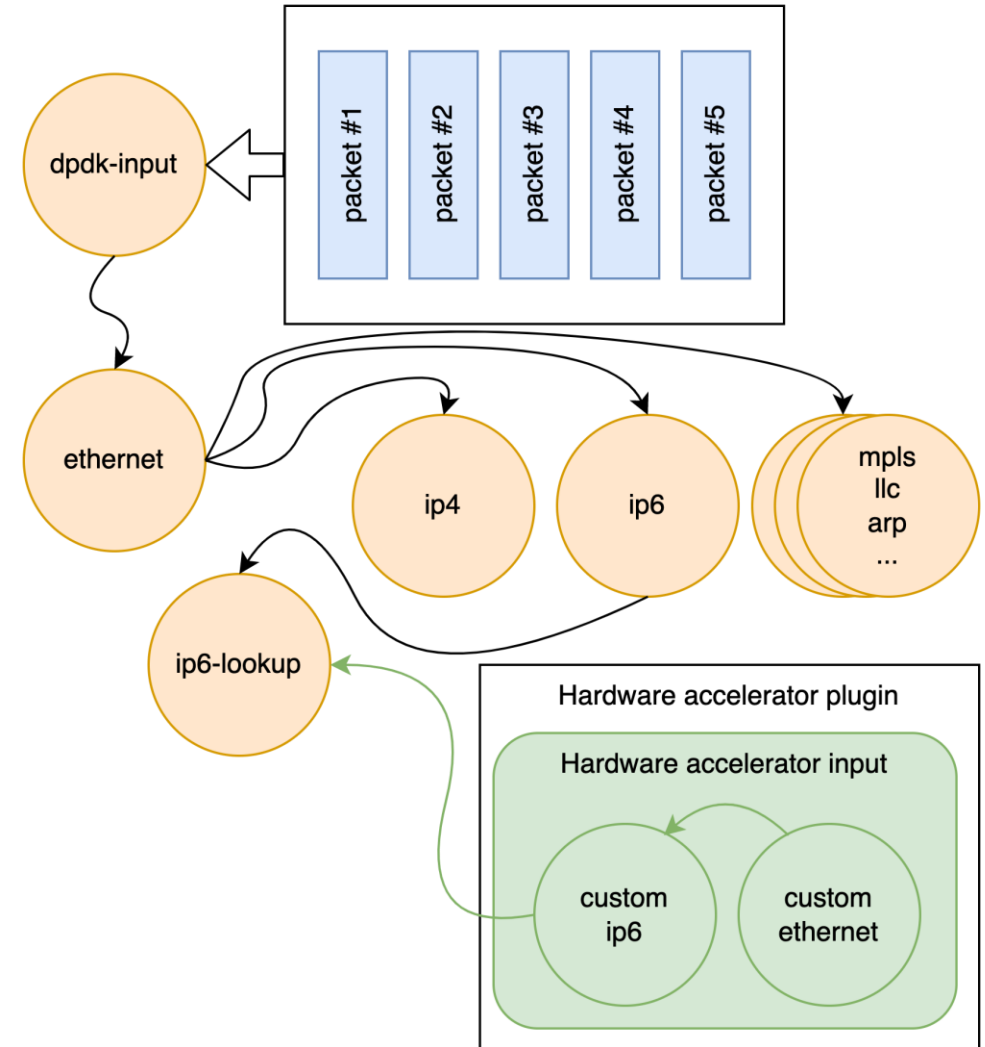
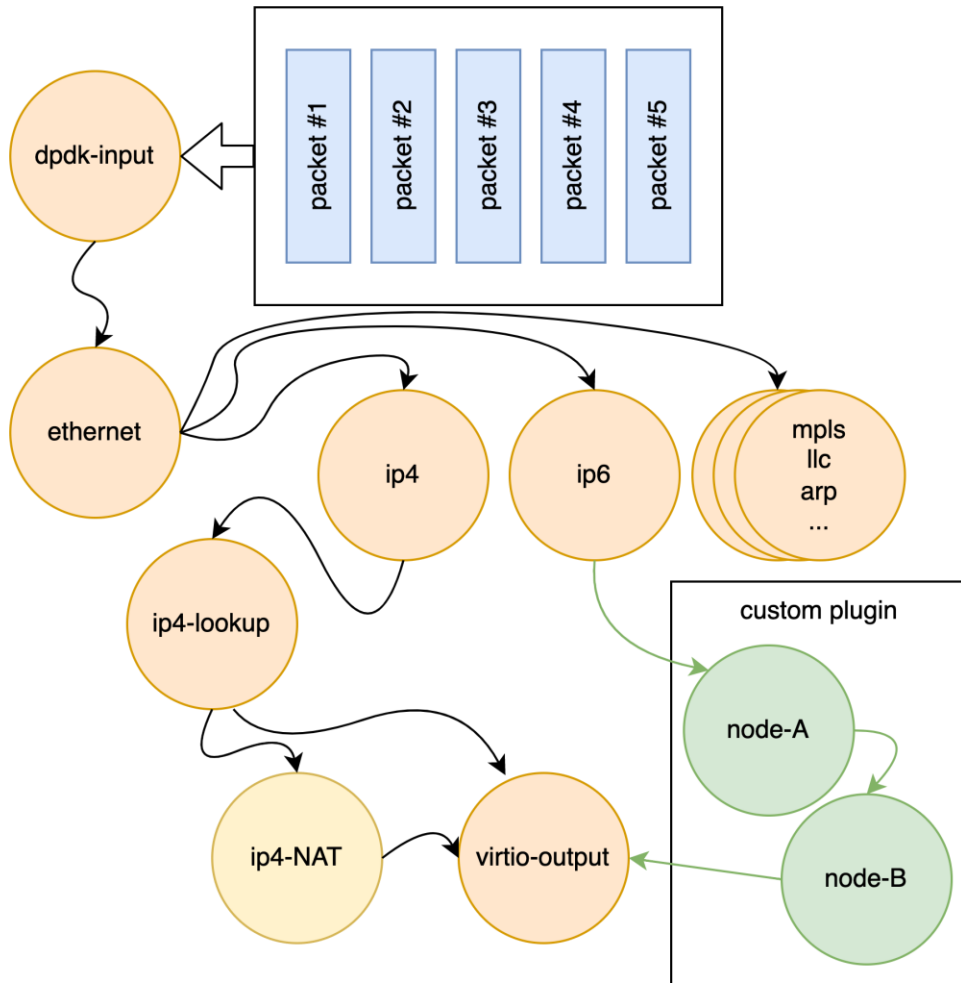


2.2 FD.io VPP(2)



2.2 FD.io VPP(3)

Ref <https://fd.io/technology/>



2.2 FD.io VPP(4) – Applications Area(1)



Routers/Universal CPE etc.

FD.io VPP supports entry hardware options from number of hardware vendors for building Customer Premise Equipmen FD.io based commercial options are available from vendors such as Netgate with TNSR, Cisco with the ASR 9000 and Carri Grade Services Engine, PANTHEON.tech with StoneWork Enterprise, and many more.

Commercial implementations integrate Strong Swan for IKE.

The implementations are accelerated with [DPDK](#) Cryptodev for whole platform crypto.

Featuring:

Data Plane	Control Plane
-----	-----
L2 Switch	Netconf/Yang
VLAN/ Q-inQ	BGP
L3 Router	IKEv2
NAT	
ACL (mac, ip, port)	



Broadband Network Gateway

FD.io VPP has a growing list of network traffic management and security features to support gateway uses cases such as Broadband Network Gateway.

Featuring:

Data Plane	Control Plane
-----	-----
L2 Switch	Netconf/Yang
L3 Router	BGP
Classification	
hQoS	
ACL	
TM (Policing, Metering)	



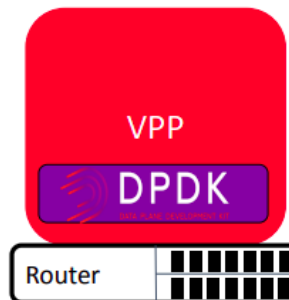
Cloud Load Balancer

FD.io VPP has a rich set of plugin's to enhance its capabilities. Cloud load-balancing is just one of number of feature enhancin plugins available to the end user.

For example: Google Maglev Implementation, Consistent Hashing, Stateful and stateless load balancing, Kube-proxy integrat

Featuring:

Data Plane	Control Plane
-----	-----
Bonding	K8s Kube-Proxy
VLAN / Q-in-Q	
NAT	
ACL (blacklist)	
TM (policing, metering)	
L4 Load Balancer	



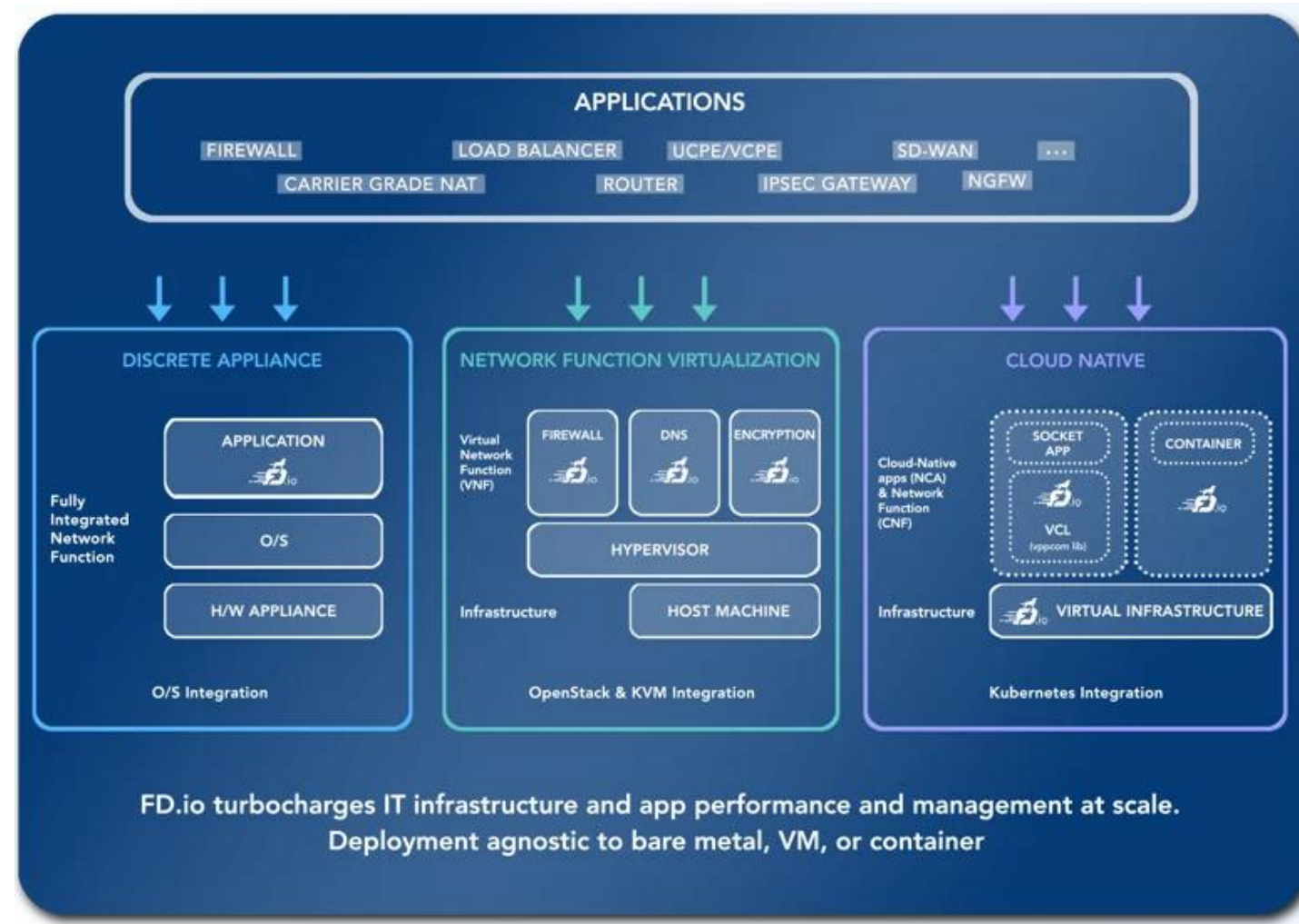
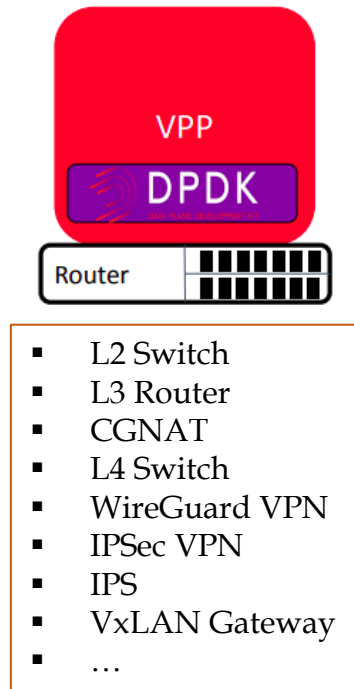
Intrusion Prevention System

FD.io VPP has four different Access Control List technologies; ranging from the simple IP-address whitelisting (called COP) to the sophisticated FD.io VPP Classifiers.

Featuring:

Data Plane	Control Plane
-----	-----
L2 Switch	Netconf/Yang
L3 Router	BGP
Classification	
NAT (mac, ip, port)	
ACL (mac, ip, port)	

2.2 FD.io VPP(4) – Applications Area(2)



Ref) <https://www.netgate.com/resources/articles-vector-packet-processing>

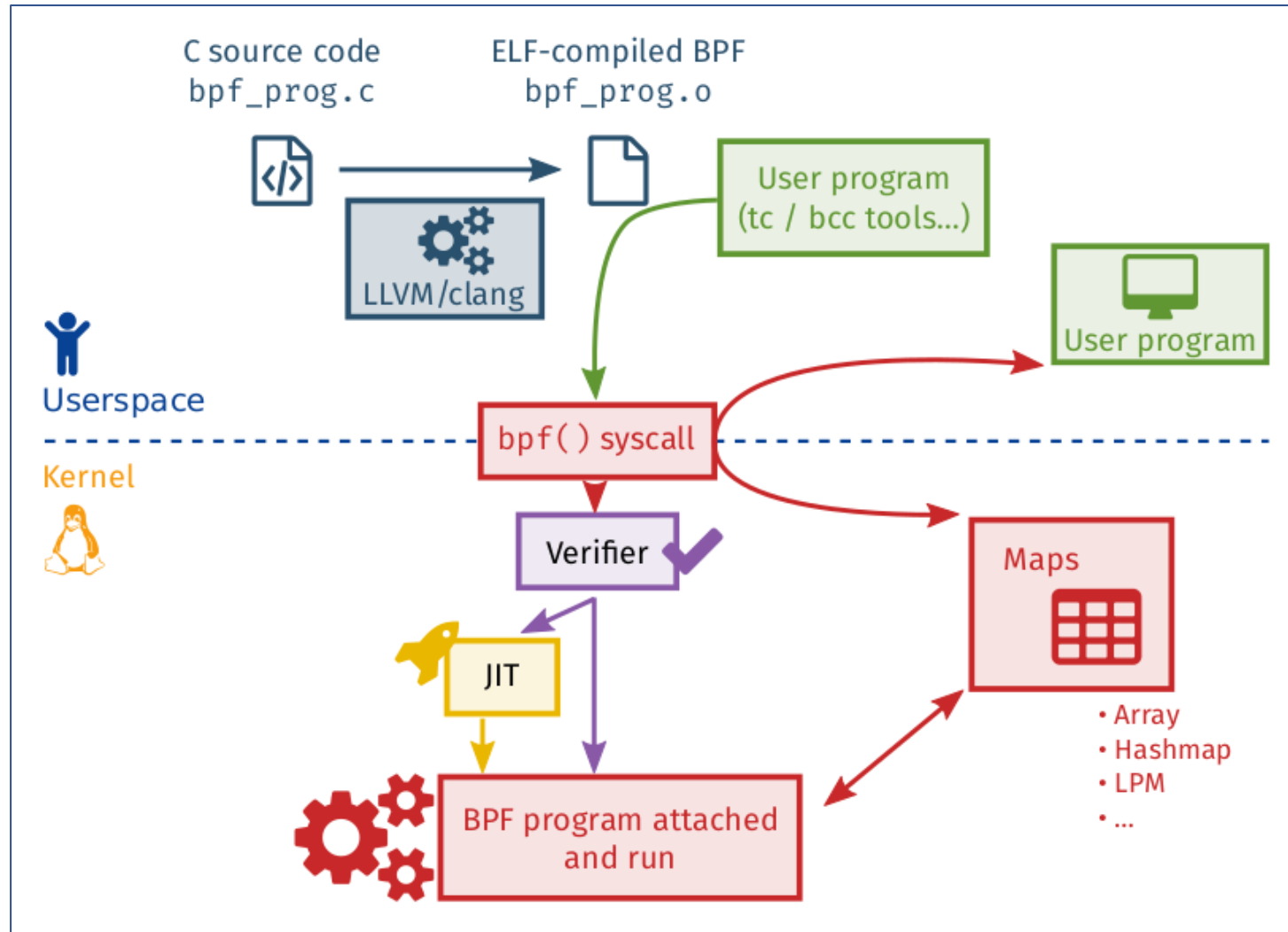
2.3 eBPF XDP

eBPF XDP



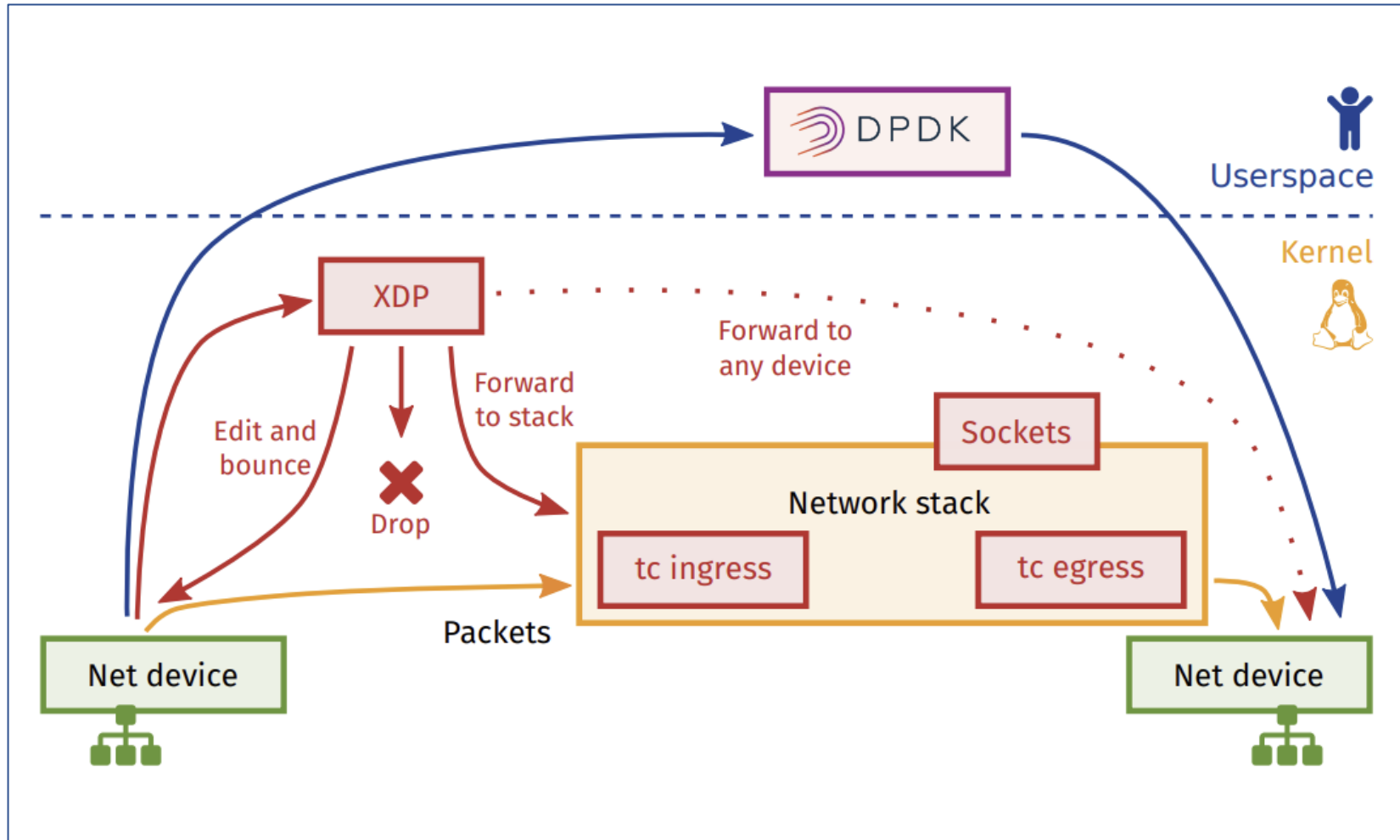
<https://ebpf.io/>

2.3 eBPF XDP(1) – eBPF Concept



Ref) https://media.frnog.org/FRnOG_28/FRnOG_28-3.pdf

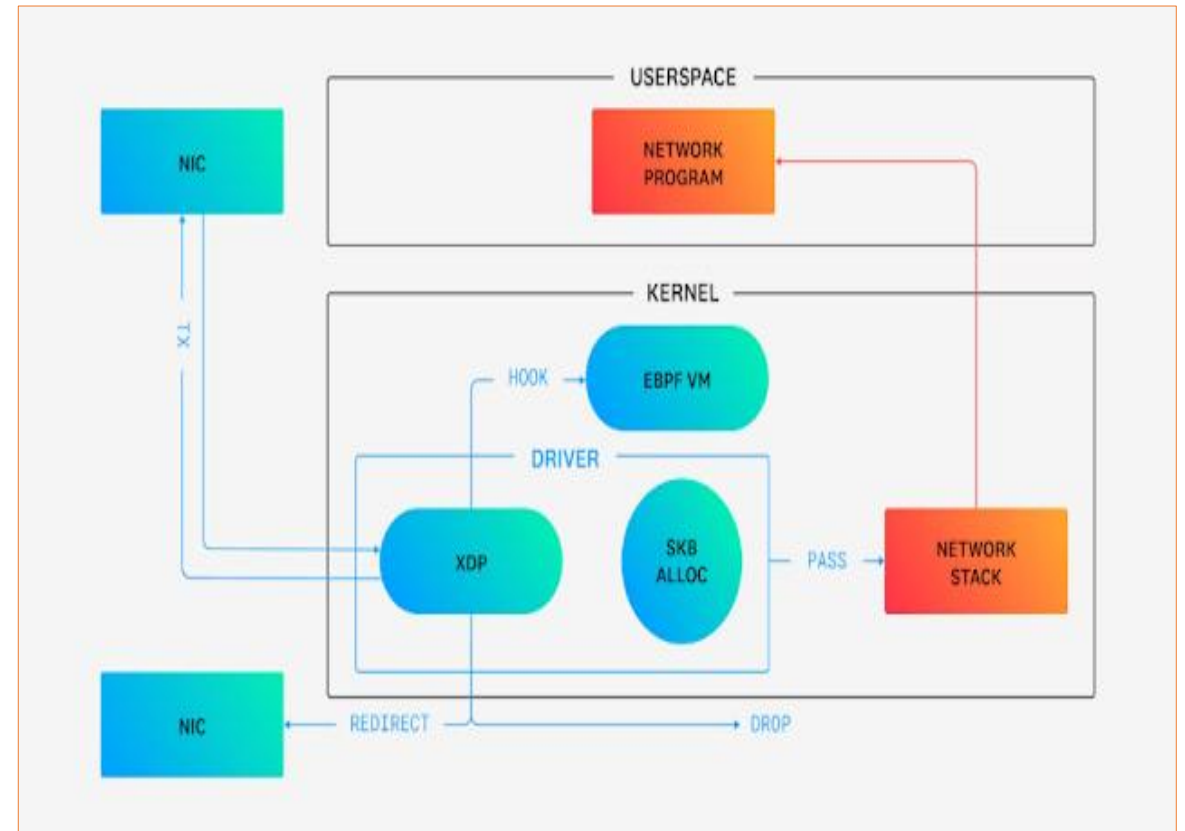
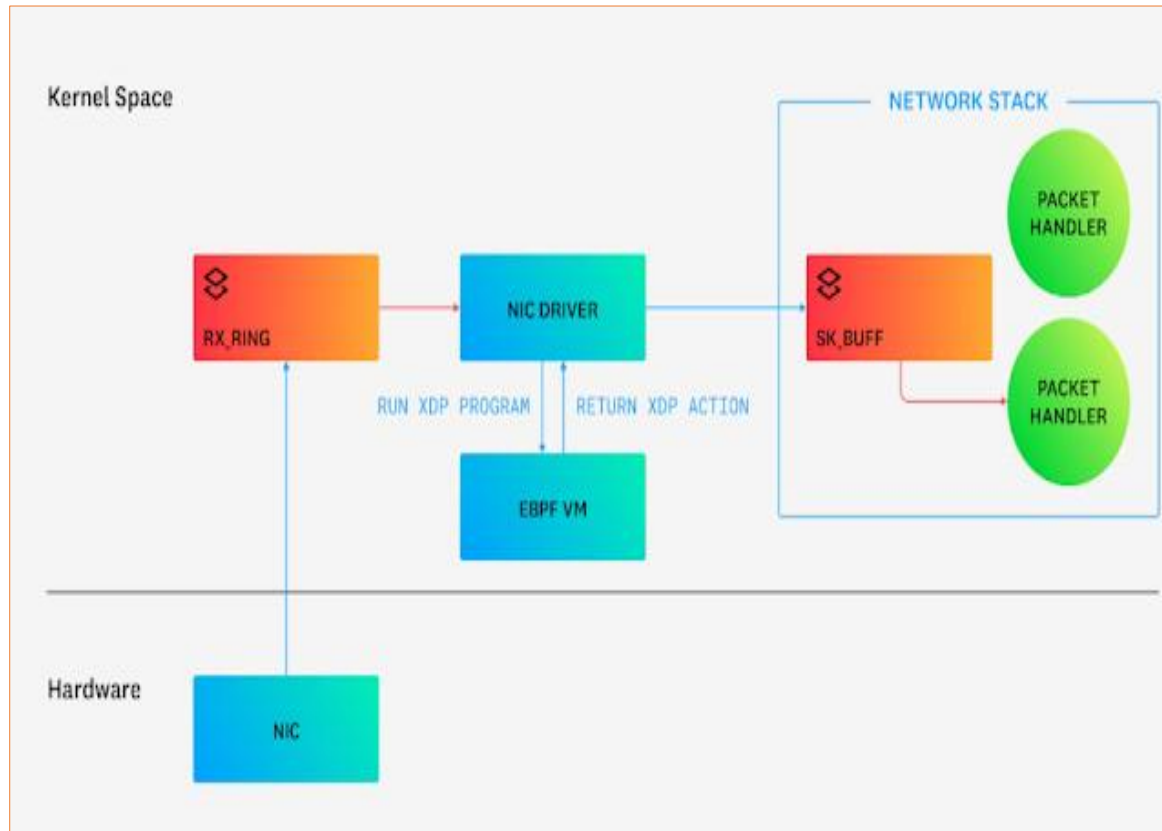
2.3 eBPF XDP(2) – XDP Concept(1)



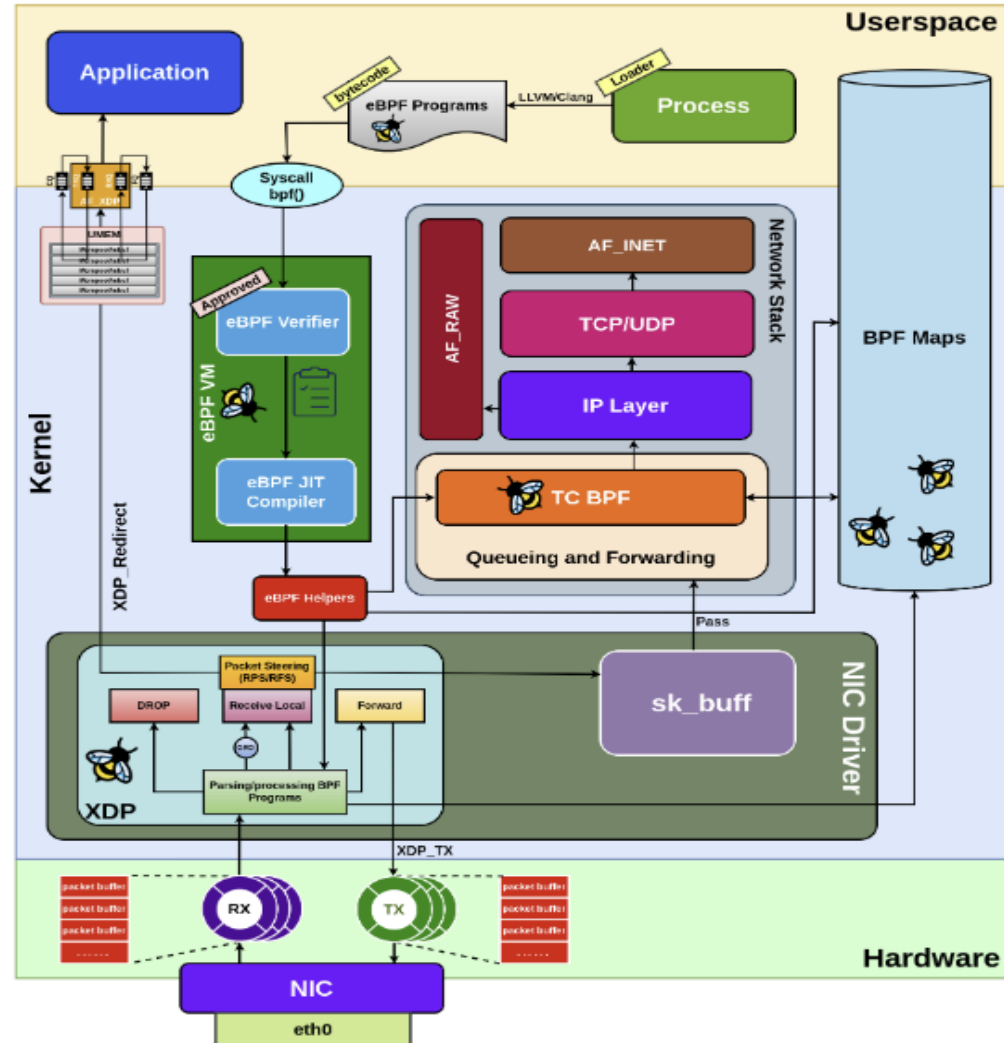
Ref) https://media.frnwg.org/FRnOG_28/FRnOG_28-3.pdf

2.3 eBPF XDP(2) – XDP Concept(2)

Ref) <https://www.datadoghq.com/blog/xdp-intro/>

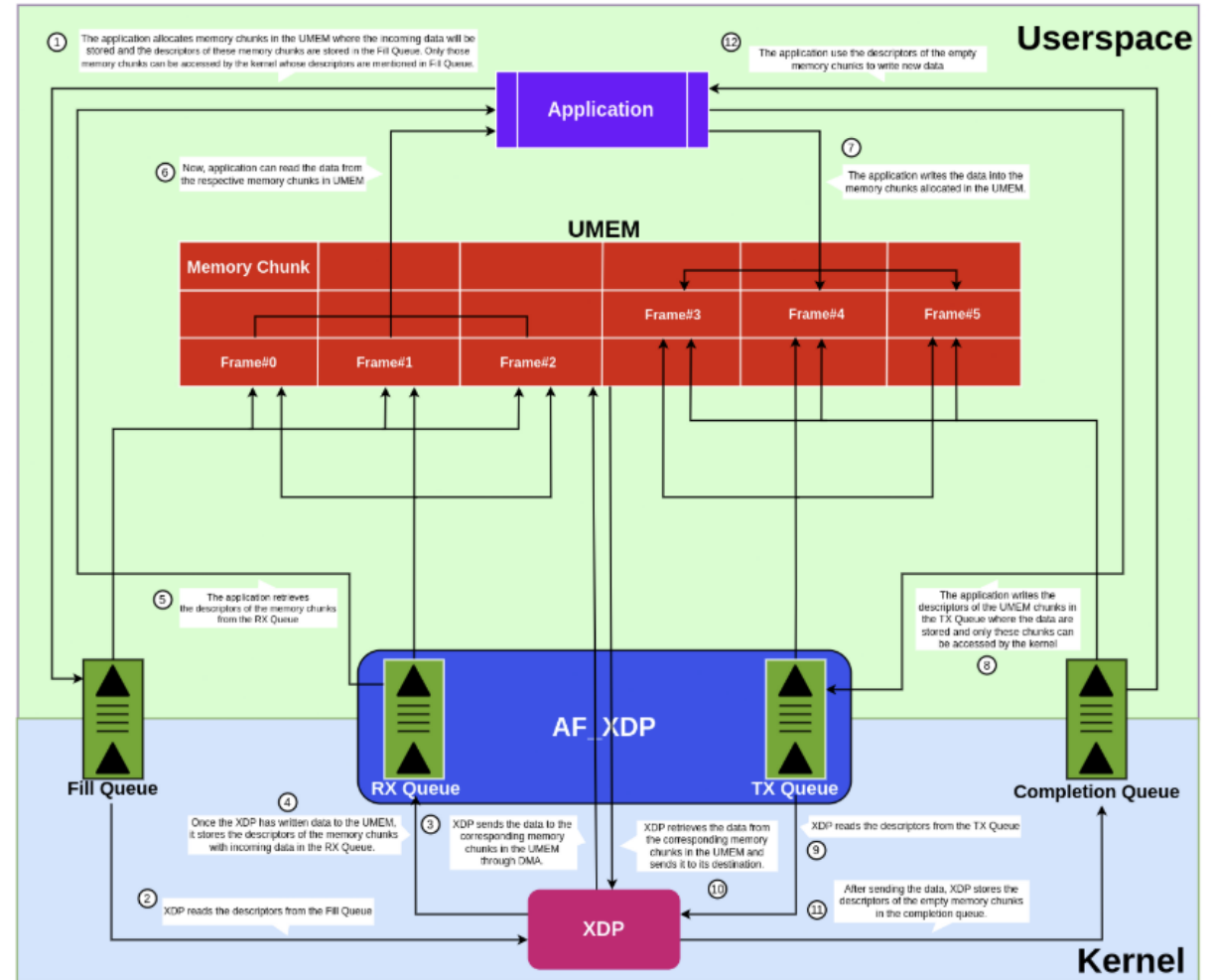
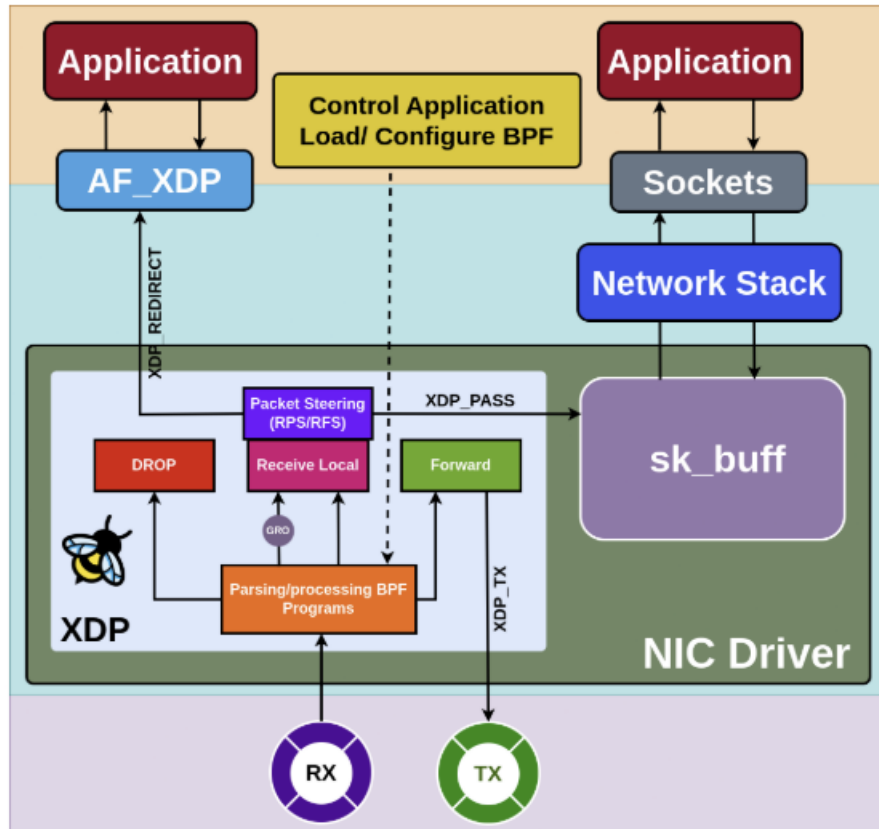


2.3 eBPF XDP(3) – XDP & AF_XDP(1)



Ref) <https://docs.ngkore.com/blogs/afxdp/afxdp.html#hardware-requirements>

2.3 eBPF XDP(3) - XDP & AF_XDP(2)

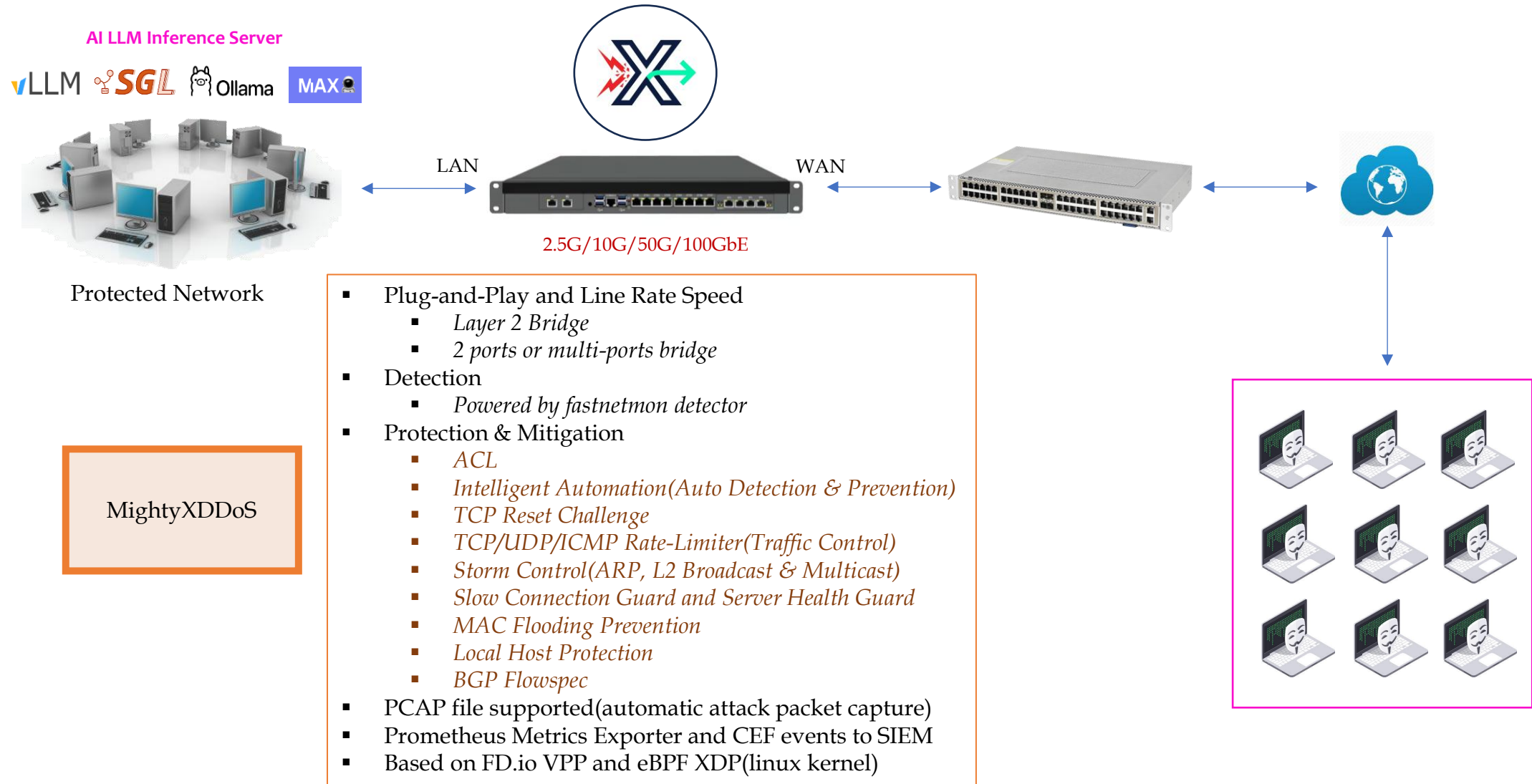


3. MightyXDDoS™

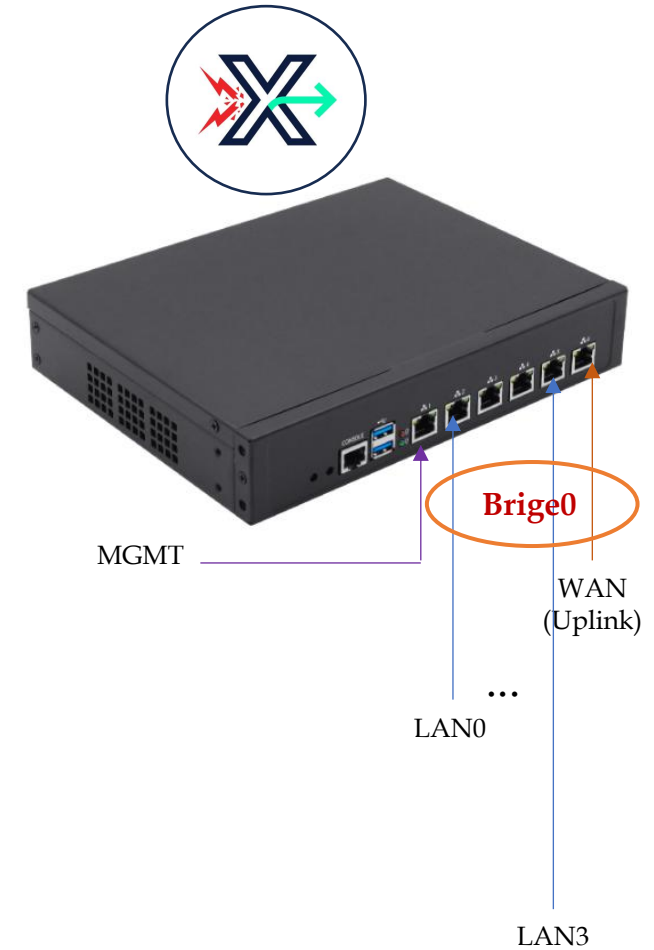
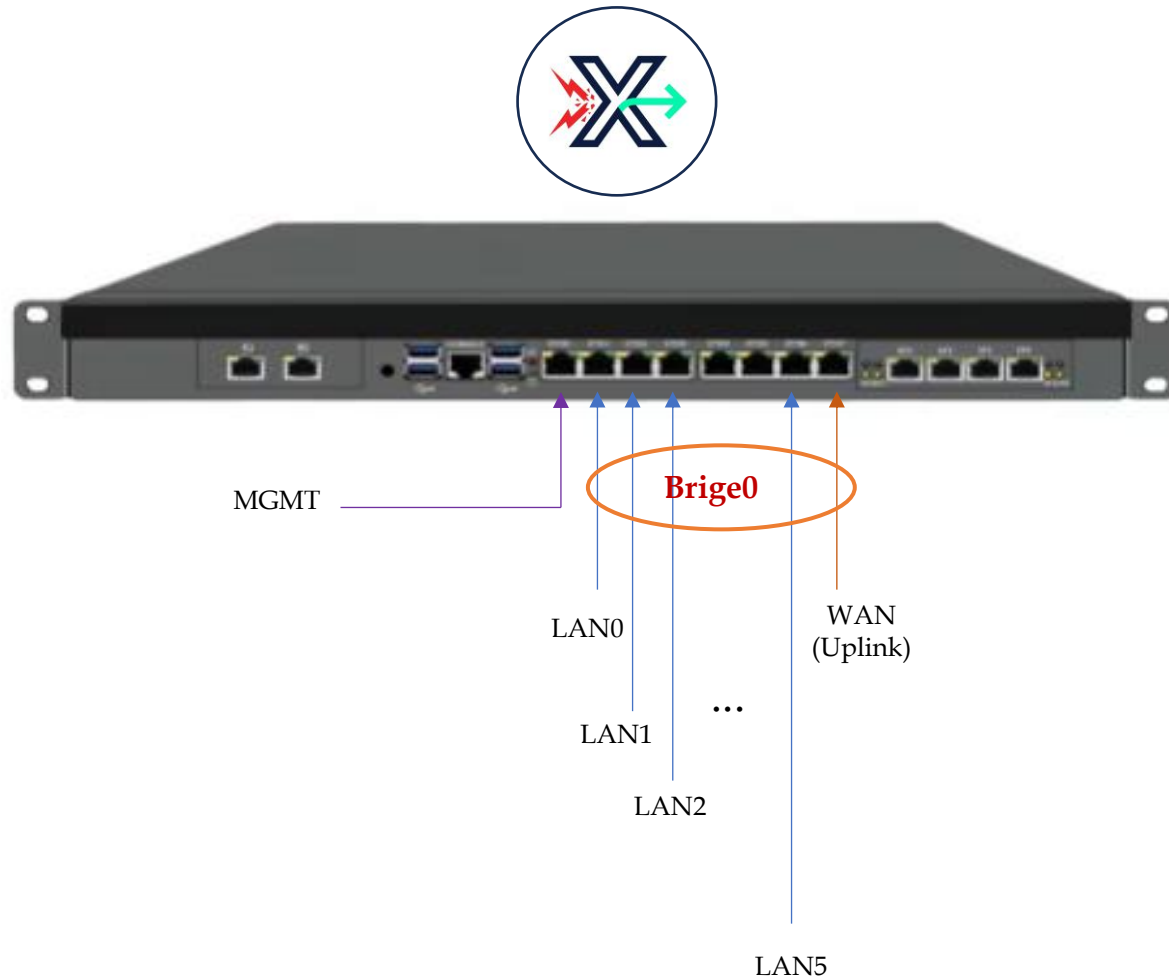


XDDoS Mitigator

3. MightyXDDoS(1) – Inline Mitigator Box(1)

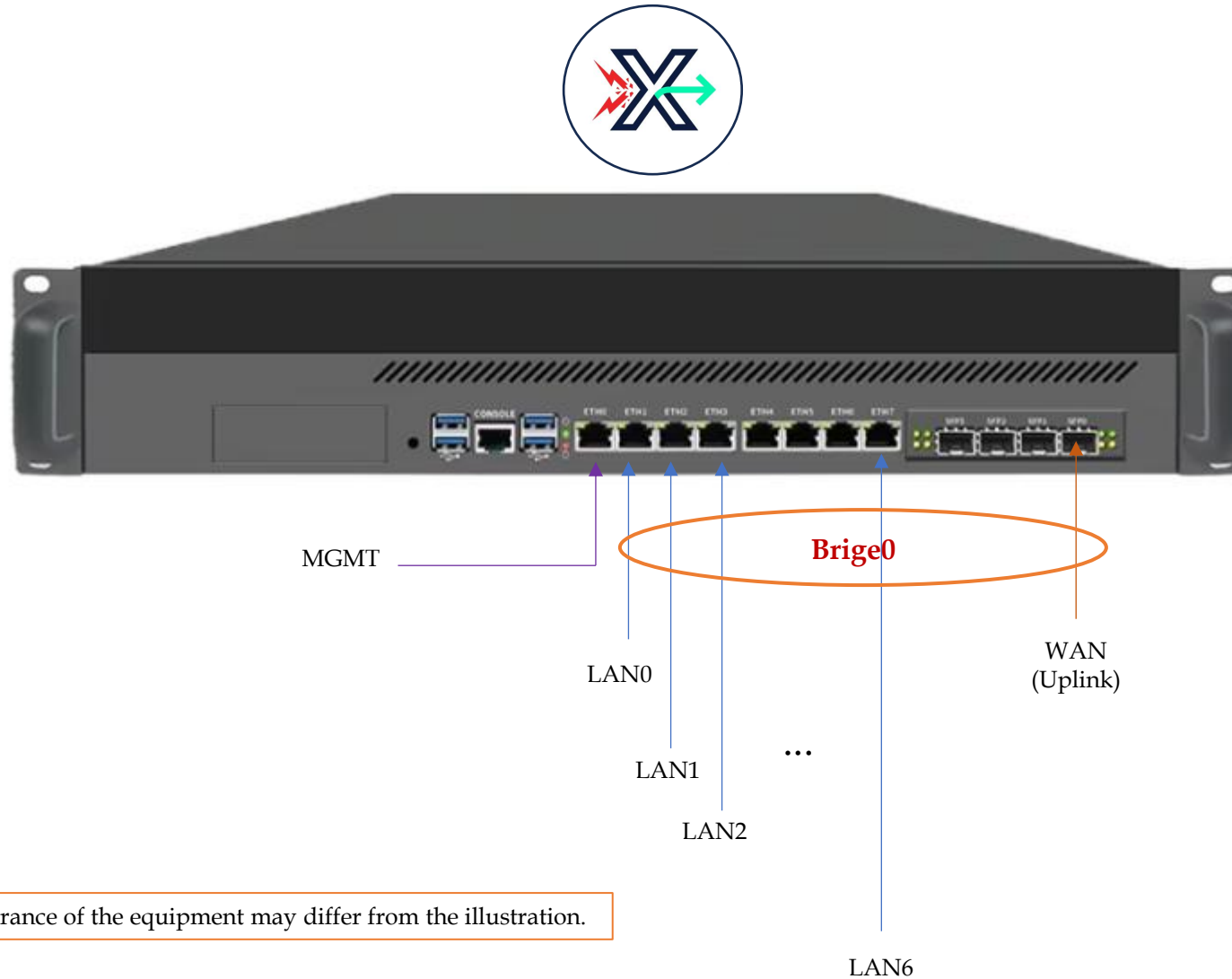


3. MightyXDDoS(1) – Inline Mitigator Box(2)

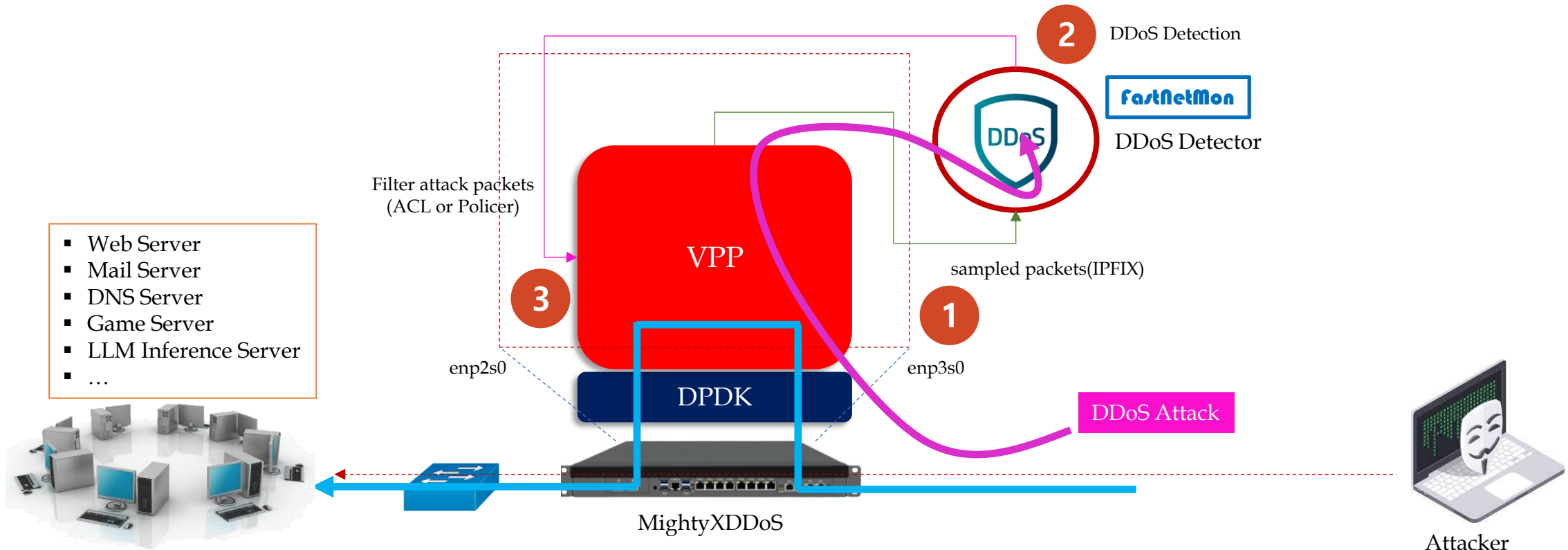


Note: The appearance of the equipment may differ from the illustration.

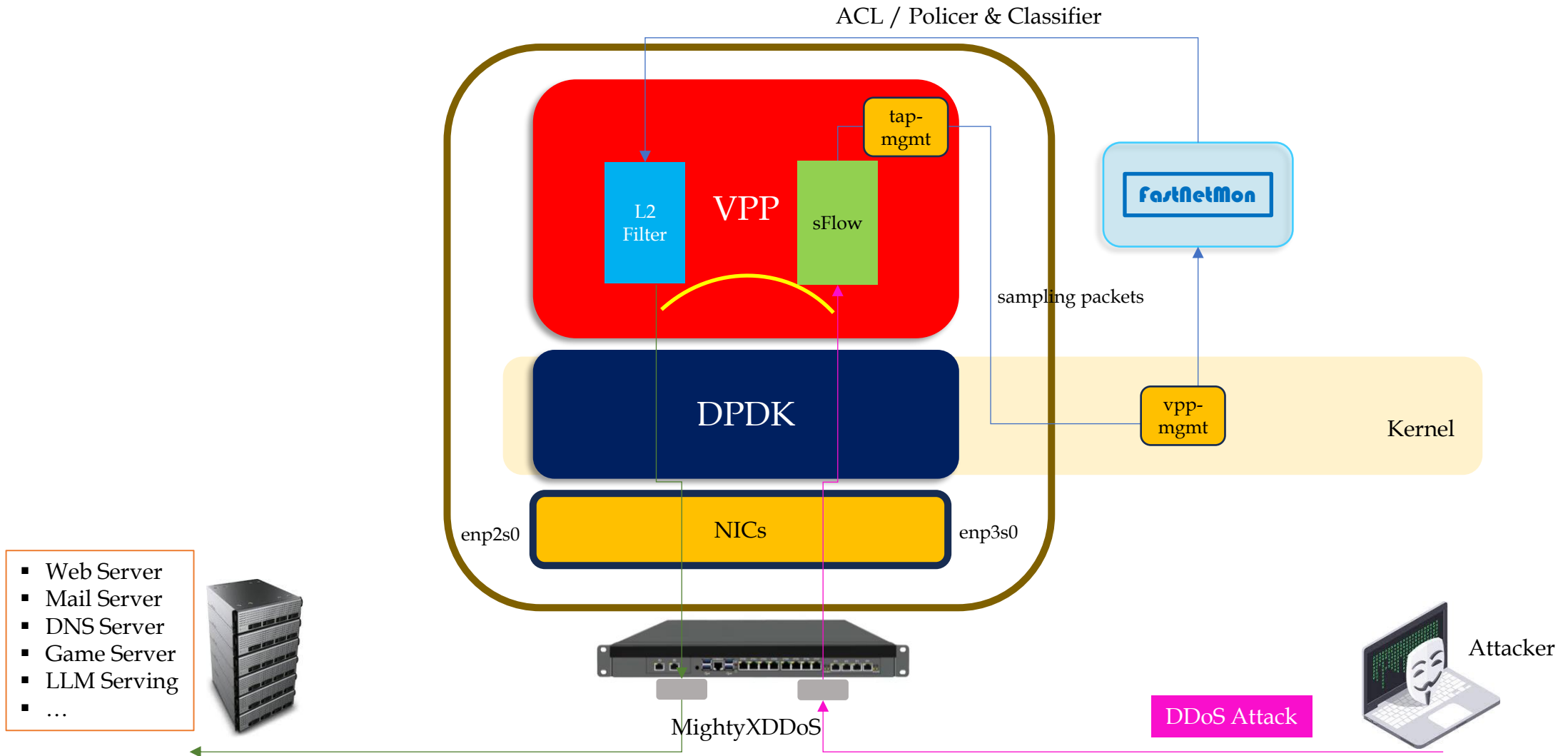
3. MightyXDDoS(1) – Inline Mitigator Box(3)



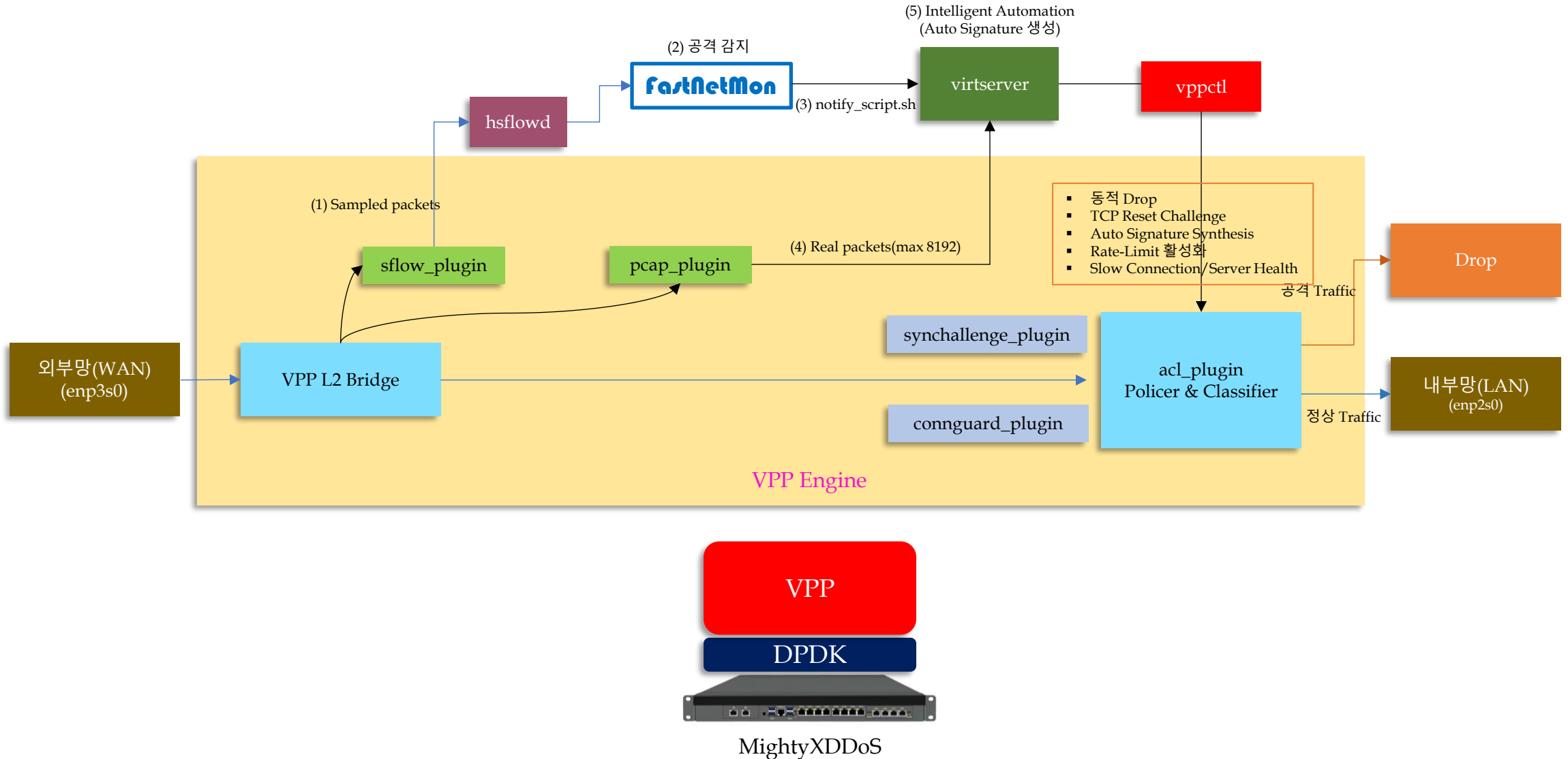
3. MightyXDDoS(2) – VPP Mitigator(1)



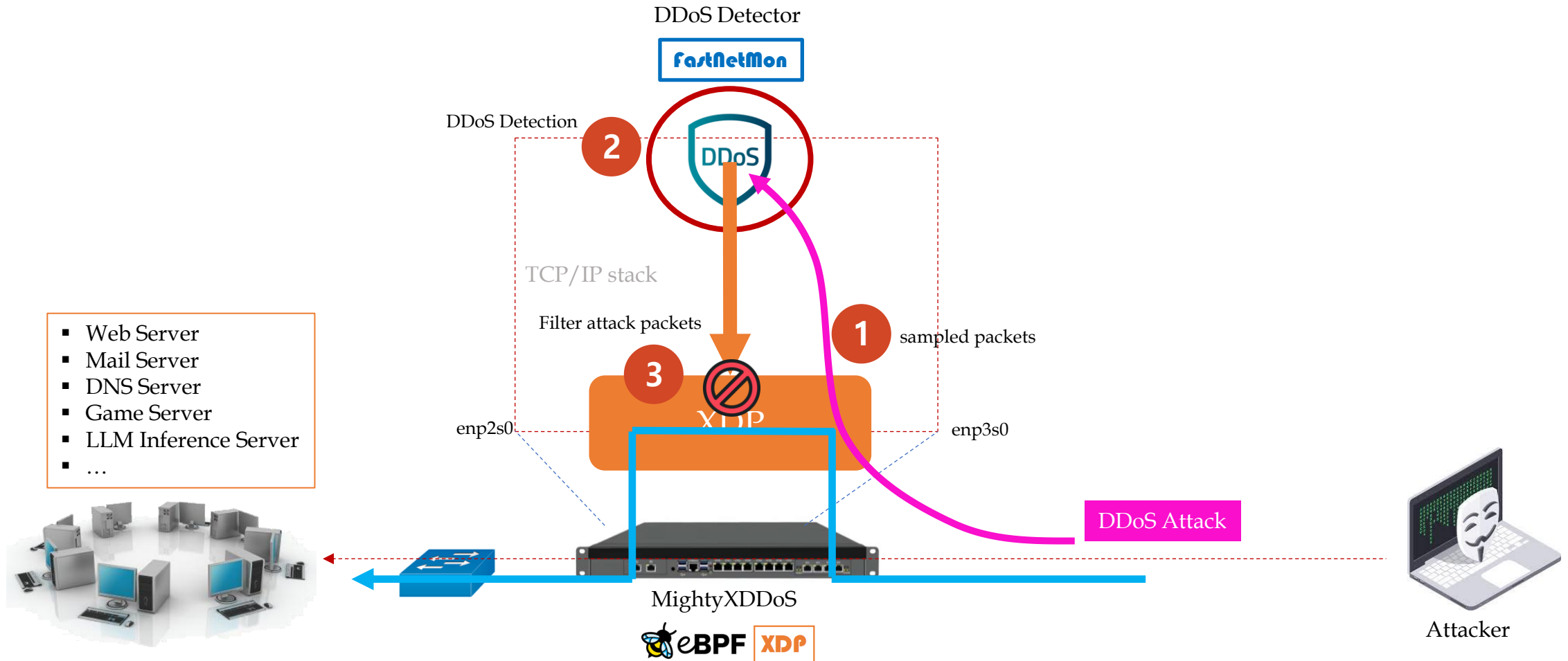
3. MightyXDDoS(2) – VPP Mitigator(2)



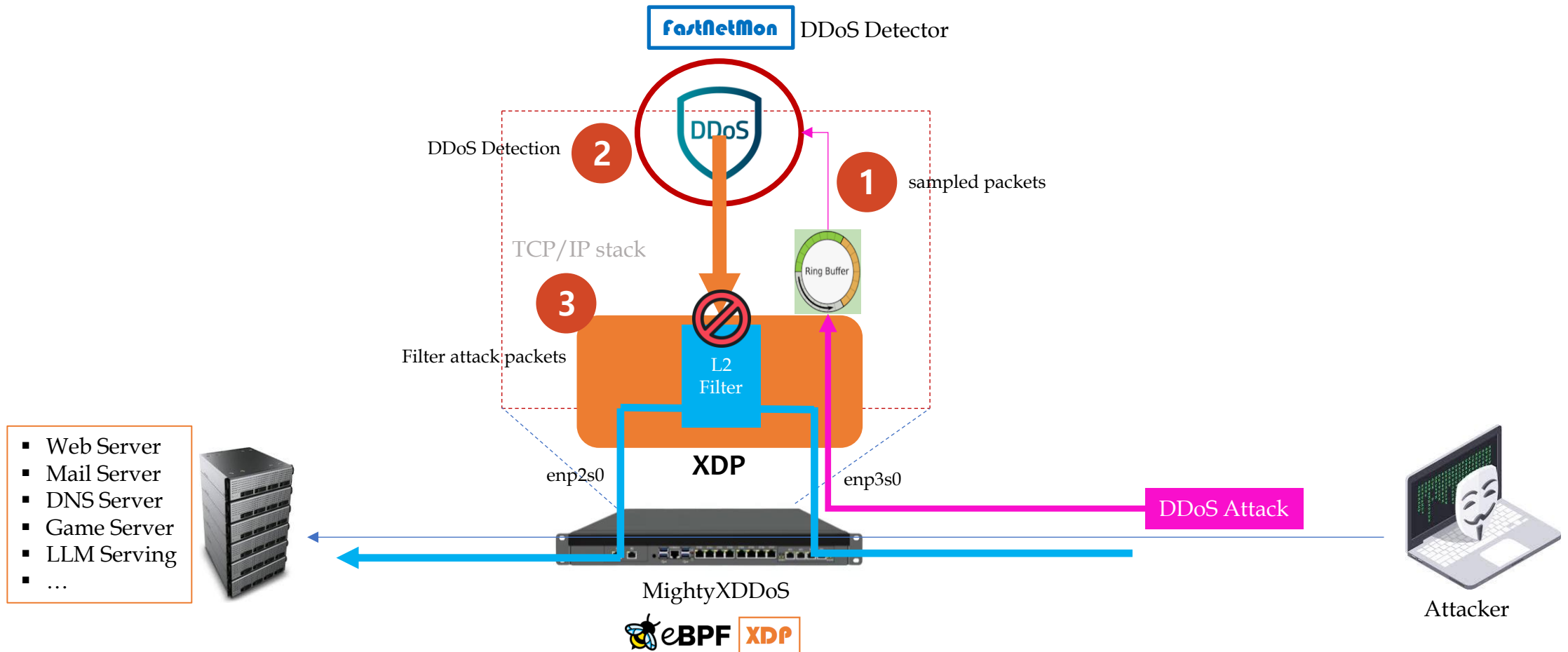
3. MightyXDDoS(2) – VPP Mitigator(3)



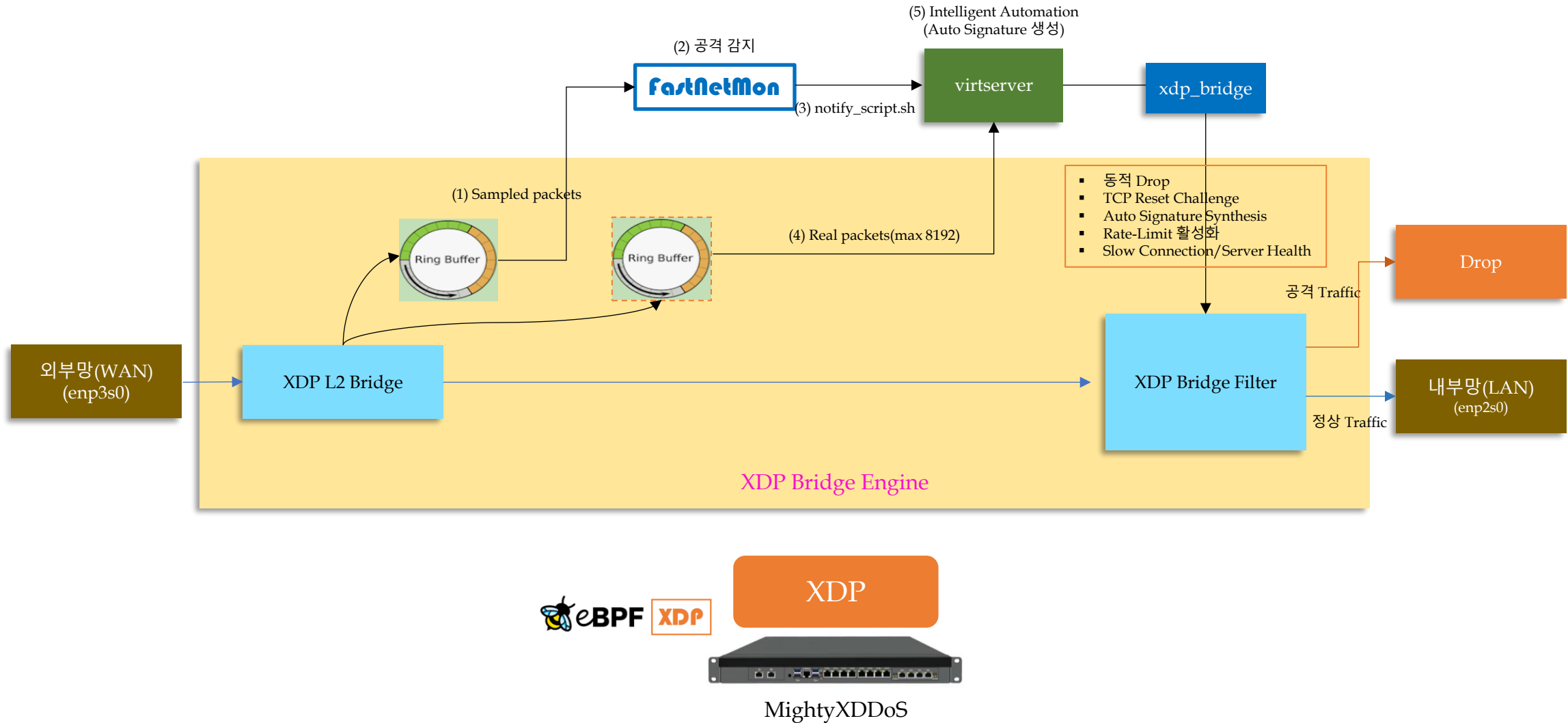
3. MightyXDDoS(3) – XDP Mitigator(1)



3. MightyXDDoS(3) – XDP Mitigator(2)

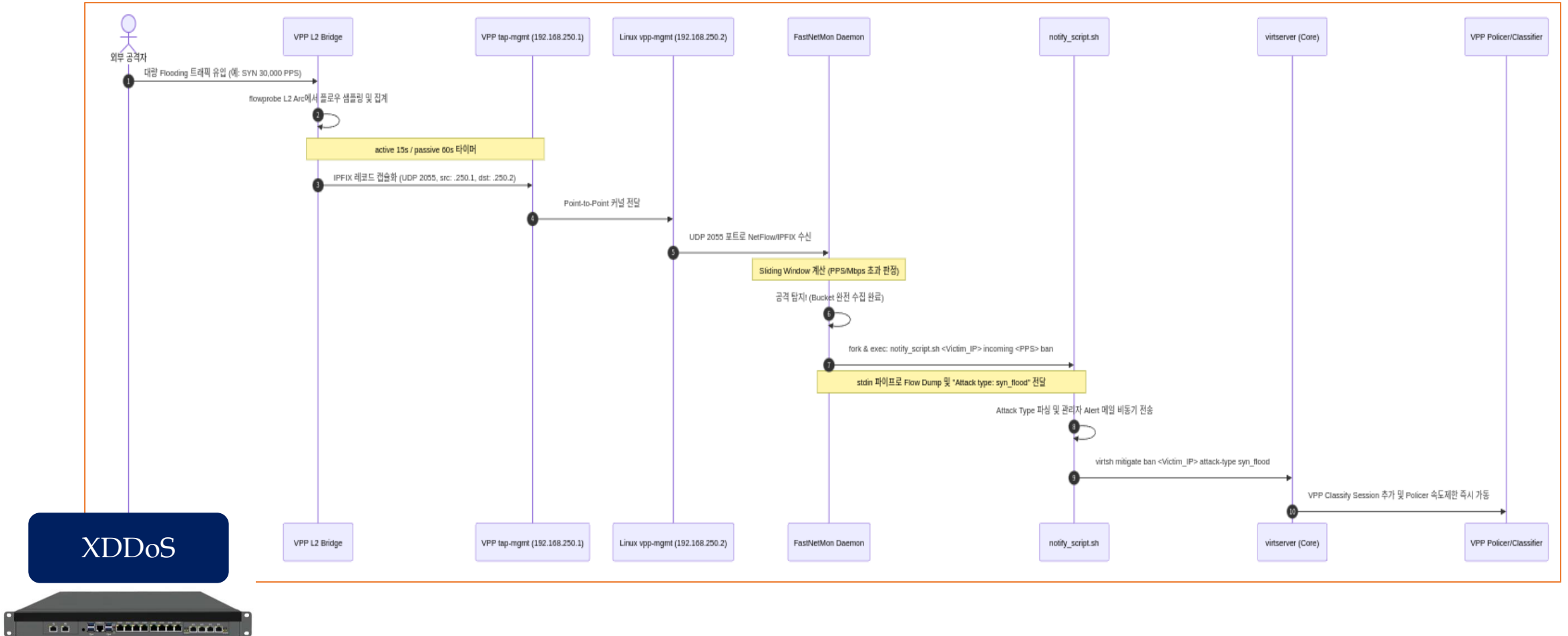


3. MightyXDDoS(3) – XDP Mitigator(3)



3. MightyXDDoS(4) – w/ FastNetMon Detector(1)

fastNetMon



3. MightyXDDoS(4) – w/ FastNetMon Detector(2)

The screenshot shows the MightyXDDoS Detector web interface. The left sidebar contains navigation links: admin (Security Administrator), Dashboard (System Monitor, Traffic Monitor), Detector (selected), View Attackers, and Mitigator (Policy, Flooding Attack, Broadcast Filter, Blocked Attacker, BGP Flowspec, Statistics). The main content area displays the Detector status with four key metrics: Detection Engine (ACTIVE), Live Traffic State (TRAFFIC DETECTED), Trigger Baselines (20000 pps / 800 Mbps), and Attack Incidents (20). Below these, the Detector Monitor (Live Telemetry) section shows a real-time stream of FastNetMon Engine data. The stream output is as follows:

```
FastNetMon 1.2.10 Try Advanced edition: https://fastnetmon.com/product-overview/
IPs ordered by: bytes
Incoming traffic
192.168.8.220      74475 pps    23 mbps    1893 flows
                  74475 pps    23 mbps    989 flows  *banned*

Outgoing traffic
                  0 pps     0 mbps     0 flows

Internal traffic
                  0 pps     0 mbps

Other traffic
                  0 pps     0 mbps

Screen updated in: 5.3e-05 sec
Traffic calculated in: 0 sec 67 microseconds
Not processed packets: 0 pps
```

The screenshot shows the MightyXDDoS Detector web interface, specifically the Recent Attack Sessions section. The left sidebar is the same as the previous screenshot. The main content area displays a table of recent attack sessions, with a search filter for victim IP or file. The table has columns for Target Victim IP, Incident Timestamp, Forensics File, and Action. The data is as follows:

TARGET VICTIM IP	INCIDENT TIMESTAMP	FORENSICS FILE	ACTION
192.168.8.220	17 09 26 11:52:36	192.168.8.220_17_09_26_11:52:36.txt	Inspect
192.168.8.220	17 09 26 11:52:26	192.168.8.220_17_09_26_11:52:26.txt	Inspect
192.168.8.220	17 09 26 11:46:56	192.168.8.220_17_09_26_11:46:56.txt	Inspect

Below the table, the Attack Details section for the selected incident (192.168.8.220_17_09_26_11:52:36.txt) is shown. The details are as follows:

```
IP: 192.168.8.220
Attack type: syn_flood
Initial attack power: 27875 packets per second
Peak attack power: 27875 packets per second
Attack direction: incoming
Attack protocol: tcp
Total incoming traffic: 8 mbps
Total outgoing traffic: 0 mbps
Total incoming pps: 27875 packets per second
Total outgoing pps: 0 packets per second
Total incoming flows: 366 flows per second
Total outgoing flows: 0 flows per second
Incoming ip fragmented traffic: 0 mbps
```

3. MightyXDDoS(5) – Mitigation Types

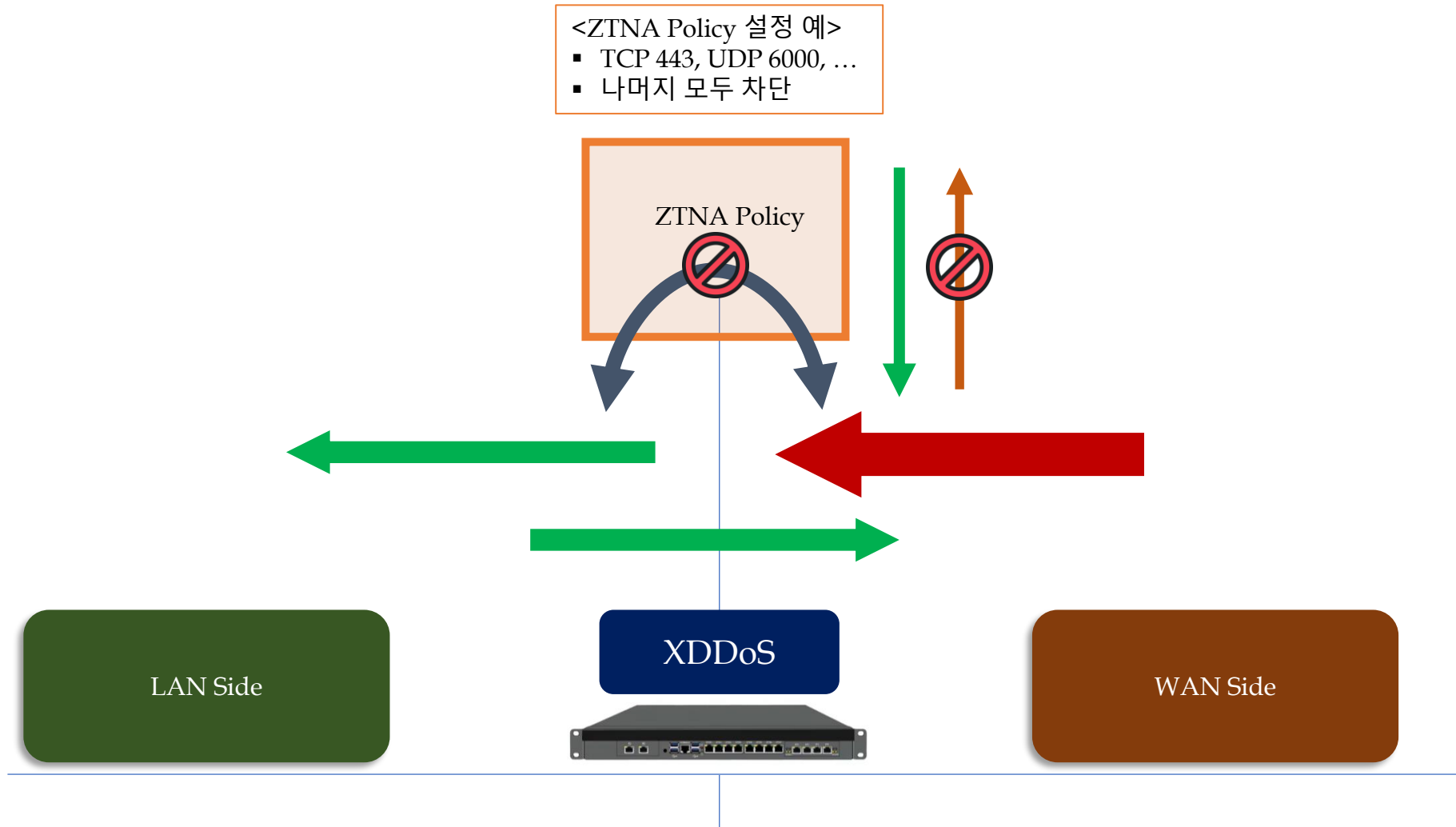
- **[1] ZTNA Policy**
 - 허용할 Port를 제외하고는 모두 차단
 - Line speed packet forwarding
- **[2] TCP DDoS Attack Protection**
 - TCP Reset Challenge(초기 1분 미만) + Auto-Signature Synthesis(패턴 차단)
 - TCP Rate-Limit(초기 1분 미만) + Auto-Signature Synthesis(패턴 차단)
 - TCP Reset Challenge Only
 - TCP PPS Threshold 기반의 Rate-Limiting
- **[3] UDP DDoS Attack Protection**
 - UDP Rate-Limit(초기 1분 미만) + Auto-Signature Synthesis(패턴 차단)
 - UDP PPS Threshold 기반의 Rate-Limiting
- **[4] ICMP DDoS Attack Protection**
 - ICMP Rate-Limit(초기 1분 미만) + Auto-Signature Synthesis(패턴 차단)
 - ICMP PPS Threshold 기반의 Rate-Limiting
 - 기본적으로 ZTNA Policy에서 ICMP는 허용 안함으로 설정

- **[5] Box Local Attack Protection**
 - Box -> Outbound Session 허용, 나머지 모두 차단
 - Rate-Limit
- **[6] Storm Control**
 - L2 Broadcast/Multicast, ARP
- **[7] Slow Connection Guard**
 - Slowloris Attack Protection
 - Slow Request/Hold/Read
- **[8] Server Health Guard**
 - Server 과부하 상태 체크(SYN+ACK 응답률 하락 체크, RST 비율 체크 등)
 - 서버 응답 데이터량 대비 요청 수 비율 변화 체크
- **[9] BGP Flowspec Protection**
 - 회선의 85%를 초과하는 트래픽 유입 시, BGP Flowspec 기능 동작

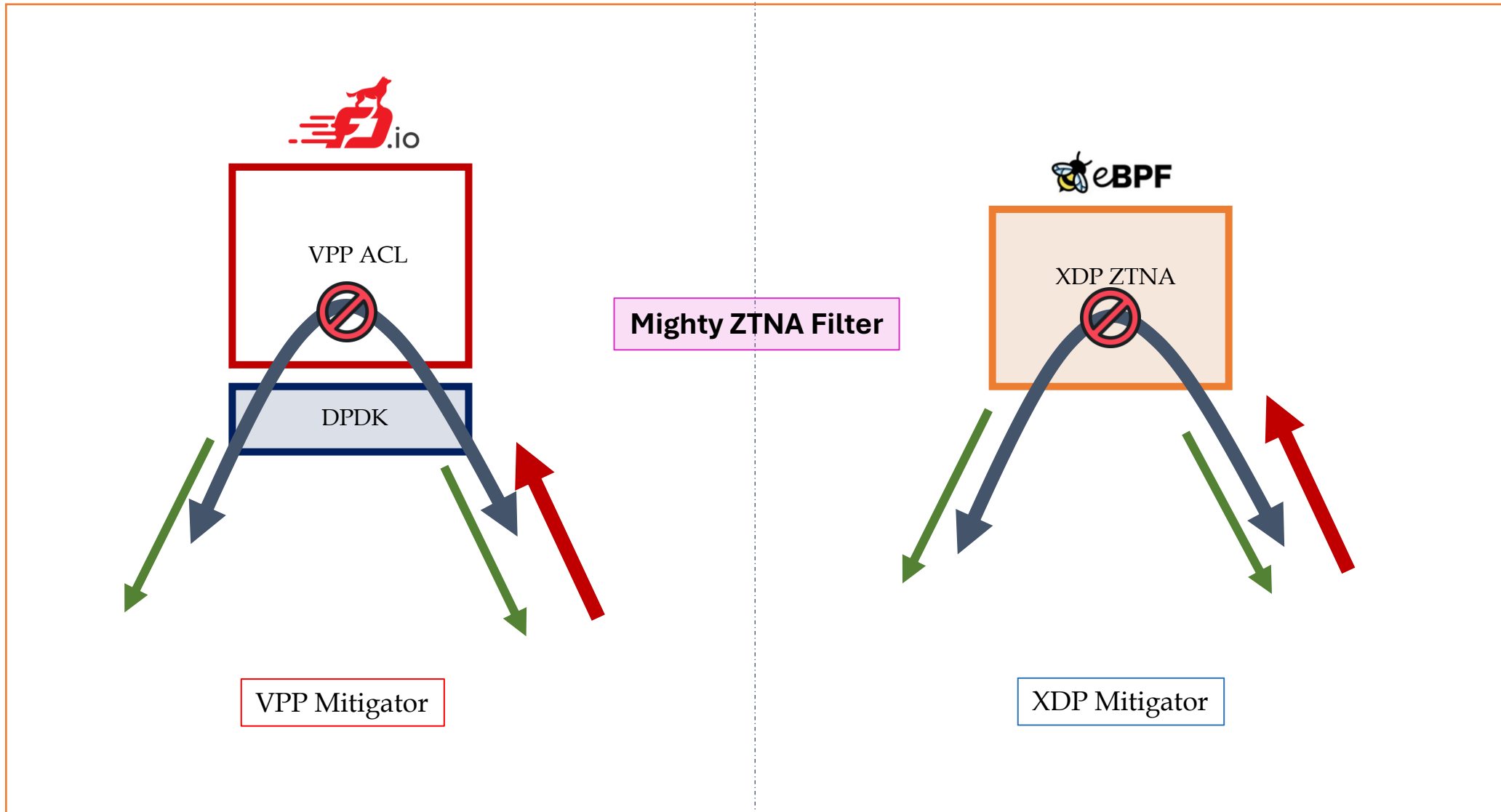
XDDoS



3. MightyXDDoS(6) – Protection Policy(1)



3. MightyXDDoS(6) – Protection Policy(2)



3. MightyXDDoS(6) – Protection Policy(3)

The screenshot shows the 'Policy' configuration page in the MightyXDDoS interface. The left sidebar contains navigation links for Dashboard, System Monitor, Traffic Monitor, Detector, View Attackers, Mitigator, Policy (selected), Flooding Attack, Broadcast Filter, Blocked Attacker, and Statistics. The main content area is titled 'Policy' and features a 'VPP Mode' button and a 'Logout' button. The 'ACL / Mitigator Interface Binding Scope' section is active, displaying two options: 'All (LAN & WAN)' (Default) and 'WAN Only'. The 'All (LAN & WAN)' option is selected, and its description states: 'Binds to every bridge member interface, LAN included – a LAN-originated attack (e.g. a compromised internal host) is also detected and blocked, not just WAN-originated ones. Slightly higher per-interface ACL overhead.' The 'WAN Only' option is also described. Below these options, a 'Save' button is visible. The 'WAN Service Port Policy' section is also visible, showing the 'Enable WAN Service Port Policy' toggle set to 'ENABLED'.

ACL / Mitigator Interface Binding Scope

Controls which bridge member interfaces get the enforcement layer bound to them: manual Firewall rules, the fastnetmon/SYN-flood automatic attacker-block ACL, and the SYN-flood classify-policer session. Detector's own traffic visibility is unaffected either way – it always watches every bridge member.

All (LAN & WAN) Default

Binds to every bridge member interface, LAN included – a LAN-originated attack (e.g. a compromised internal host) is also detected and blocked, not just WAN-originated ones. Slightly higher per-interface ACL overhead.

WAN Only

Binds only to WAN-role interfaces – lower overhead, appropriate when attacks are only expected from the WAN side. LAN-originated attacks are still detected by Detector but not enforced against by this box. Requires every bridge member interface to have a LAN/WAN role assigned first.

Applies immediately to VPP. **Save**

WAN Service Port Policy ZN-TYPE, AUTO-GENERATED

Enter only a protocol and port – no source/destination/interface needed. Once enabled, only these ports are reachable from WAN into the protected network; everything else is denied by default. Traffic from LAN to the Internet is never affected by this setting. For ad-hoc manual rules with full source/destination control, use the **Firewall** page instead – that layer is evaluated first regardless of this setting.

Enable WAN Service Port Policy **ENABLED**

ALLOWED PORTS **+ Add Port Rule**

The screenshot shows the 'Policy' configuration page in the MightyXDDoS interface, specifically the 'WAN Service Port Policy' section. The left sidebar is the same as in the previous screenshot. The main content area shows the 'Enable WAN Service Port Policy' toggle set to 'ENABLED'. Below this, the 'ALLOWED PORTS' section is visible, containing a table with columns for Protocol, Port, Description, and Action. The table lists several ports: TCP 443 (https), TCP 8080 (https), UDP 6000 (test udp), ICMP Any (no ports) (icmp test), and TCP 80 (http). Each row has a red trash icon in the Action column. Below the table, a 'Save' button is visible. At the bottom of the page, there are two graphs: 'ACL Permit Packets (cumulative)' and 'ACL Deny Packets (cumulative)'. The 'ACL Permit Packets' graph shows a green line at approximately 100,000 packets. The 'ACL Deny Packets' graph shows a red line at approximately 140 packets. Both graphs are labeled 'Last updated: 오후 2:29:47'.

Enable WAN Service Port Policy **ENABLED**

ALLOWED PORTS **+ Add Port Rule**

Protocol	Port	Description	Action
TCP	443	https	
TCP	8080	https	
UDP	6000	test udp	
ICMP	Any (no ports)	icmp test	
TCP	80	http	

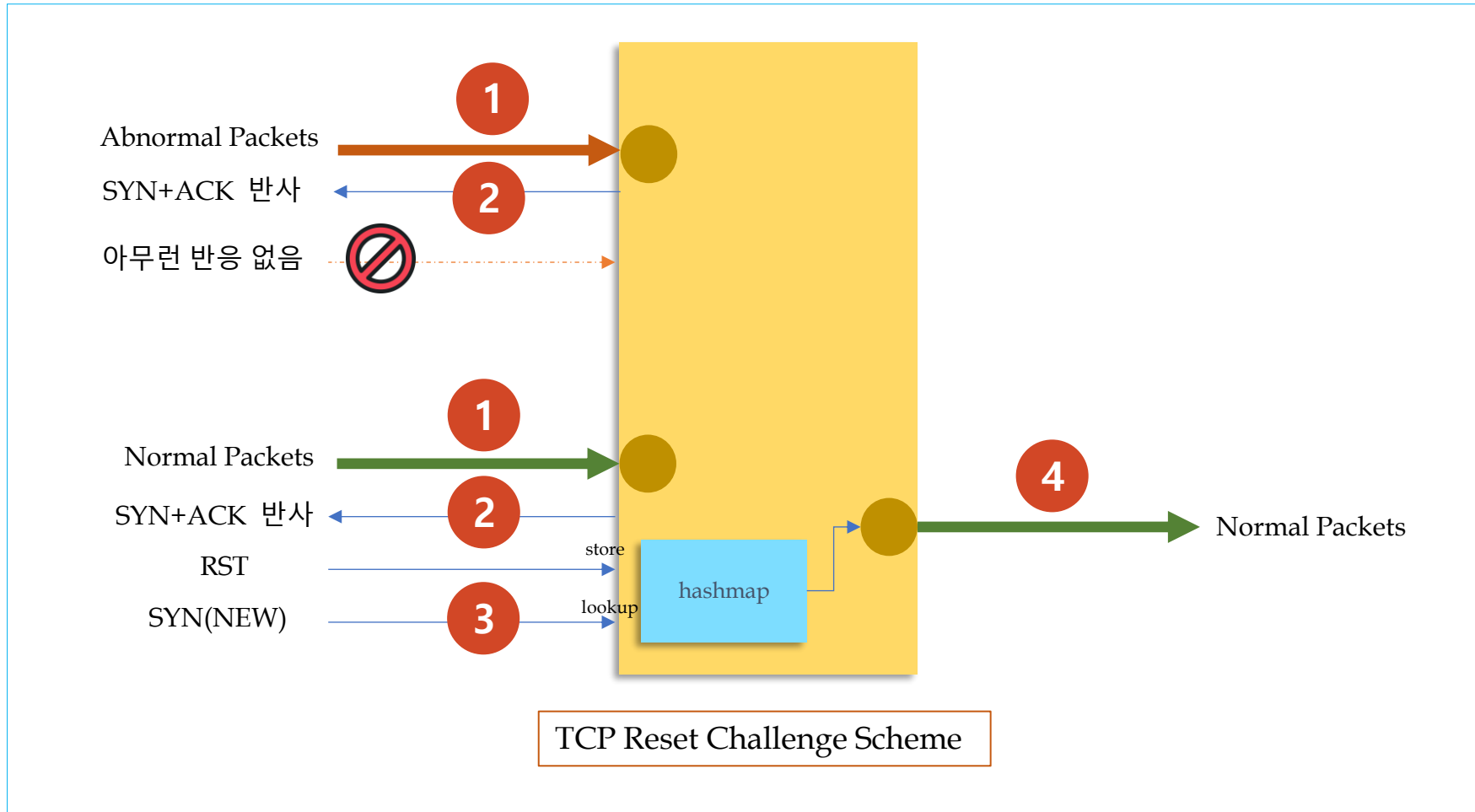
Applies immediately. **Save**

ACL Permit Packets (cumulative) Last updated: 오후 2:29:47

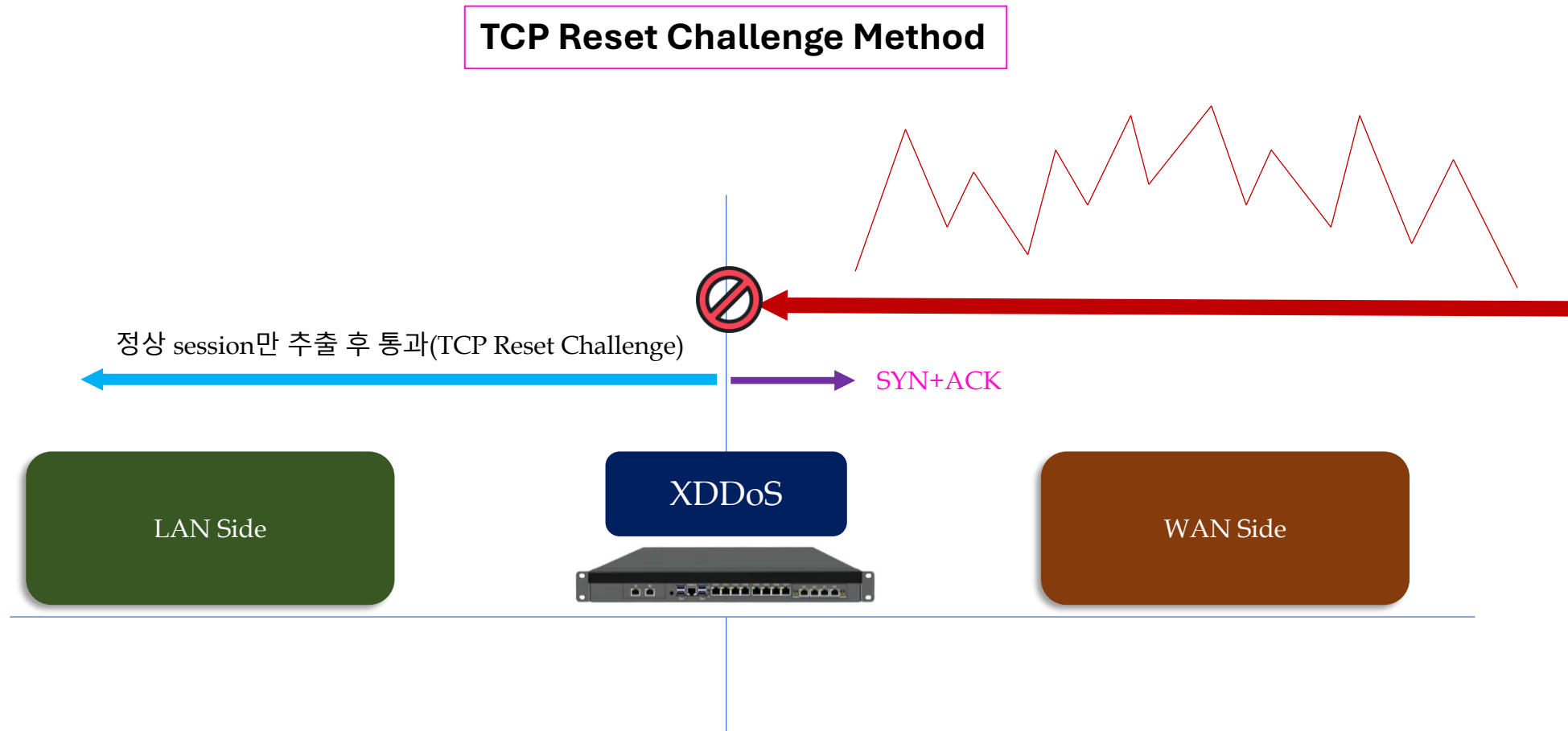
ACL Deny Packets (cumulative) Last updated: 오후 2:29:47

3. MightyXDDoS(7) – TCP Reset Challenge Scheme(1)

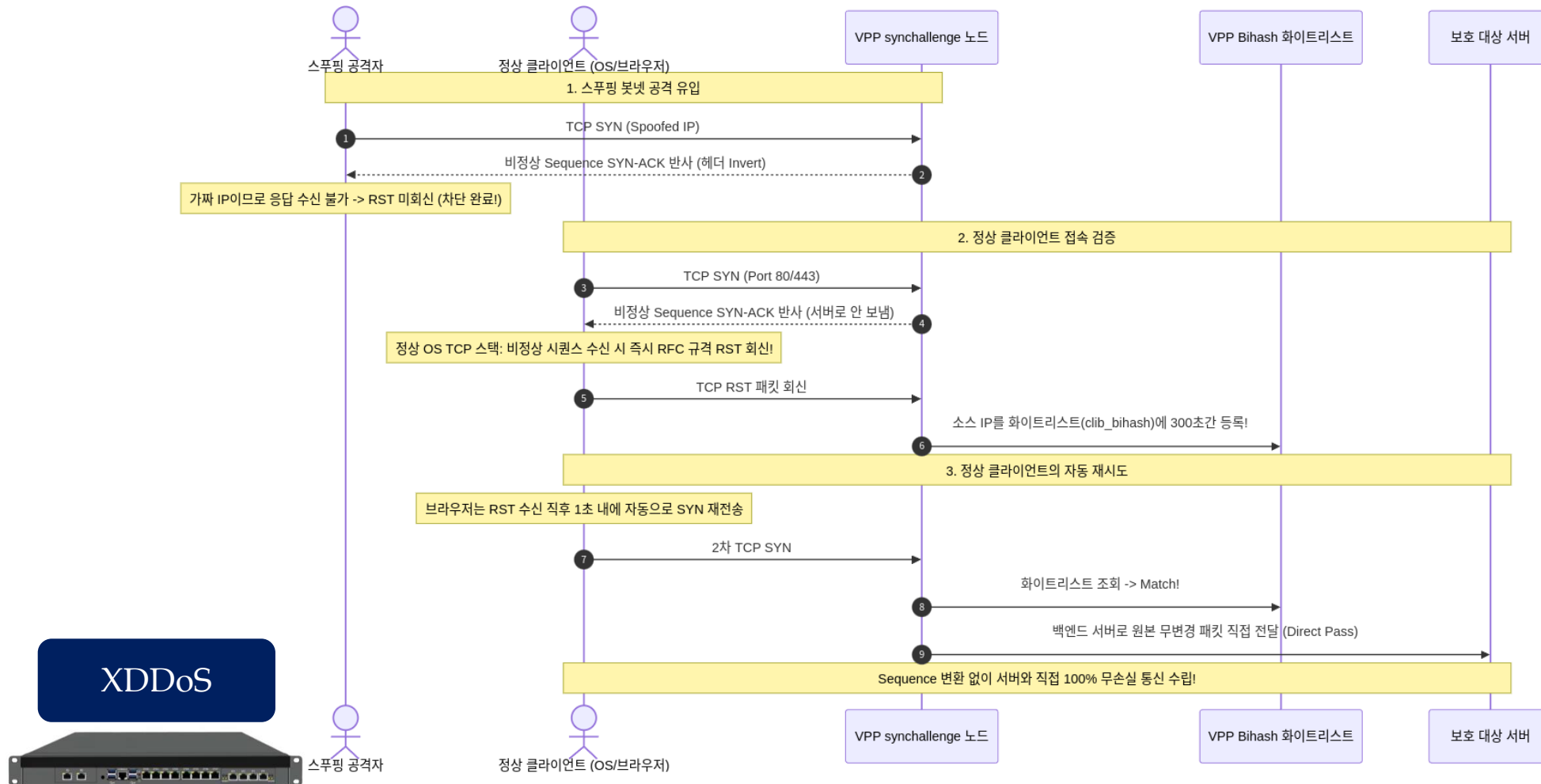
공격 패킷 차단, 정상 TCP 접속만 선별 후 통과 기법



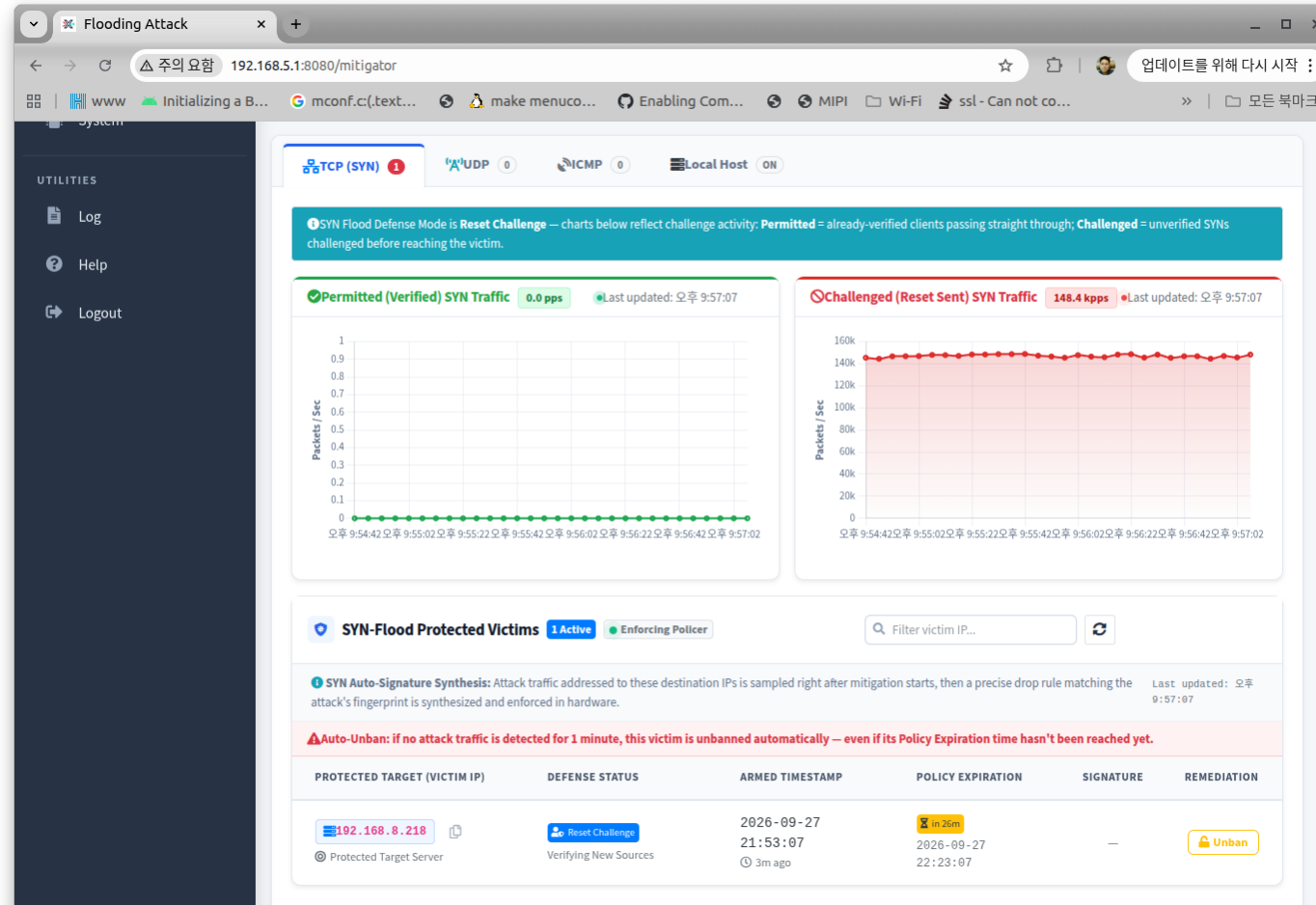
3. MightyXDDoS(7) – TCP Reset Challenge Scheme(2)



3. MightyXDDoS(7) – TCP Reset Challenge Scheme(3)

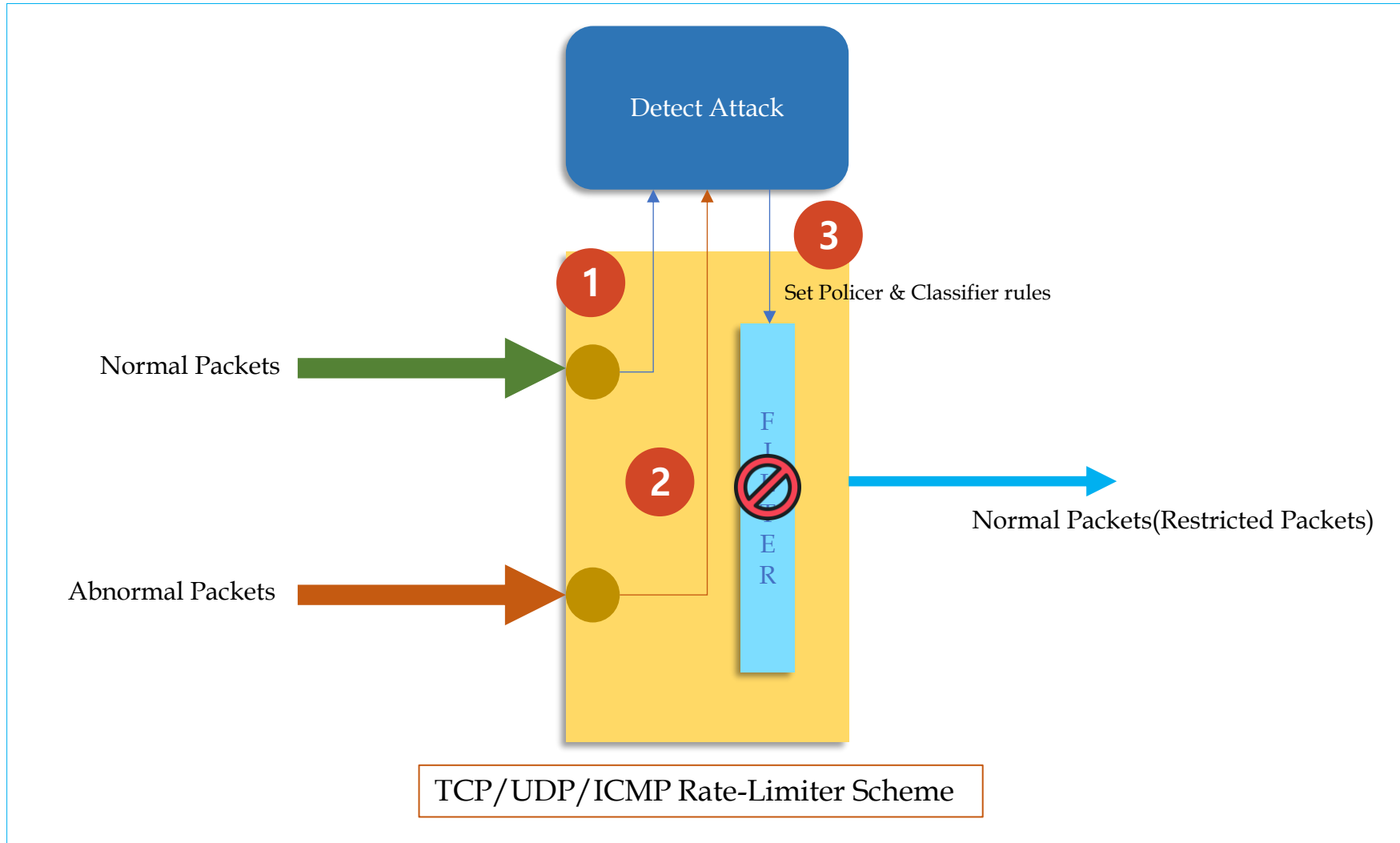


3. MightyXDDoS(7) – TCP Reset Challenge Scheme(4)



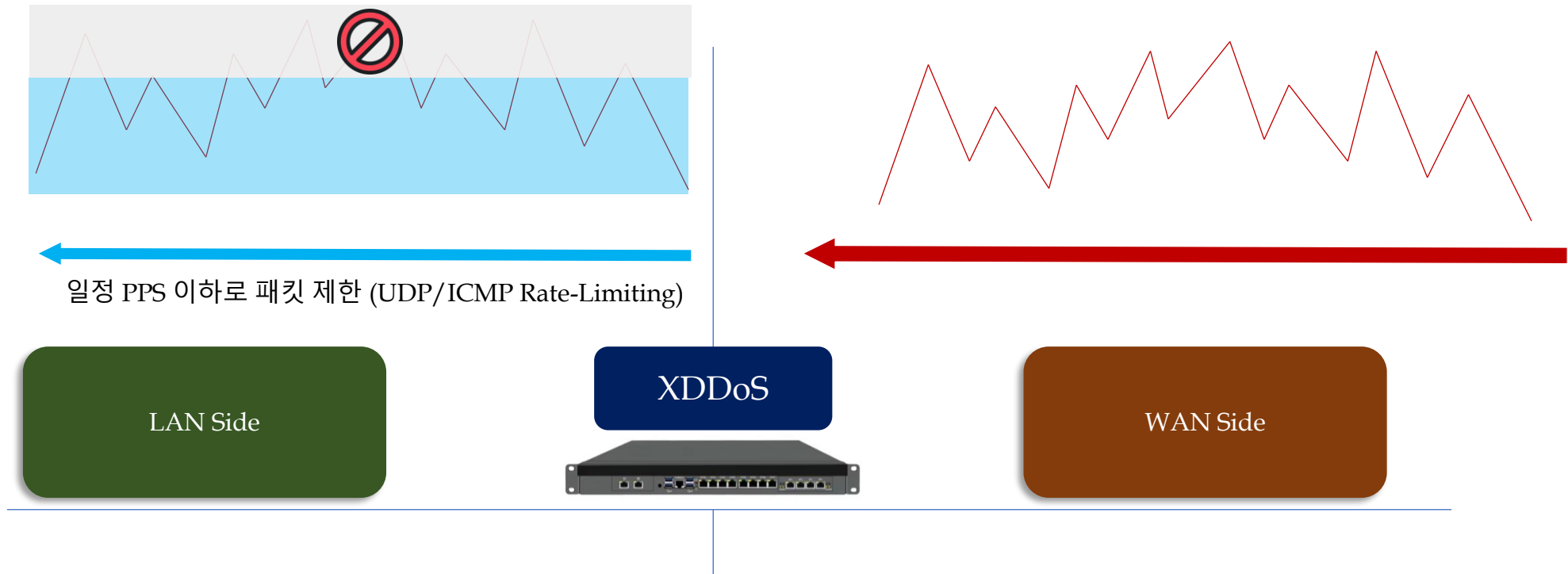
3. MightyXDDoS(8) – Rate-Limiter Scheme(1)

Traffic Control (Policer & Classifier) 기법

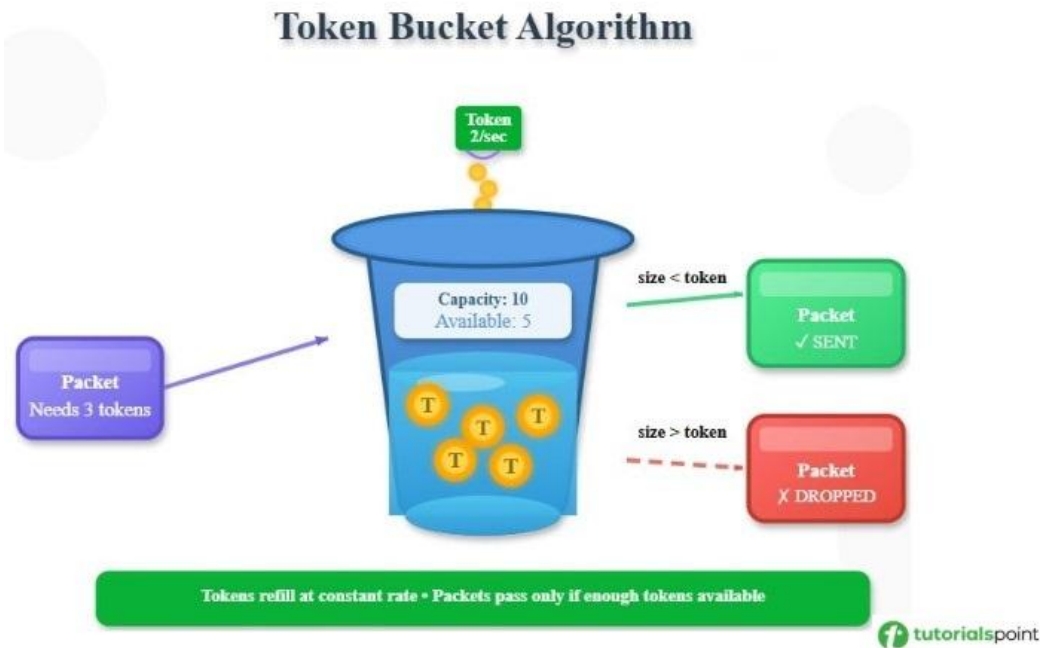


3. MightyXDDoS(8) – Rate-Limiter Scheme(2)

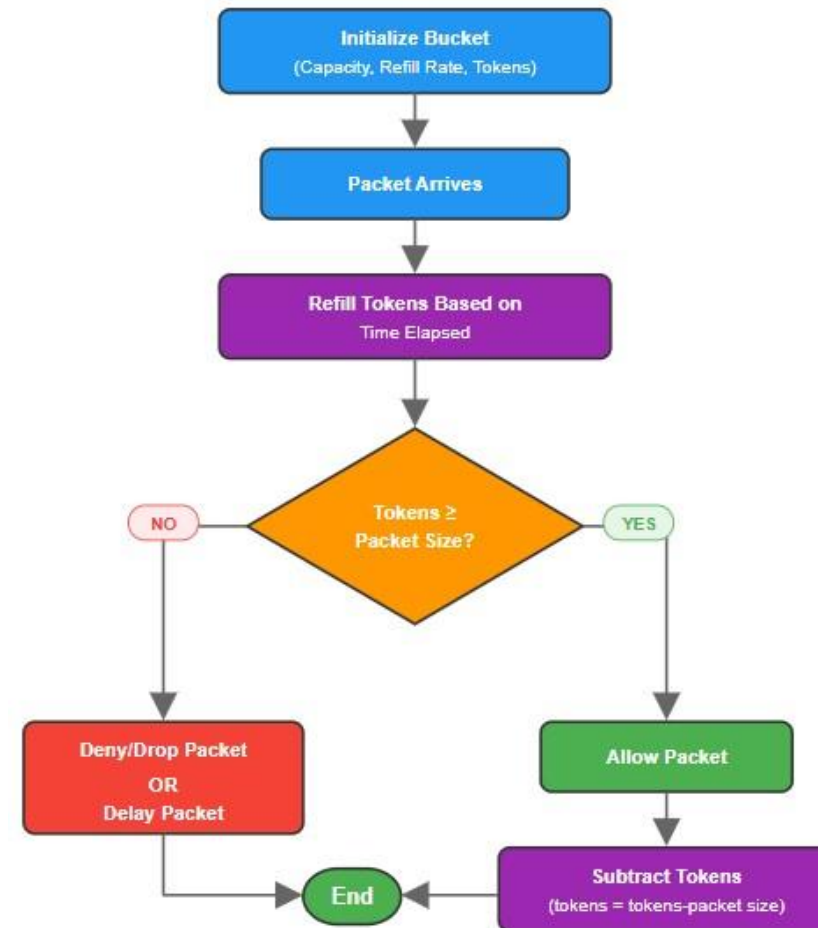
UDP/ICMP Rate-Limit Method



3. MightyXDDoS(8) – Rate-Limiter Scheme(3)



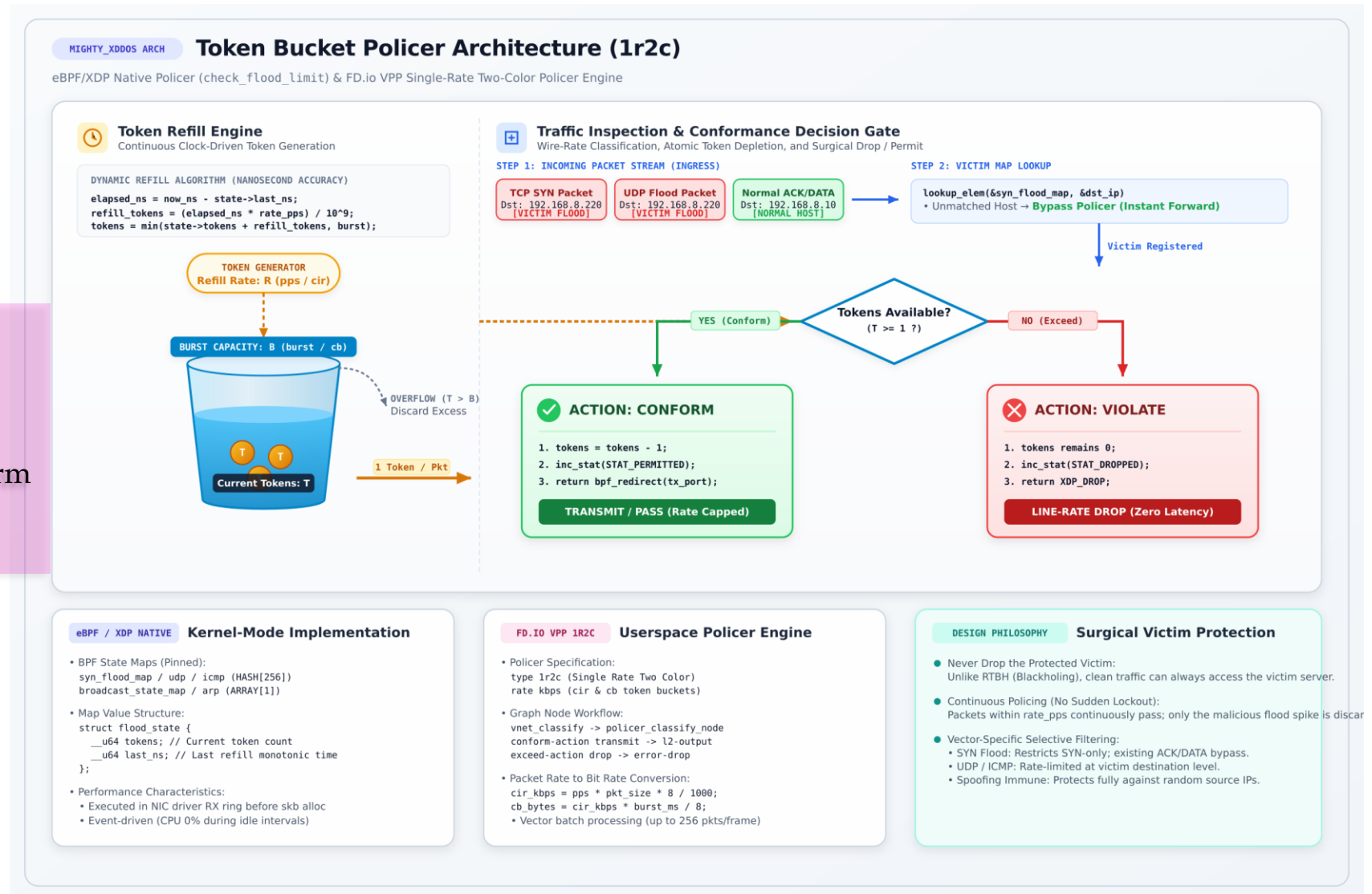
Token Bucket Algorithm Flowchart



3. MightyXDDoS(8) – Rate-Limiter Scheme(4)

Rate-Limiter

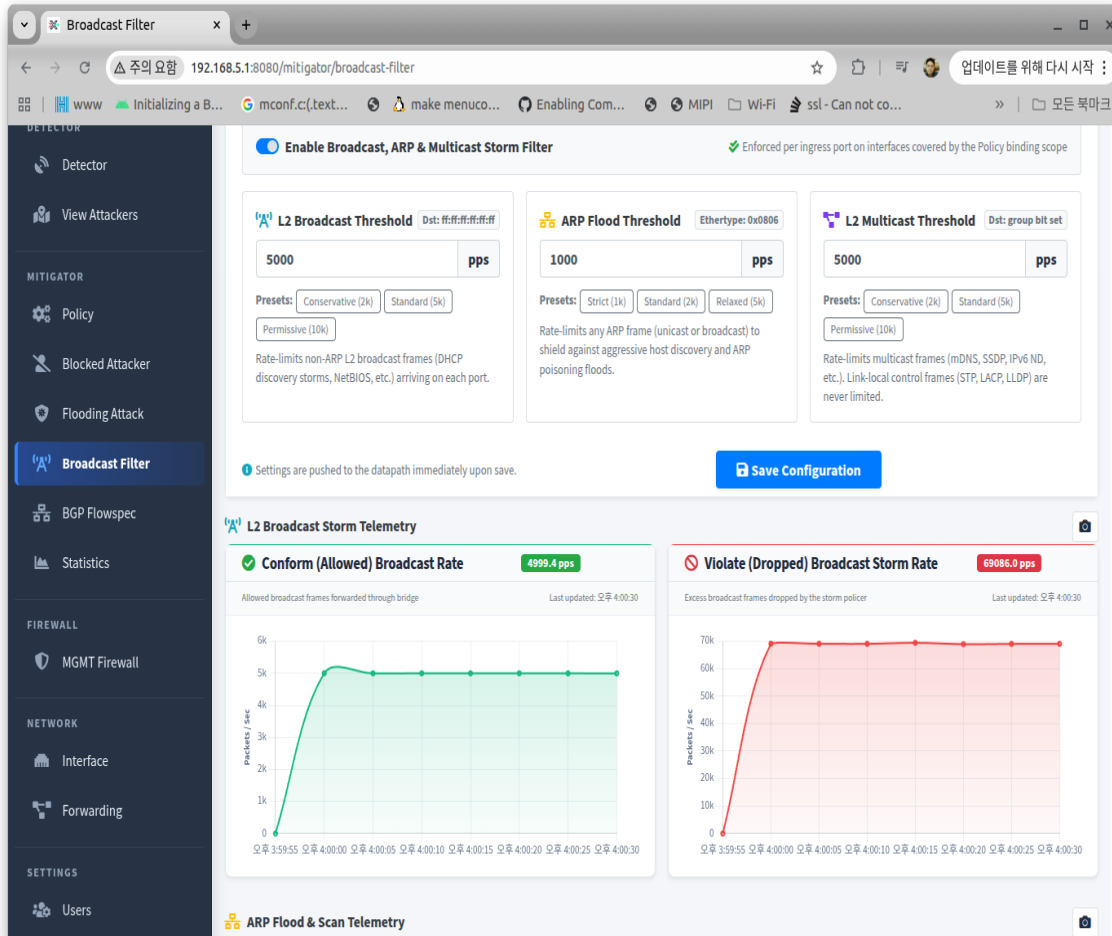
- TCP Syn Flooding
- UDP Flooding
- ICMP Flooding
- ARP Broadcast Storm
- Broadcast/Multicast Storm
- MAC Flooding
- ...



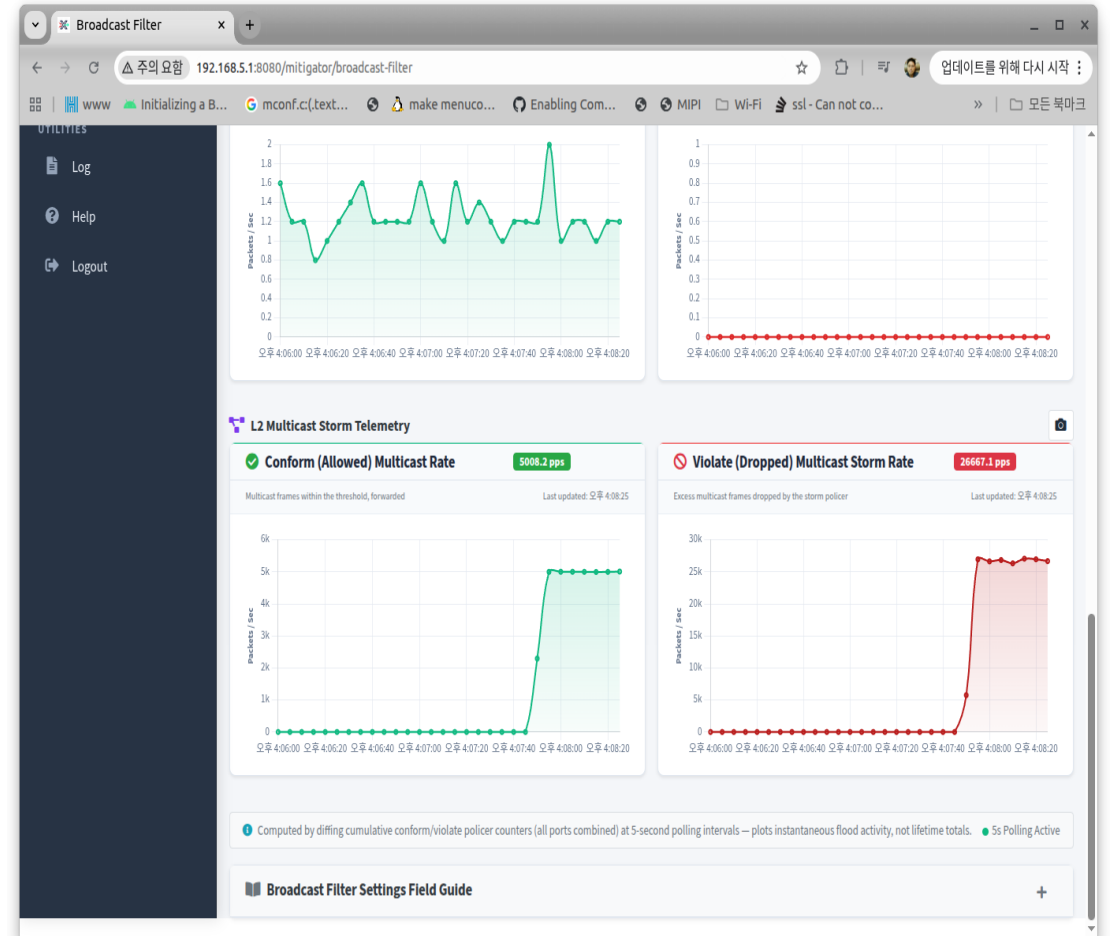
3. MightyXDDoS(8) – Rate-Limiter Scheme(5)

L2 Broadcast/Multicast, ARP Storm Filter

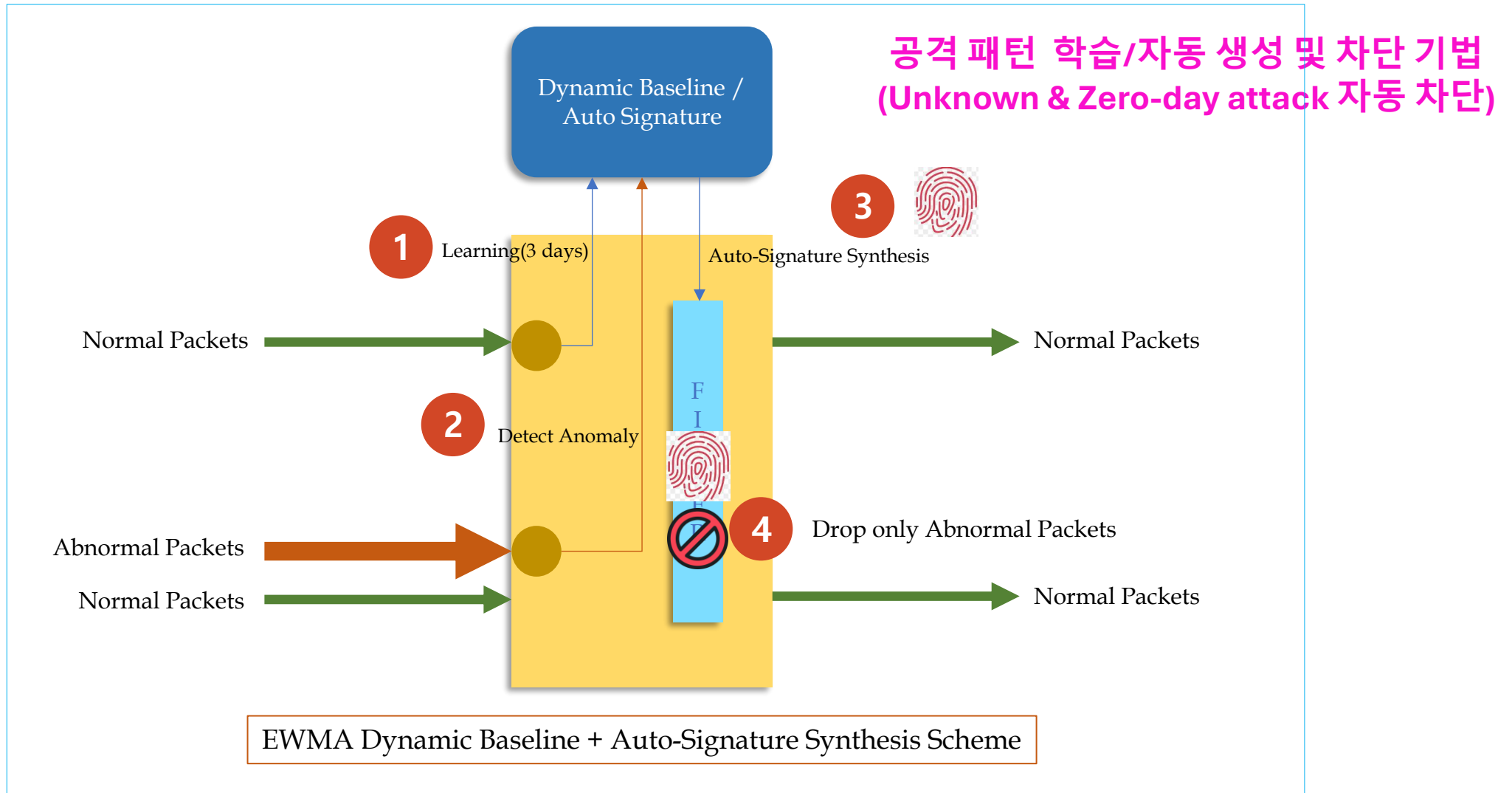
L2 Broadcast Storm Filter



L2 Multicast Storm Filter

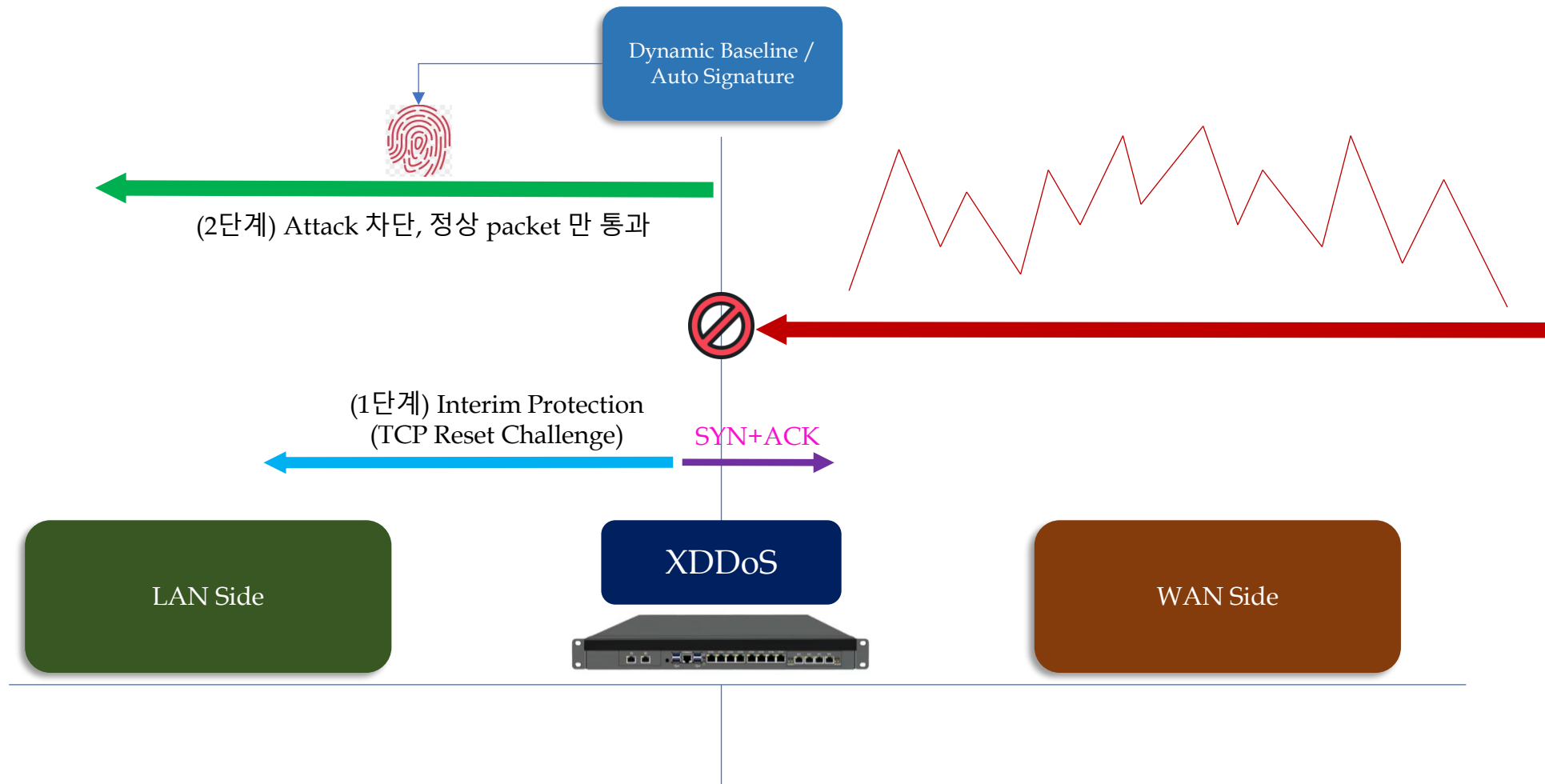


3. MightyXDDoS(9) – Intelligent Automation(1)



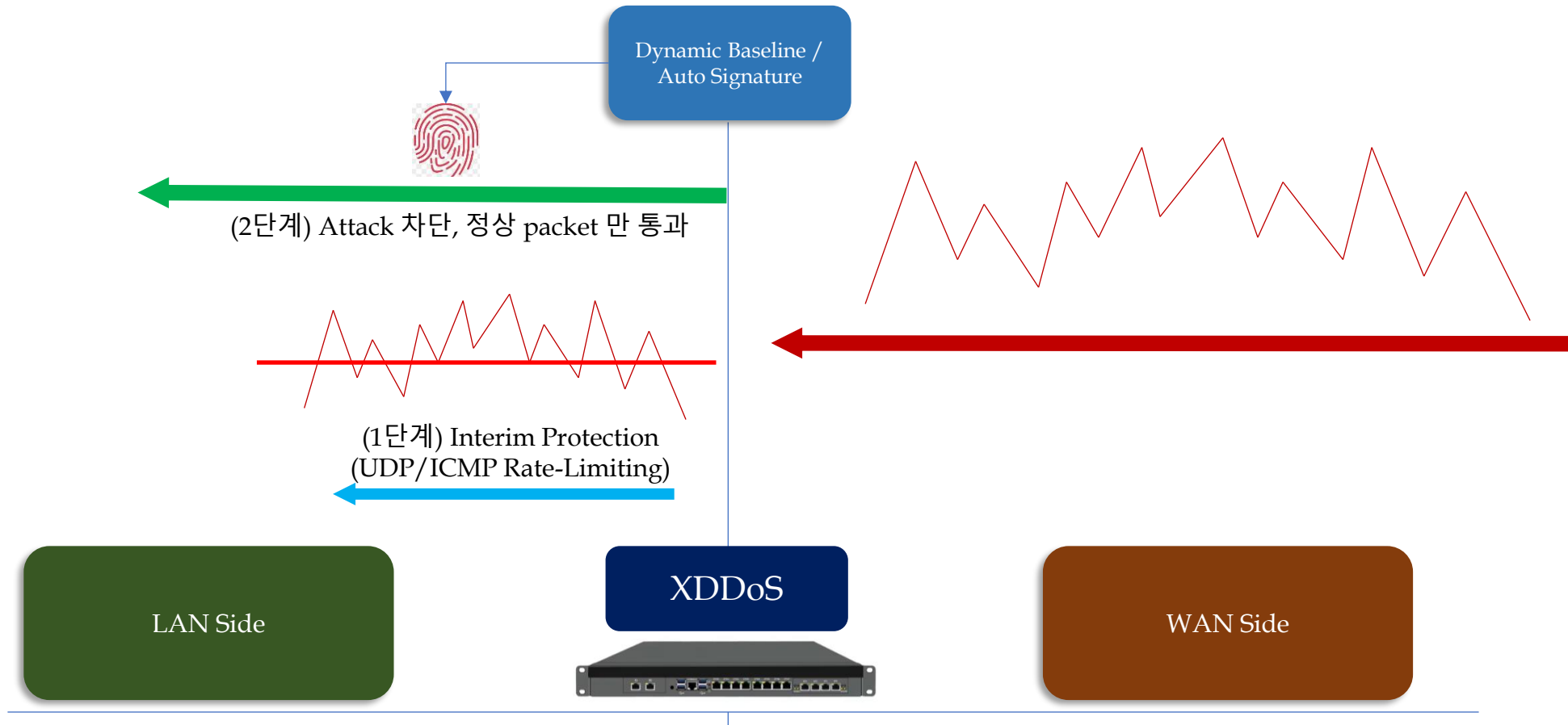
3. MightyXDDoS(9) – Intelligent Automation(2)

Mode1) Interim(Reset Challenge) + Auto-Signature Synthesis Method



3. MightyXDDoS(9) – Intelligent Automation(3)

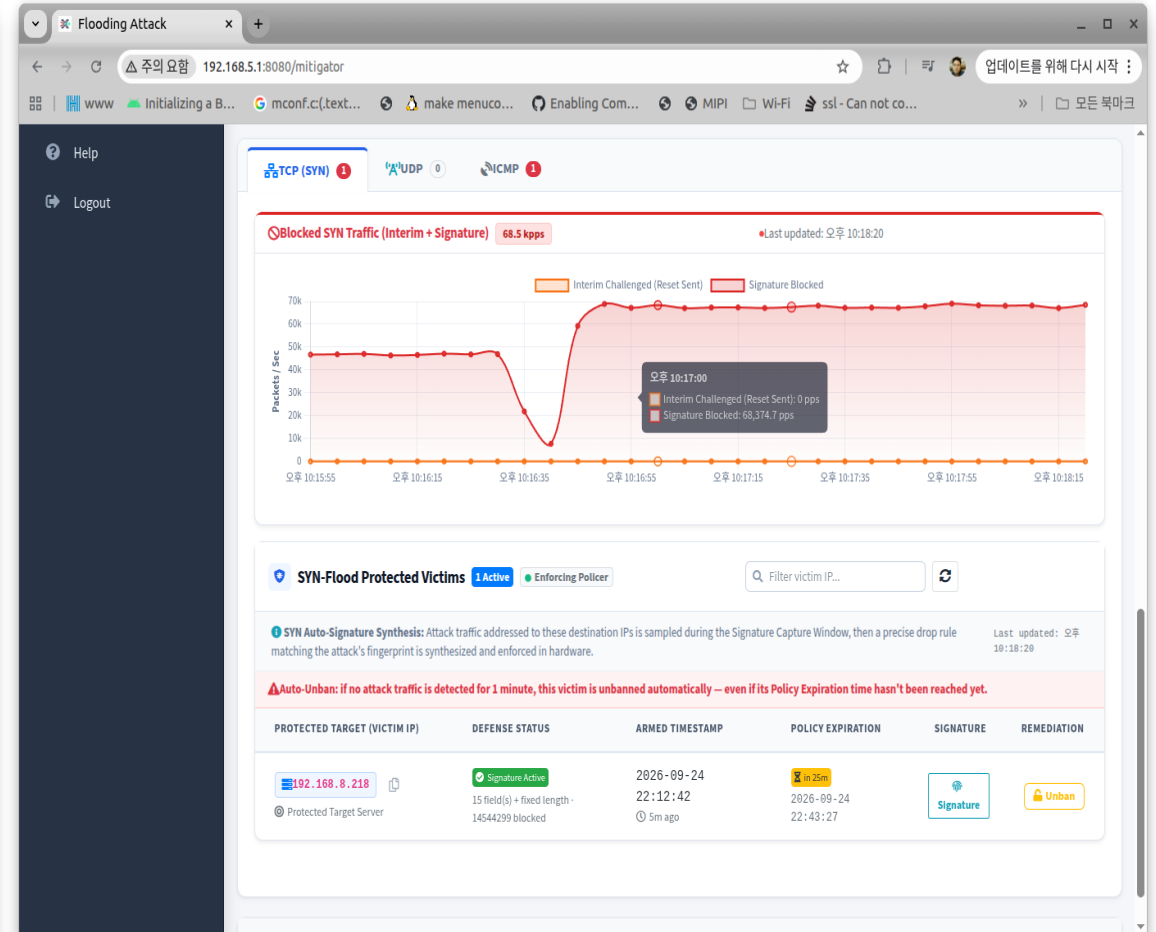
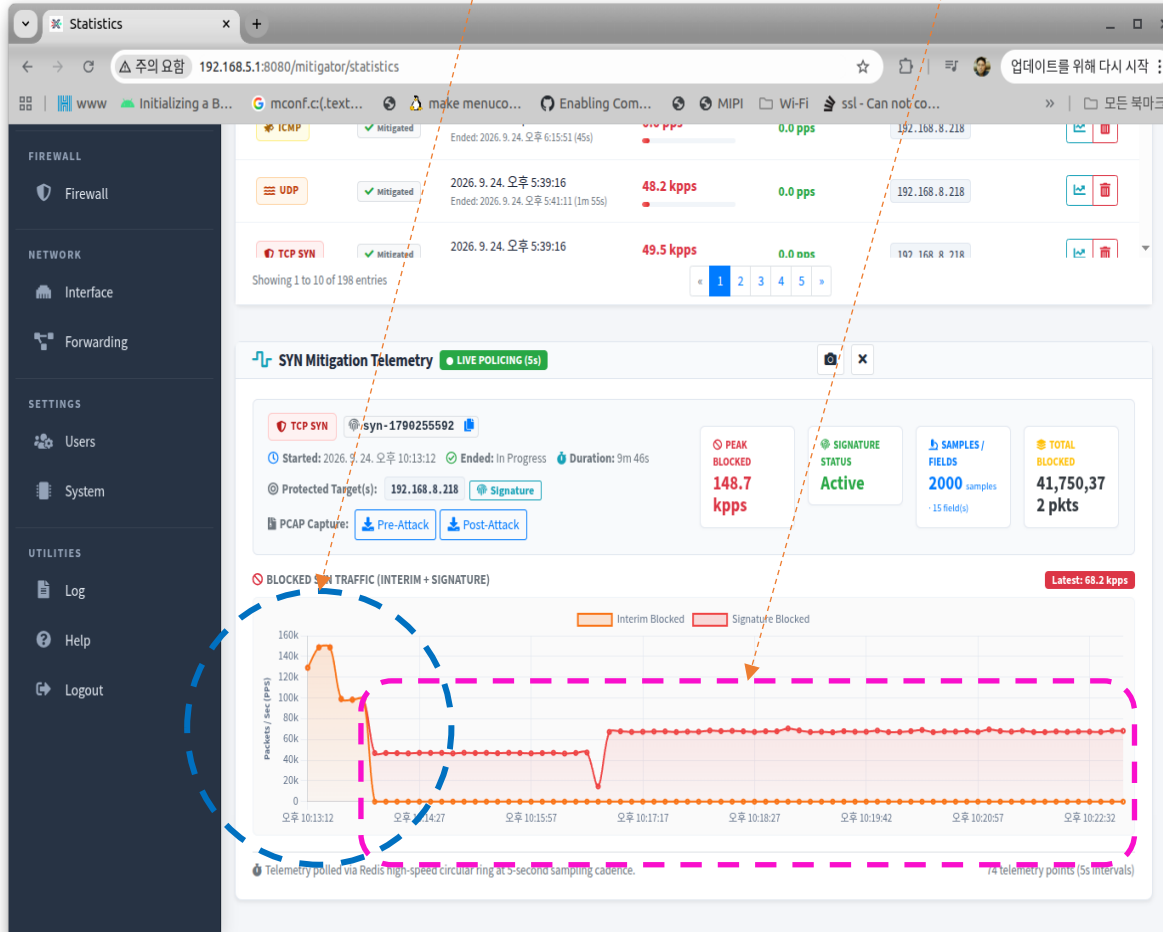
Mode2) Interim(Rate-Limit) + Auto-Signature Synthesis Method



3. MightyXDDoS(9) – Intelligent Automation(4)

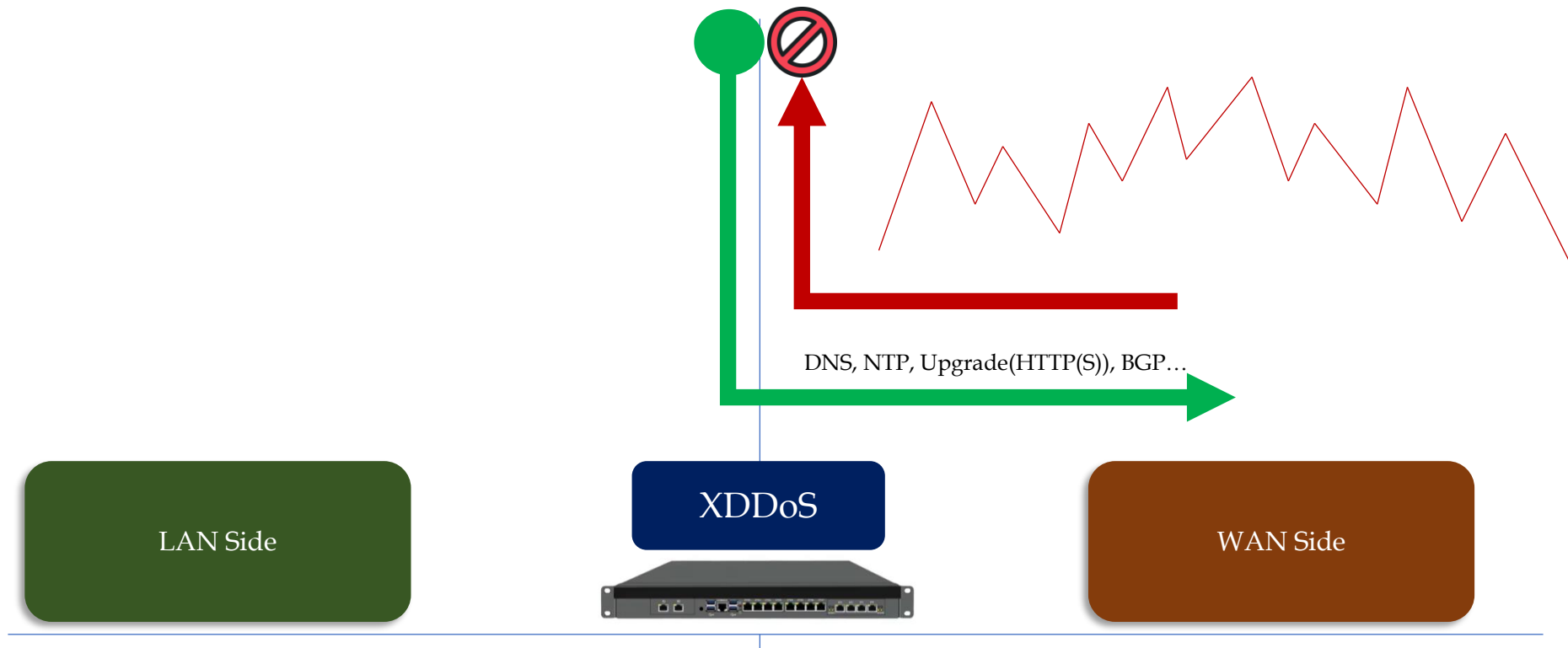
(1단계) Interim Protection
(TCP Reset Challenge)

(2단계) Auto-Signature Protection

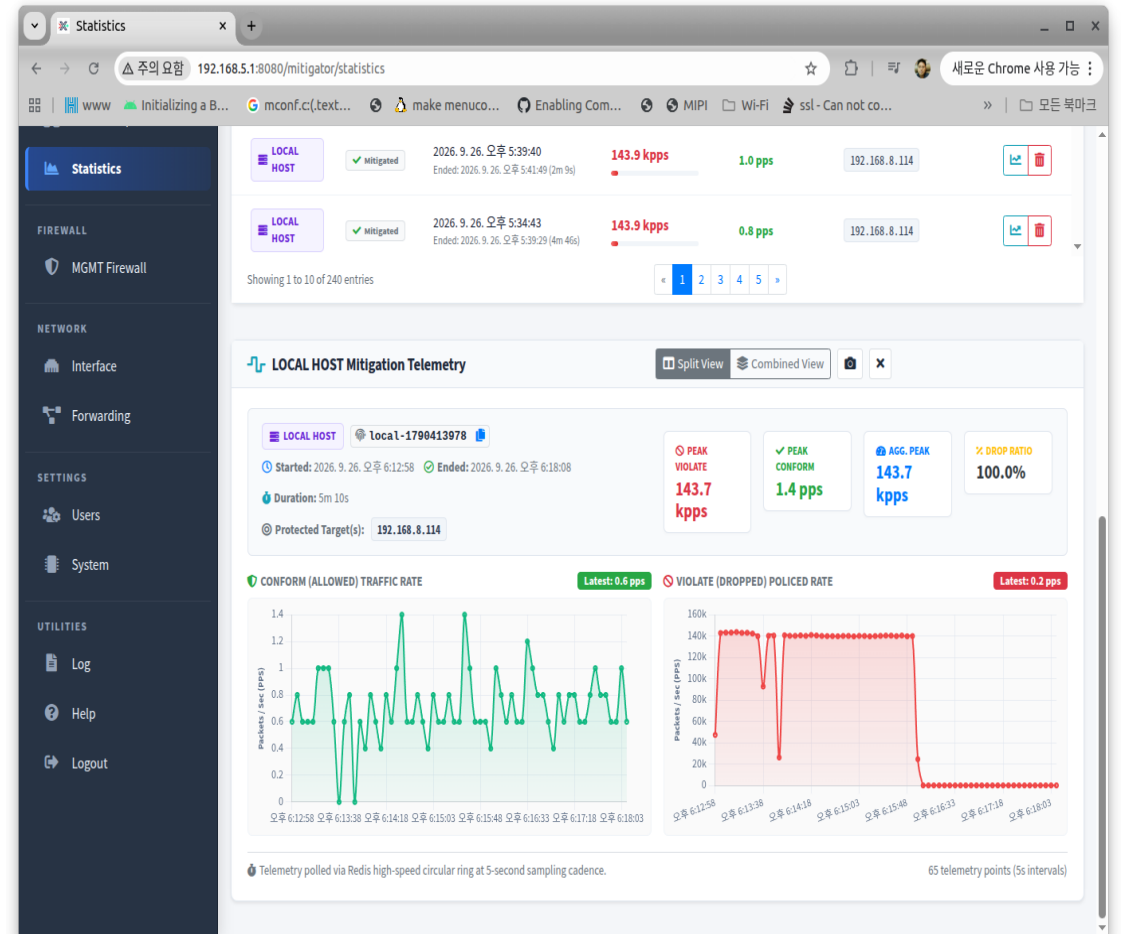
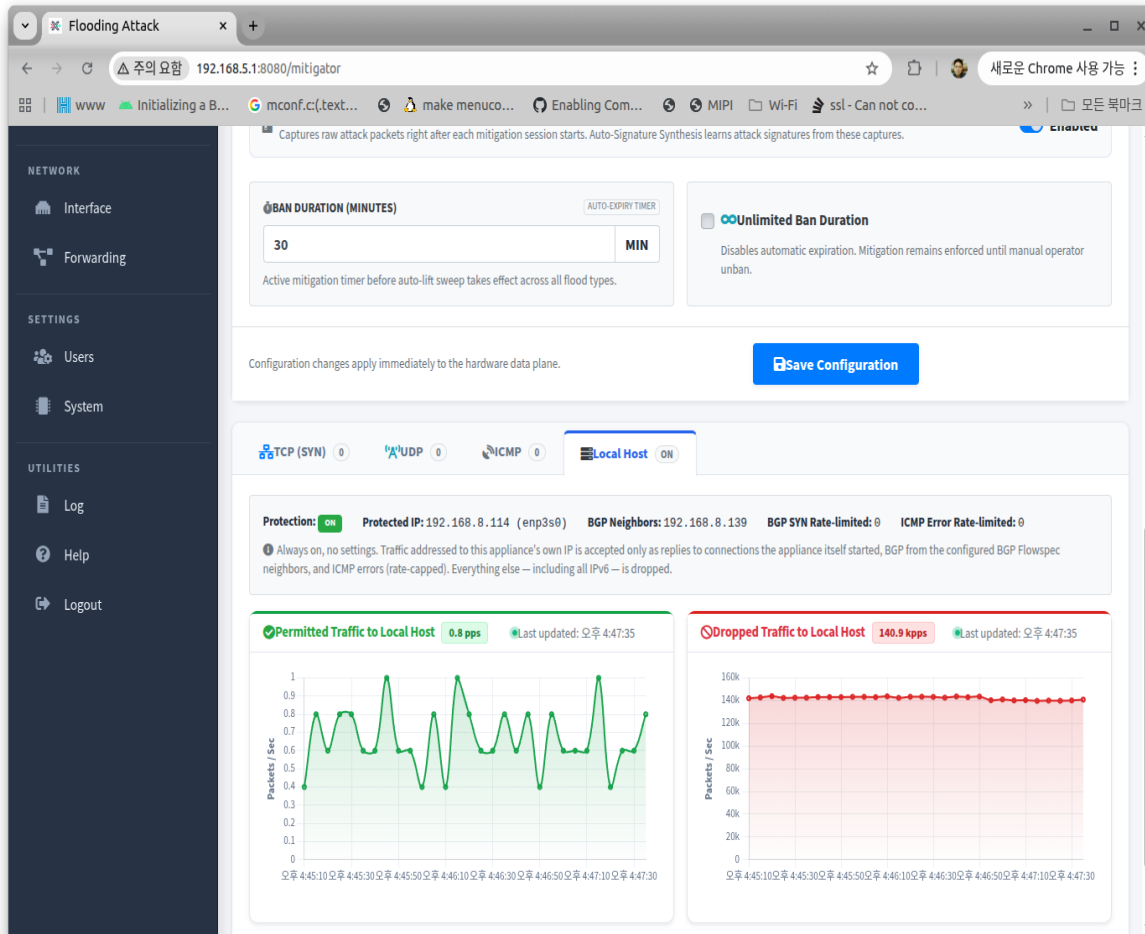


3. MightyXDDoS(10) – Local Host Mitigation(1)

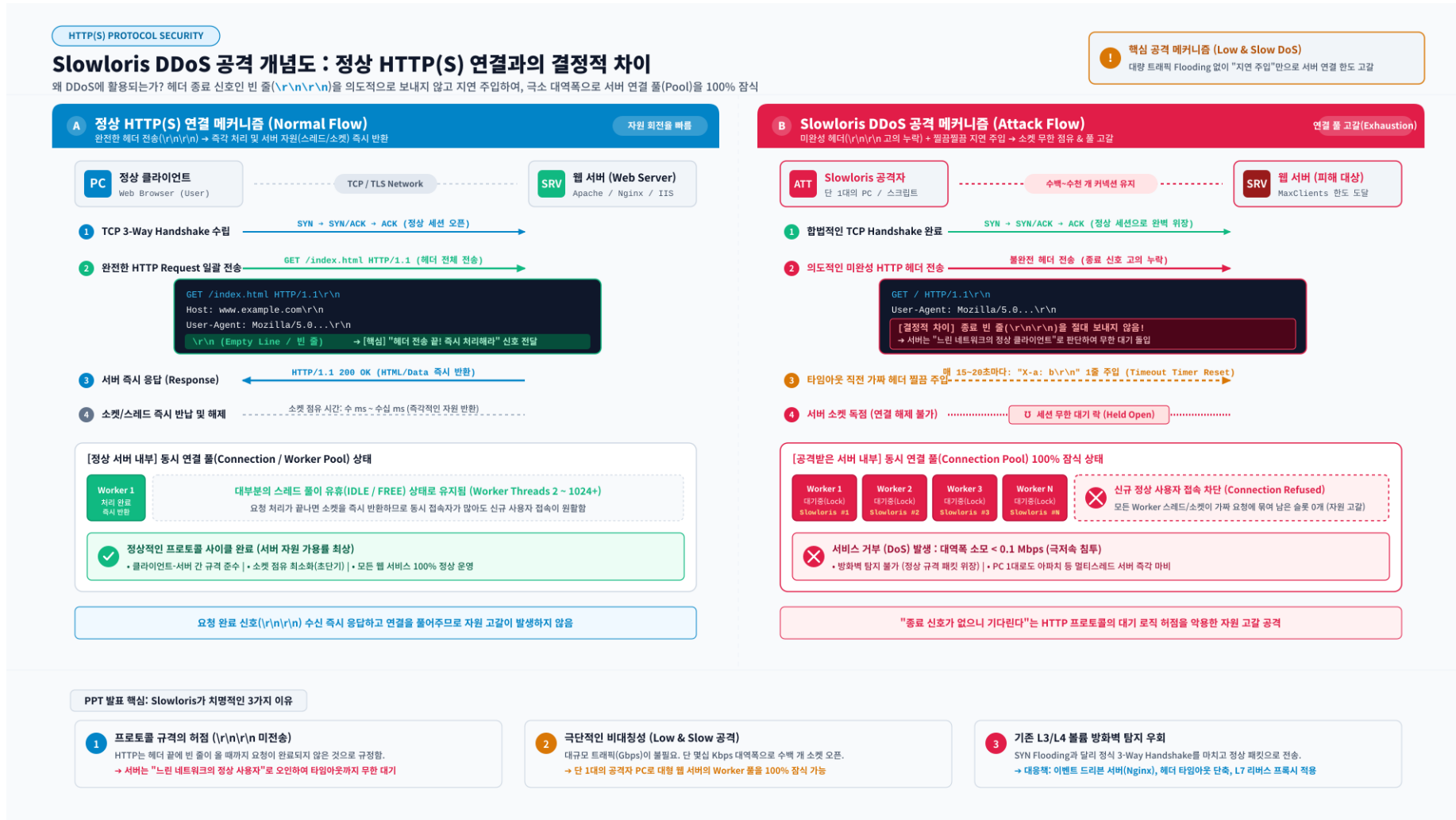
Local Host Protection Method



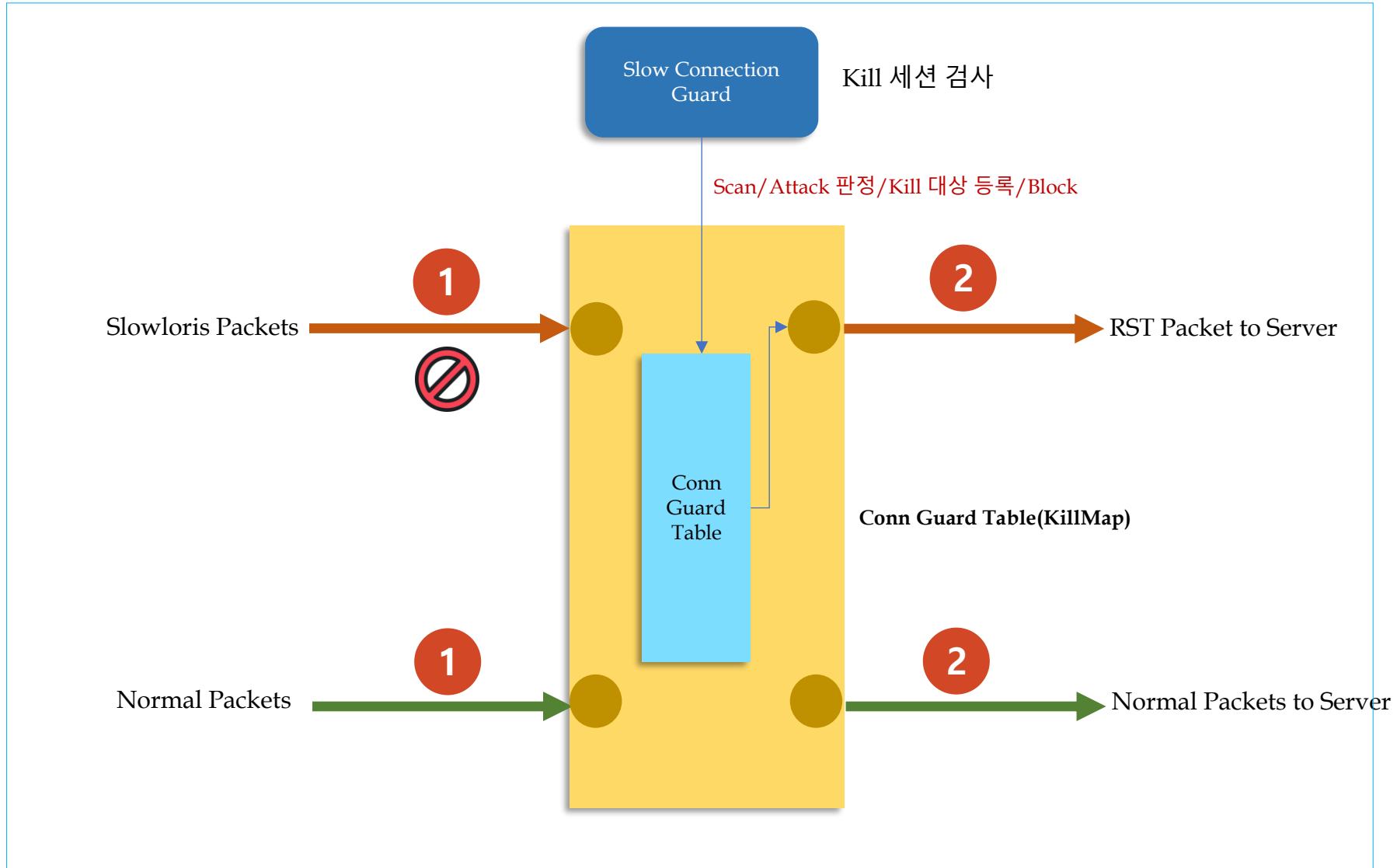
3. MightyXDDoS(10) – Local Host Mitigation(2)



3. MightyXDDoS(11) – Slow Connection Guard(1)



3. MightyXDDoS(11) – Slow Connection Guard(2)



3. MightyXDDoS(11) – Slow Connection Guard(3)

The screenshot shows the 'Application Filter' interface for the server IP 192.168.5.1:8080/mitigator/app-filter#slow. The 'Slow Connection Guard' is enabled. The interface includes a sidebar with navigation options like Detector, MITIGATOR, and FIREWALL. The main content area shows the 'Slow Connection Guard' settings, including a table of detected connections and various configuration options.

Server IP	Ports	Description	Actions
192.168.8.218	80 443	nanopi	[Edit] [Delete]

Slow Connection Guard ENABLED

☒ **Enable Slow Connection Guard** Finds connections that hold a server's workers open with very little traffic (Slowloris, slow POST, slow read).

Action

- ☒ **Monitor**
Detect and report only. Recommended first, to check the thresholds against real traffic.
- ☐ **Reset**
Reset each detected connection at the server, so its worker is freed at once.
- ☐ **Reset + Block**
Reset, and block a source that keeps getting reset or exceeds a per-source limit.

Detection

- Slow Request**
10 sec 64 bytes/s
The server has waited this long for a request the client keeps sending below this rate.
- Silent Hold**
30 sec
A connection was opened but the client has sent nothing for this long.
- Slow Read**
30 sec
The client has kept its receive window at zero for this long.

Concurrent Connections per Source
256 conns
Per protected service. Sources behind a large NAT may need a higher value.

New Connections per Source
100 per sec
Per protected service.

Block Source (Reset + Block)
5 resets in 60 sec
block for 10 min
The block duration also applies to Server Health blocks.

Save

The screenshot shows the 'Application Filter' interface for the server IP 192.168.5.1:8080/mitigator/app-filter#slow. The 'Slow Connection Guard' is enabled. The interface includes a sidebar with navigation options like NETWORK, SETTINGS, and UTILITIES. The main content area shows the 'Slow Connection Guard' status and recent detections.

Reported Only 0.0 conn/s Last updated: 오후 5:06:46

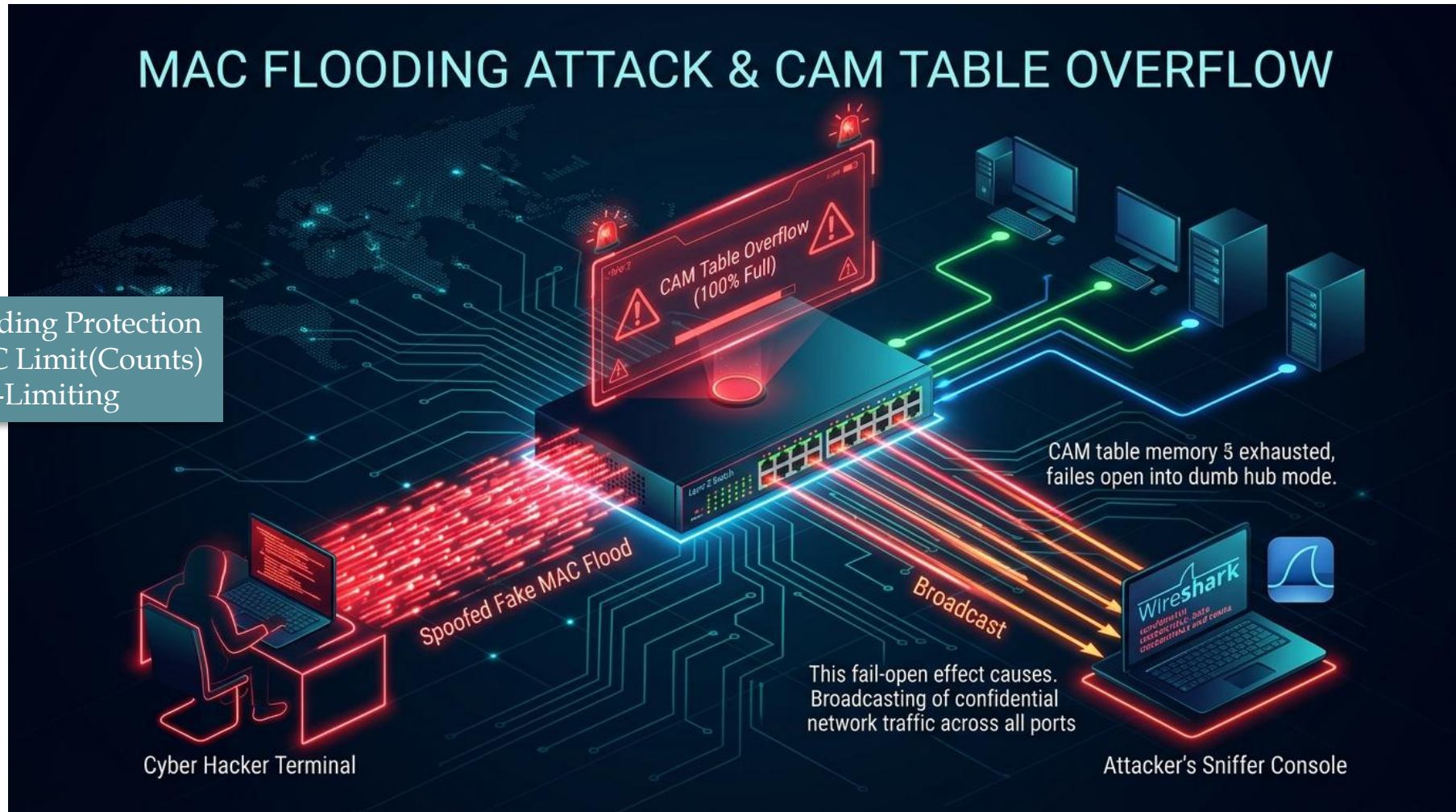
Connections Reset 0.0 conn/s Last updated: 오후 5:06:46

Recent Detections Grouped by source, service and detection type. Last updated: 오후 5:06:46

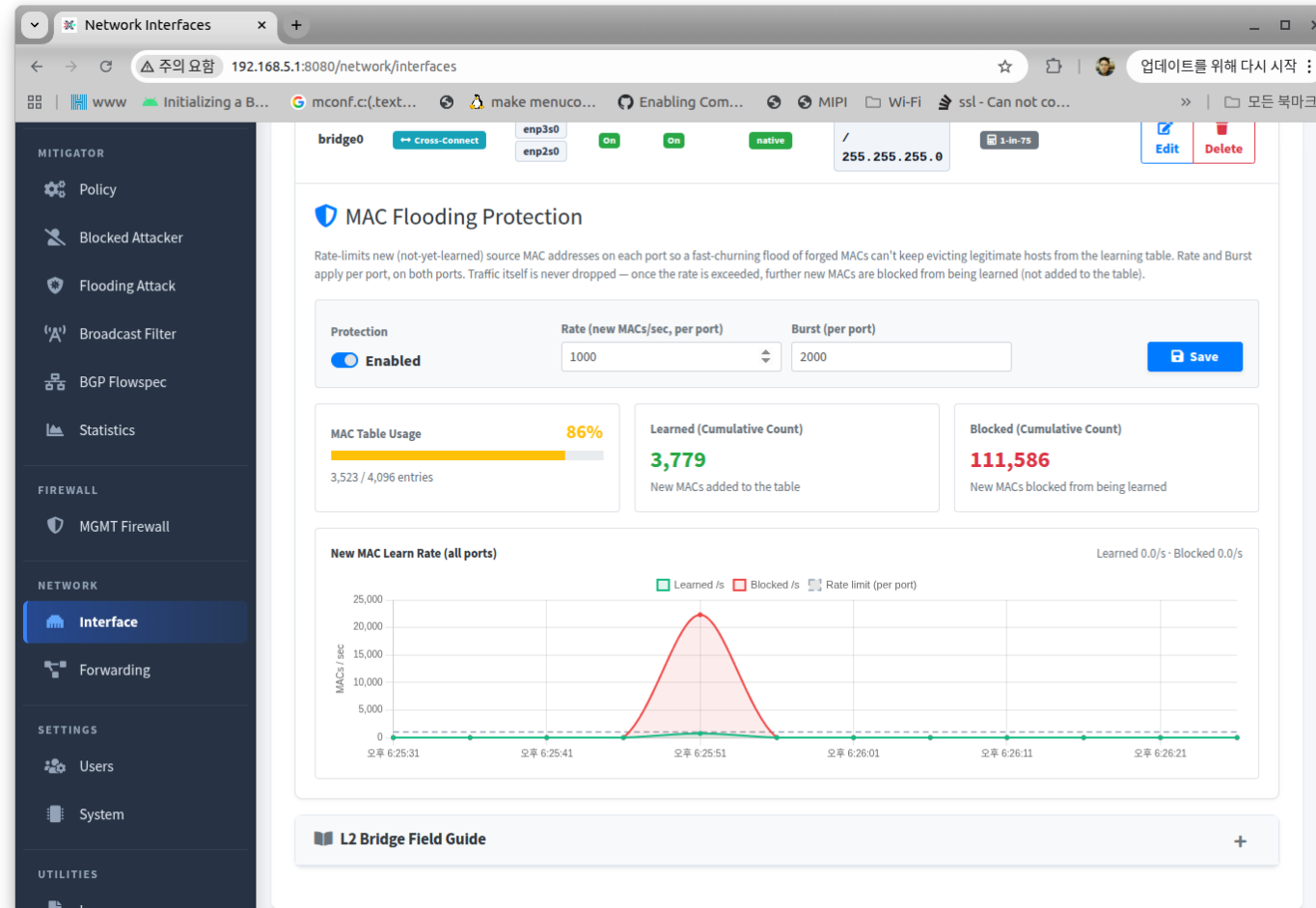
Last Seen	Source	Service	Detection	Count	Result	Last Detail	First Seen
2026. 9. 29. 오후 5:06:28	192.168.8.139	192.168.8.218:80	Slow Request	1,242	Reset 1,242	10 s 2026. 9. 29. 오후 4:55:55	

3. MightyXDDoS(12) – MAC Flooding Protection(1)

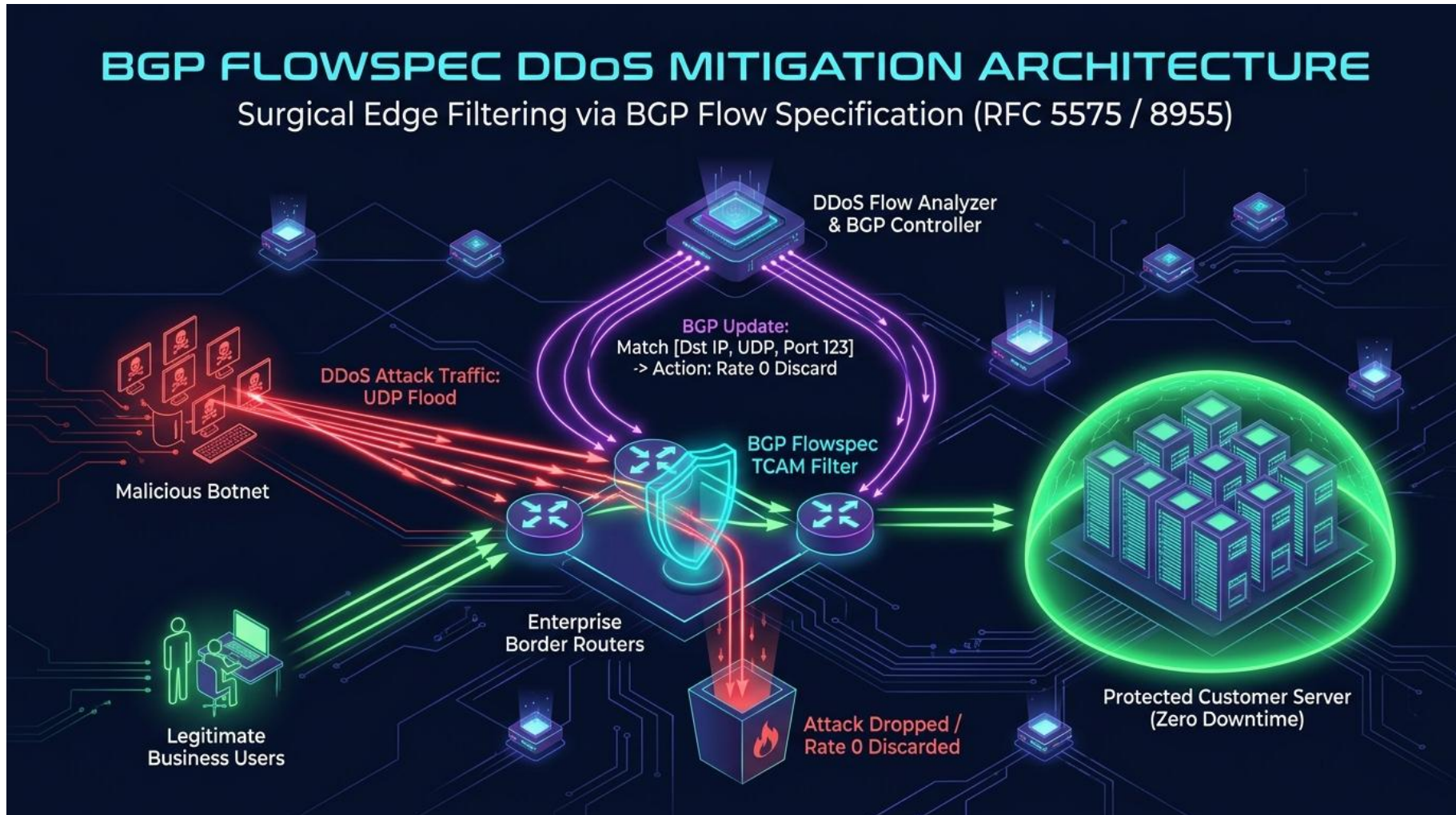
- MAC Flooding Protection
 - ✓ MAC Limit(Counts)
 - ✓ Rate-Limiting



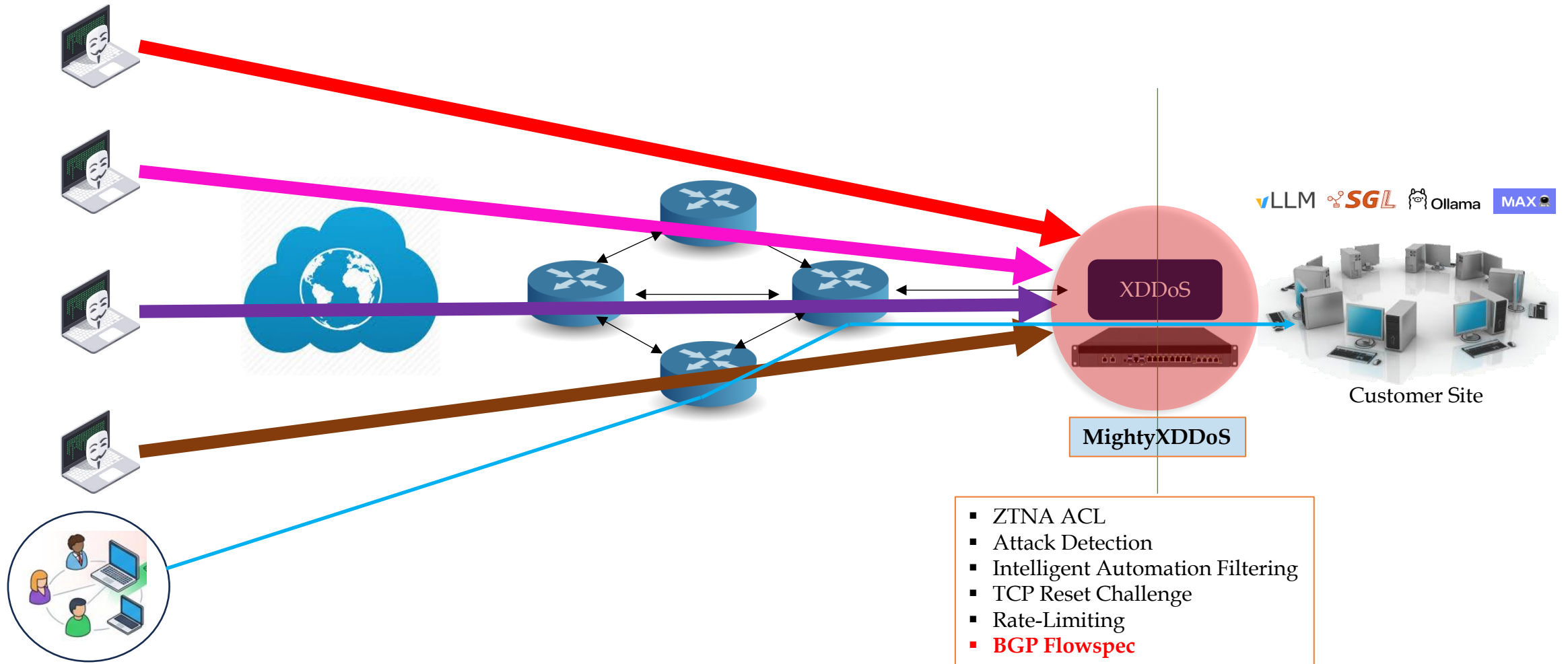
3. MightyXDDoS(12) – MAC Flooding Protection(2)



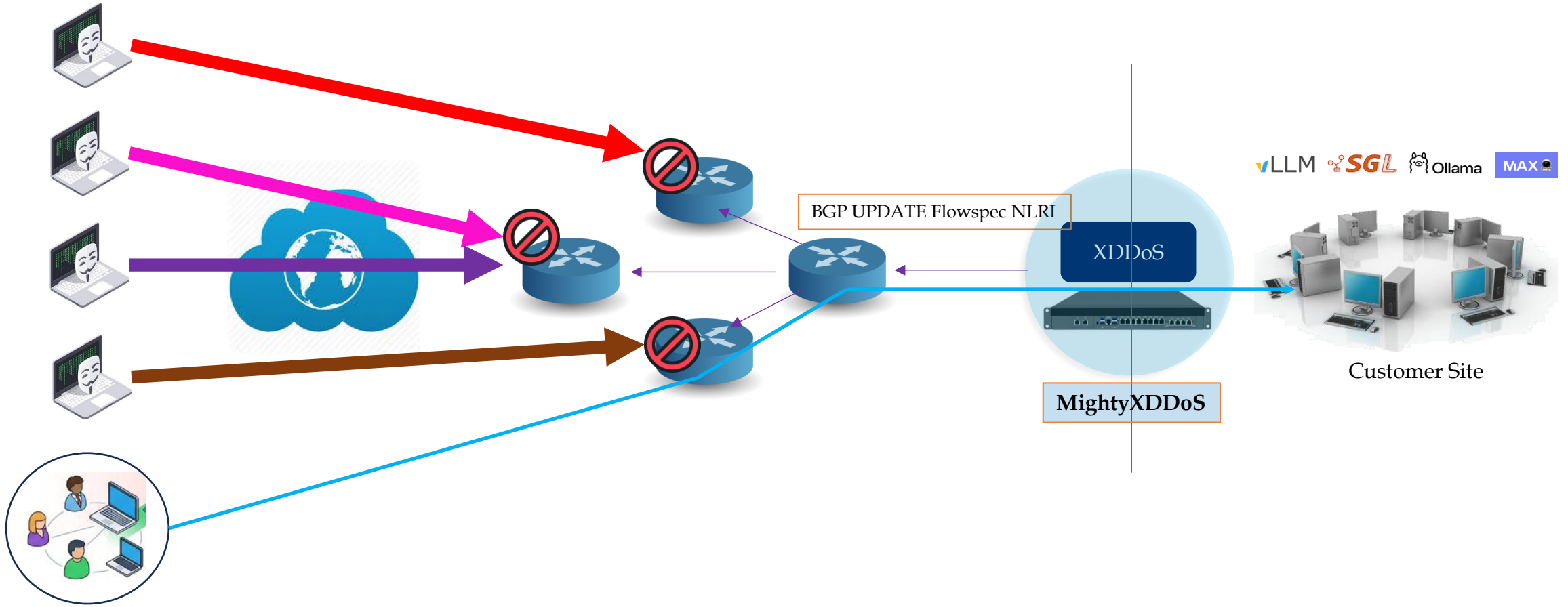
3. MightyXDDoS(13) – BGP Flowspec(1)



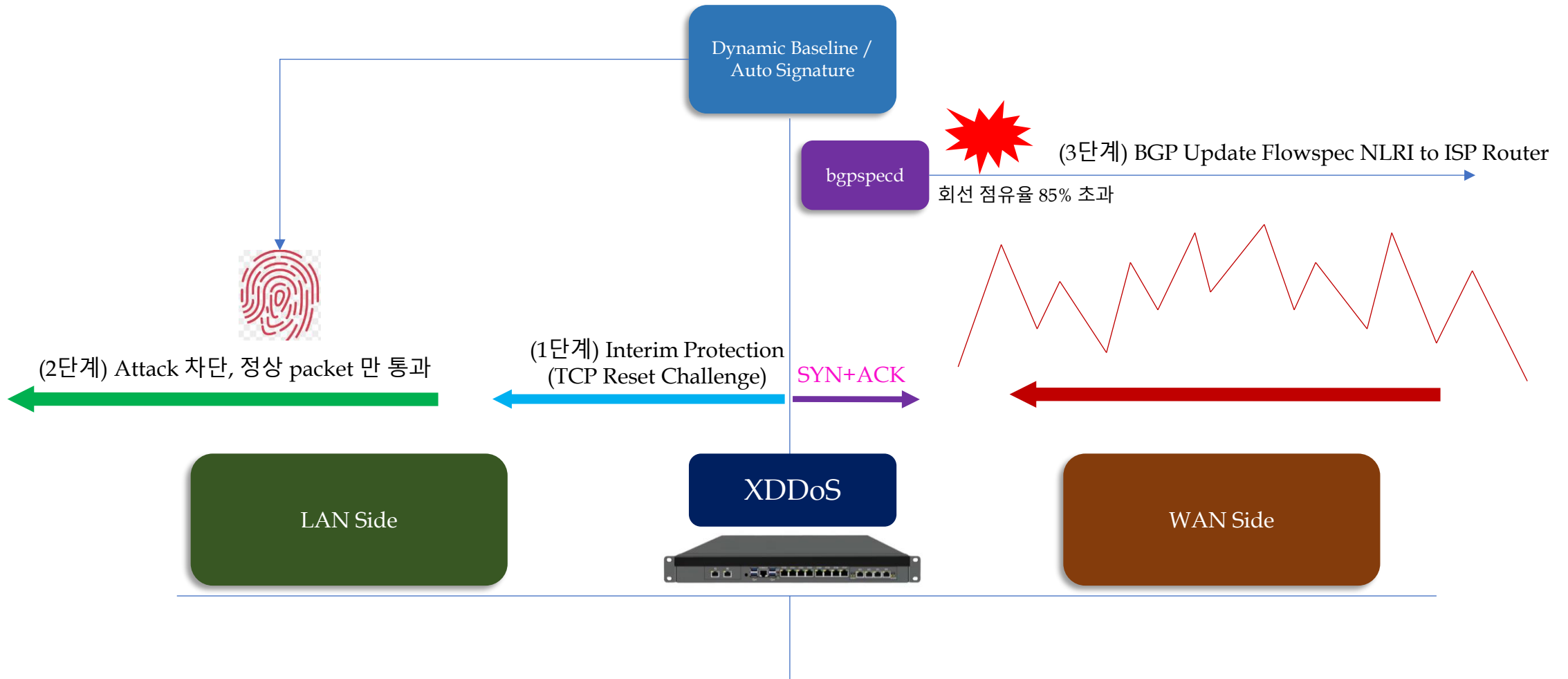
3. MightyXDDoS(13) – BGP Flowspec(2)



3. MightyXDDoS(13) – BGP Flowspec(3)

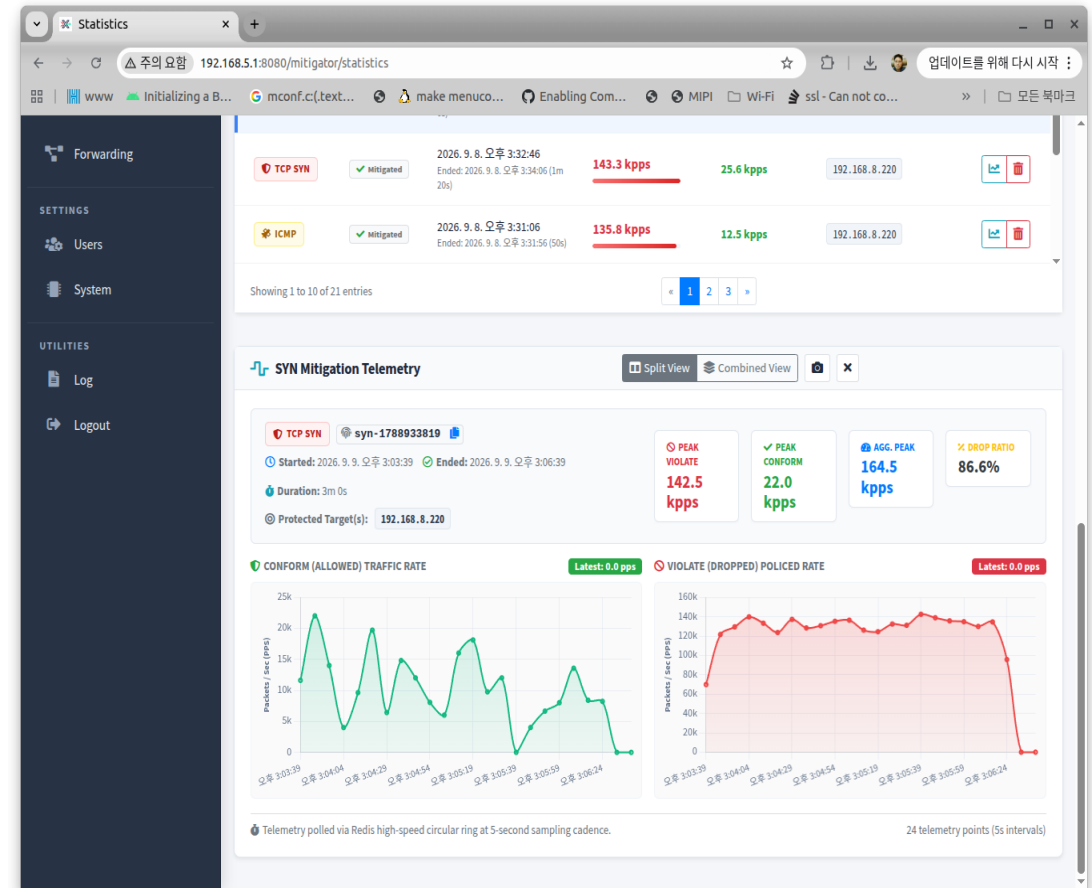
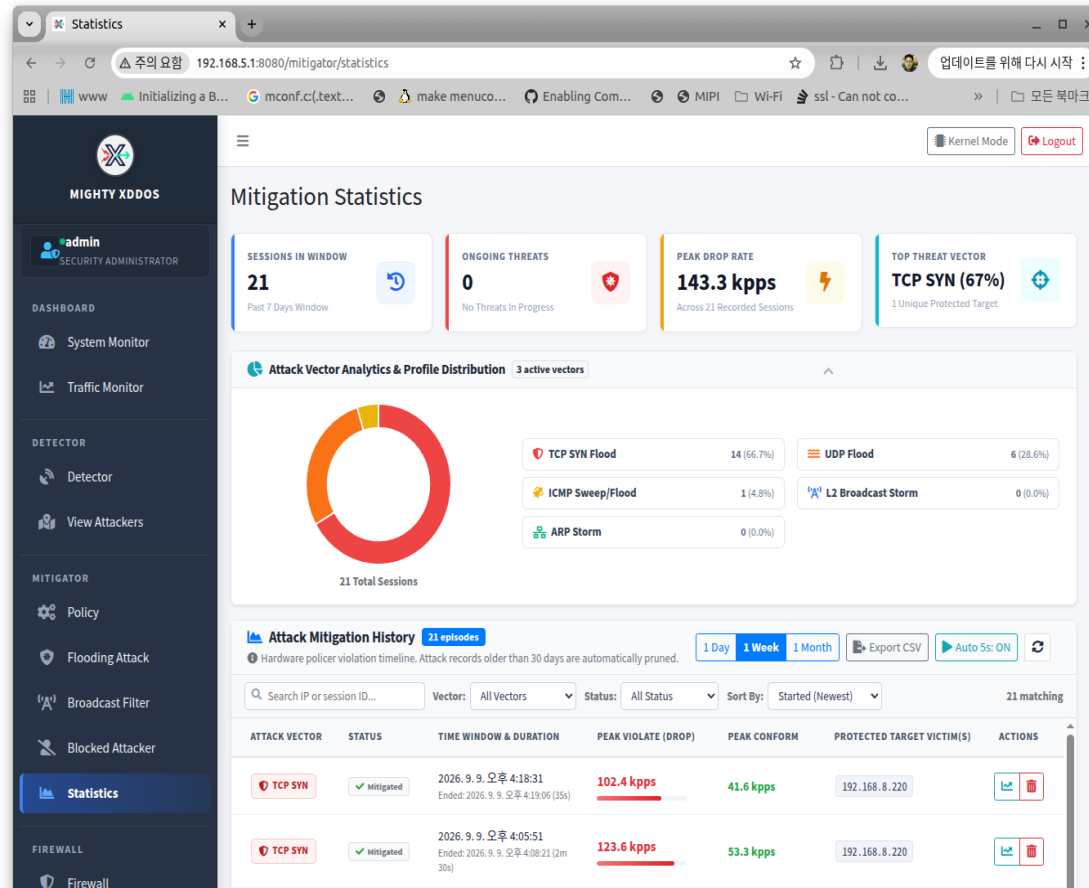


3. MightyXDDoS(13) – BGP Flowspec(4)



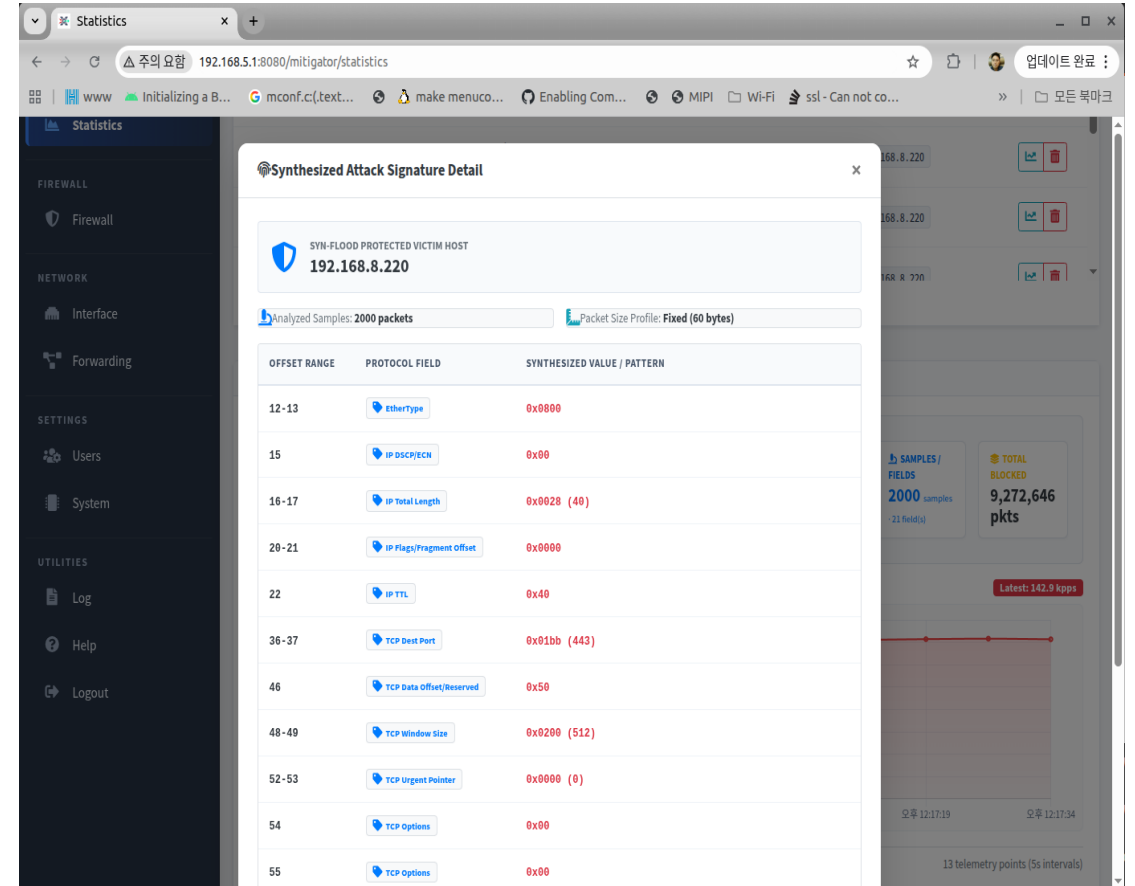
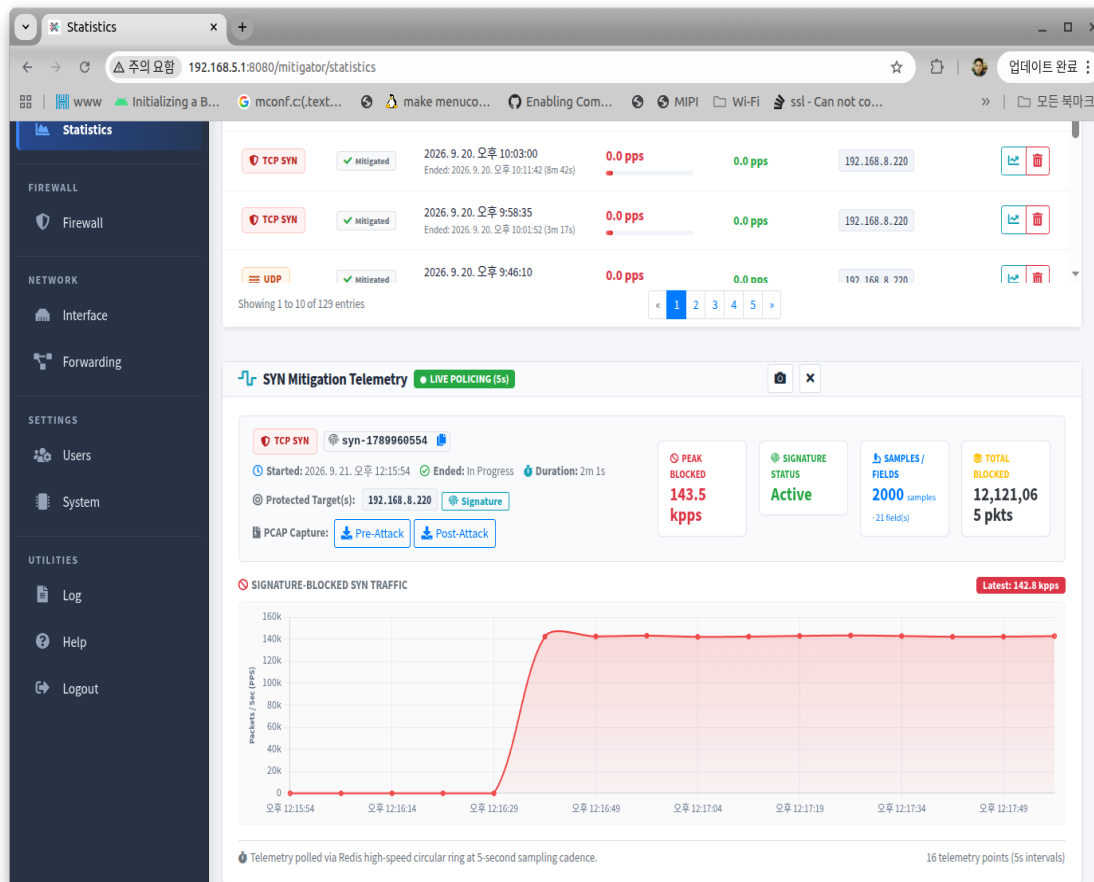
3. MightyXDDoS(14) – Statistics(1)

Defense Mode: TCP Reset Challenge Mode

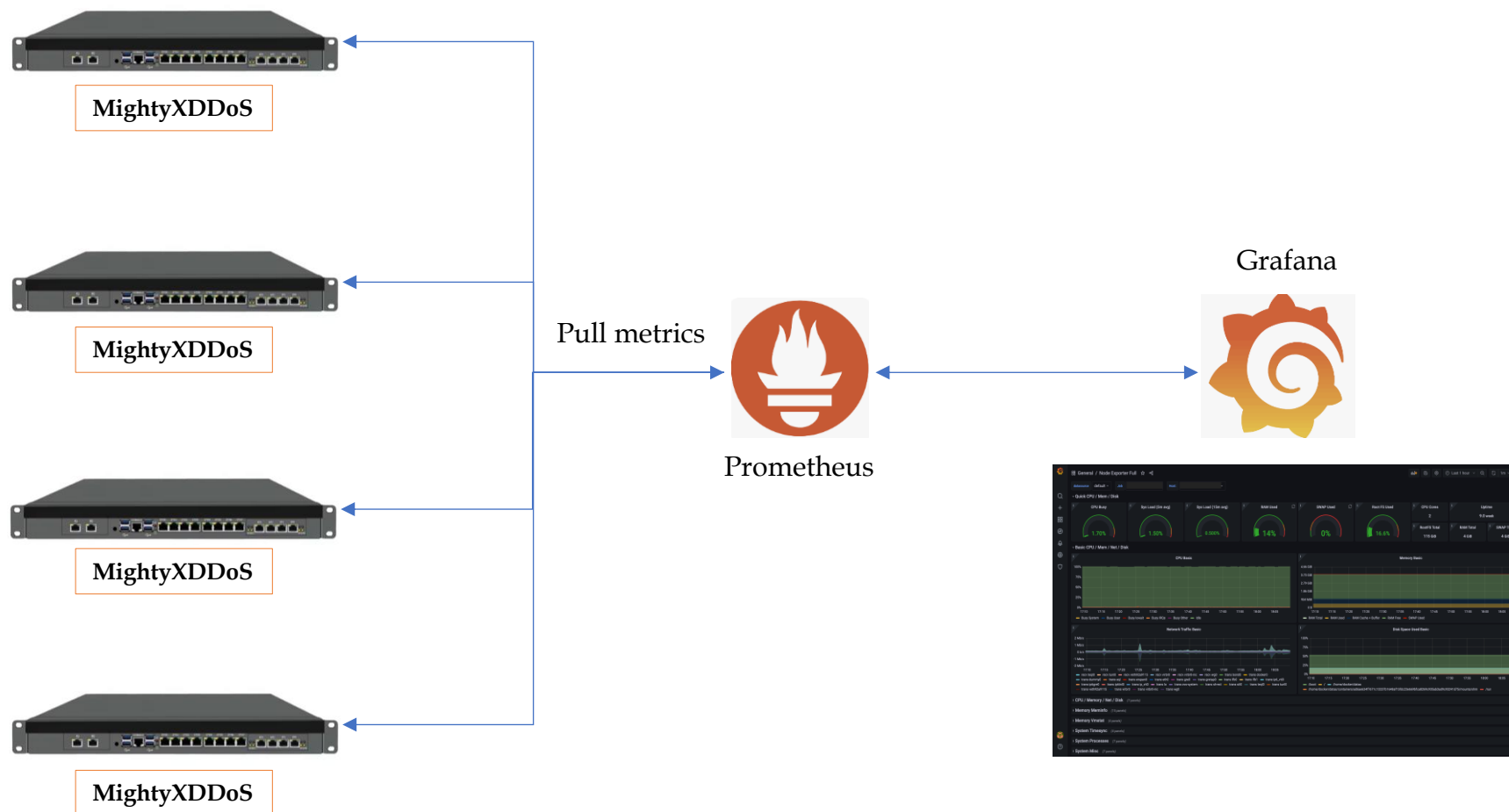


3. MightyXDDoS(14) – Statistics(2)

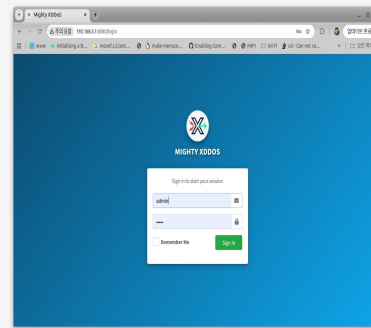
Defense Mode: Auto-Signature Synthesis Mode



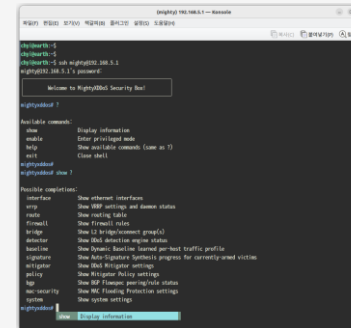
3. MightyXDDoS(15) – Prometheus Metrics Exporter



4. XDDoS User Interface

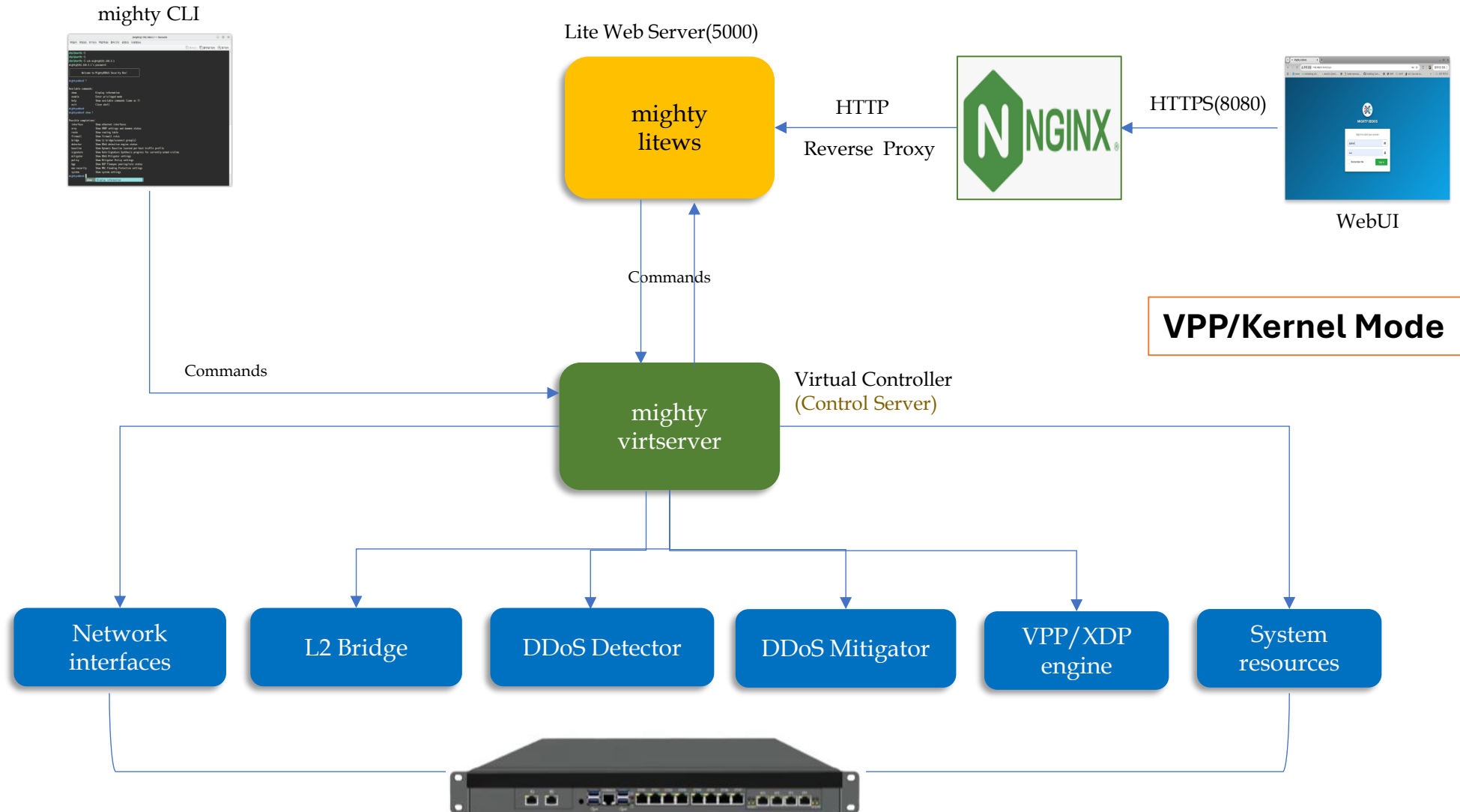


WebUI

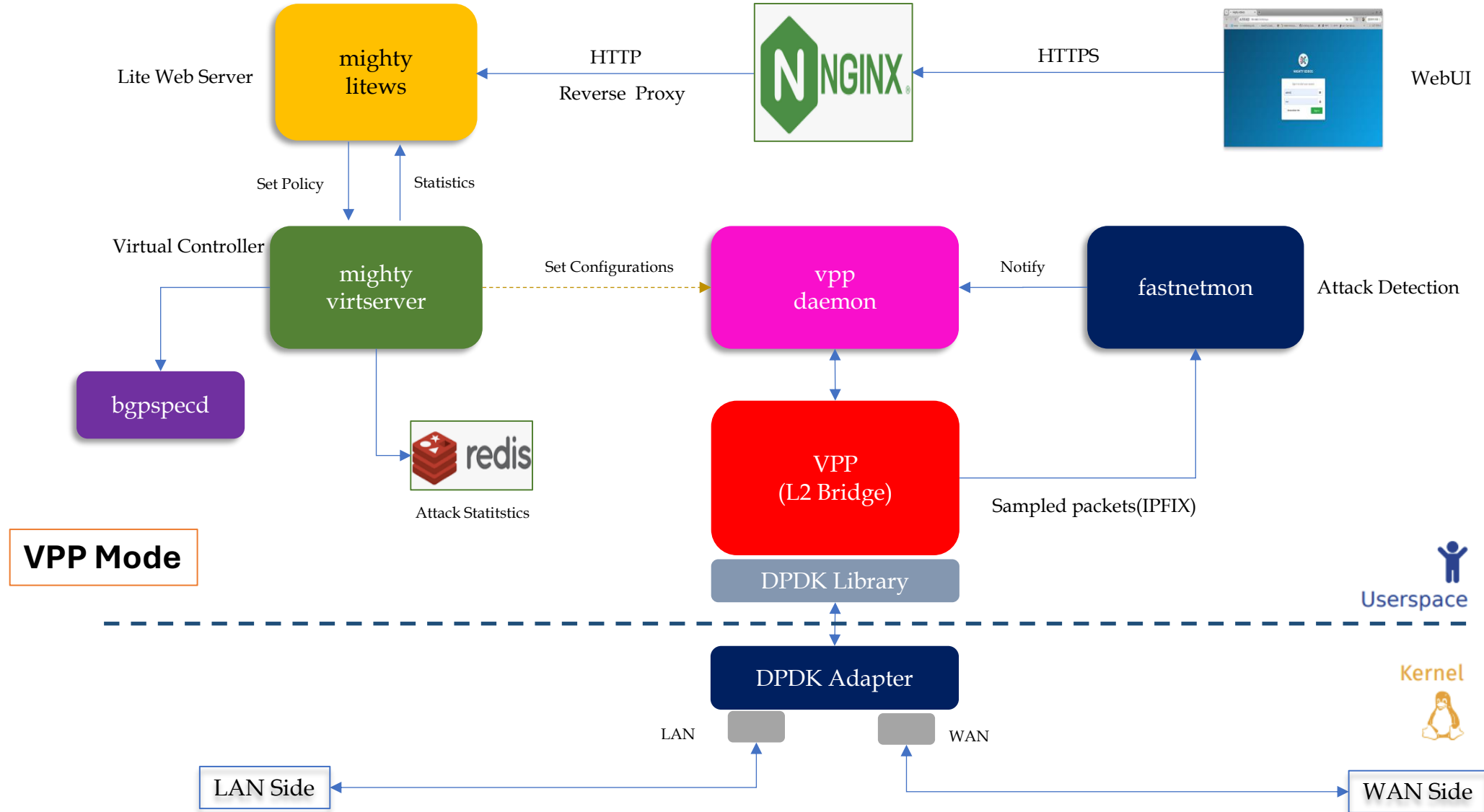


CLI

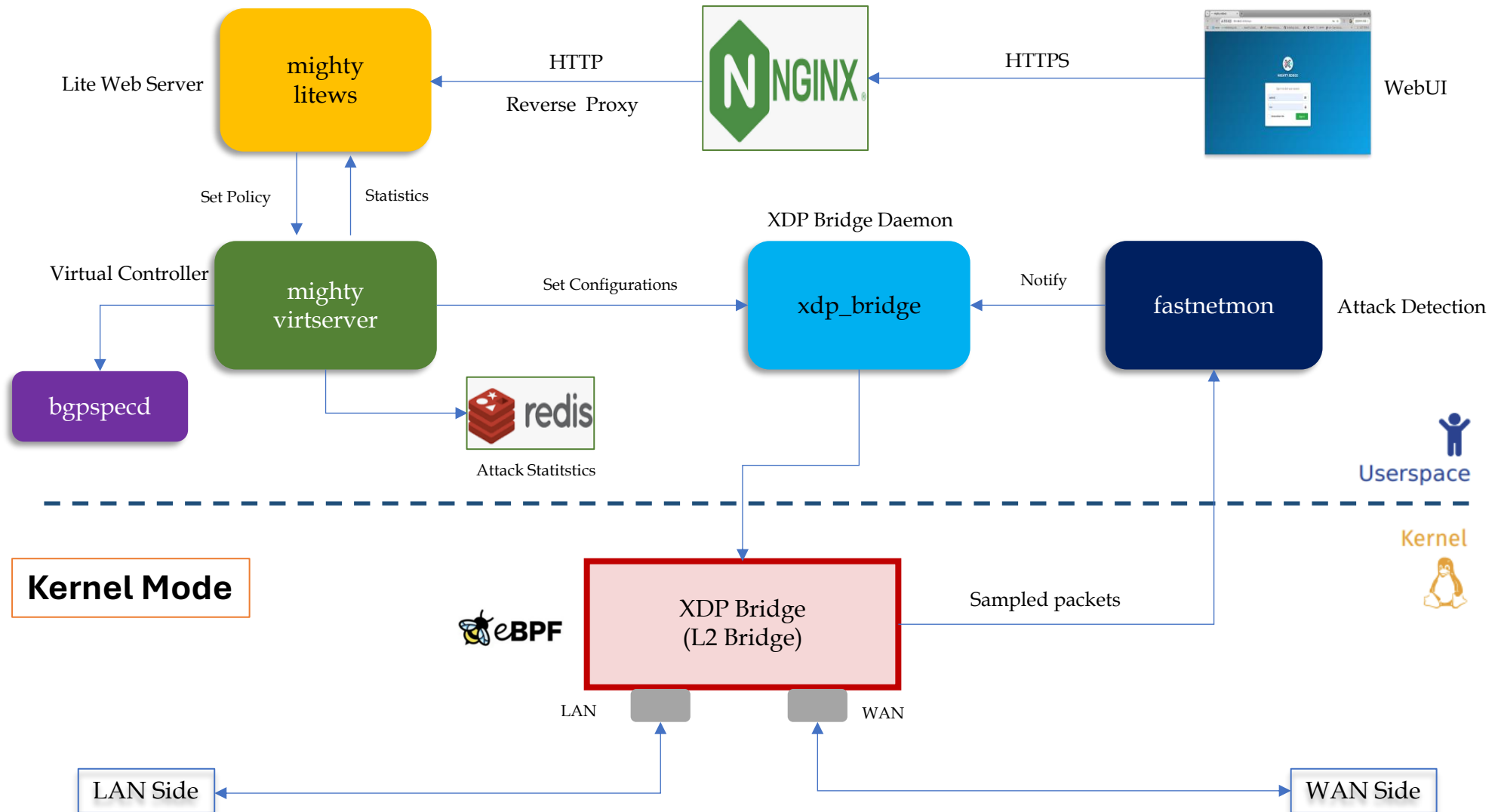
4. XDDoS User Interface(1) – Mighty Engine(1)



4. XDDoS User Interface(1) – Mighty Engine(2)

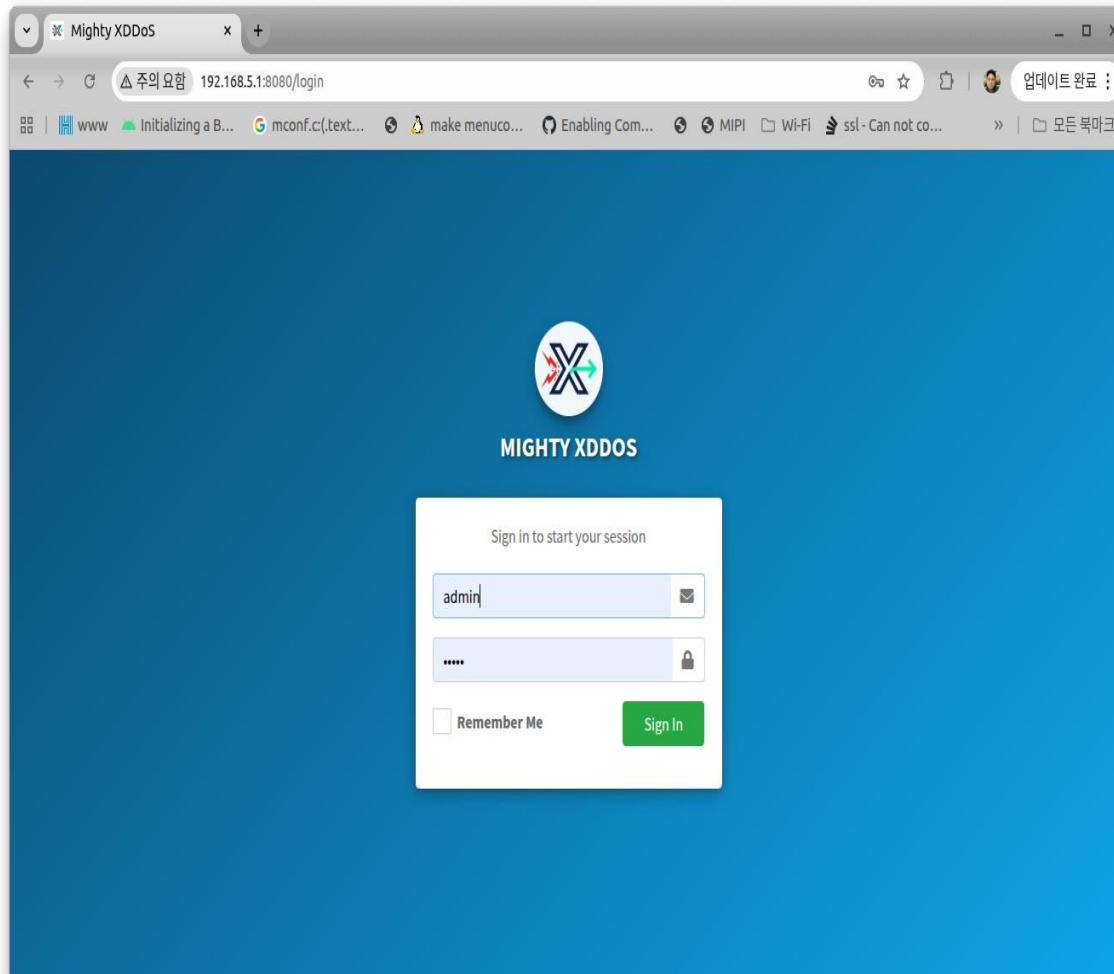


4. XDDoS User Interface(1) – Mighty Engine(3)

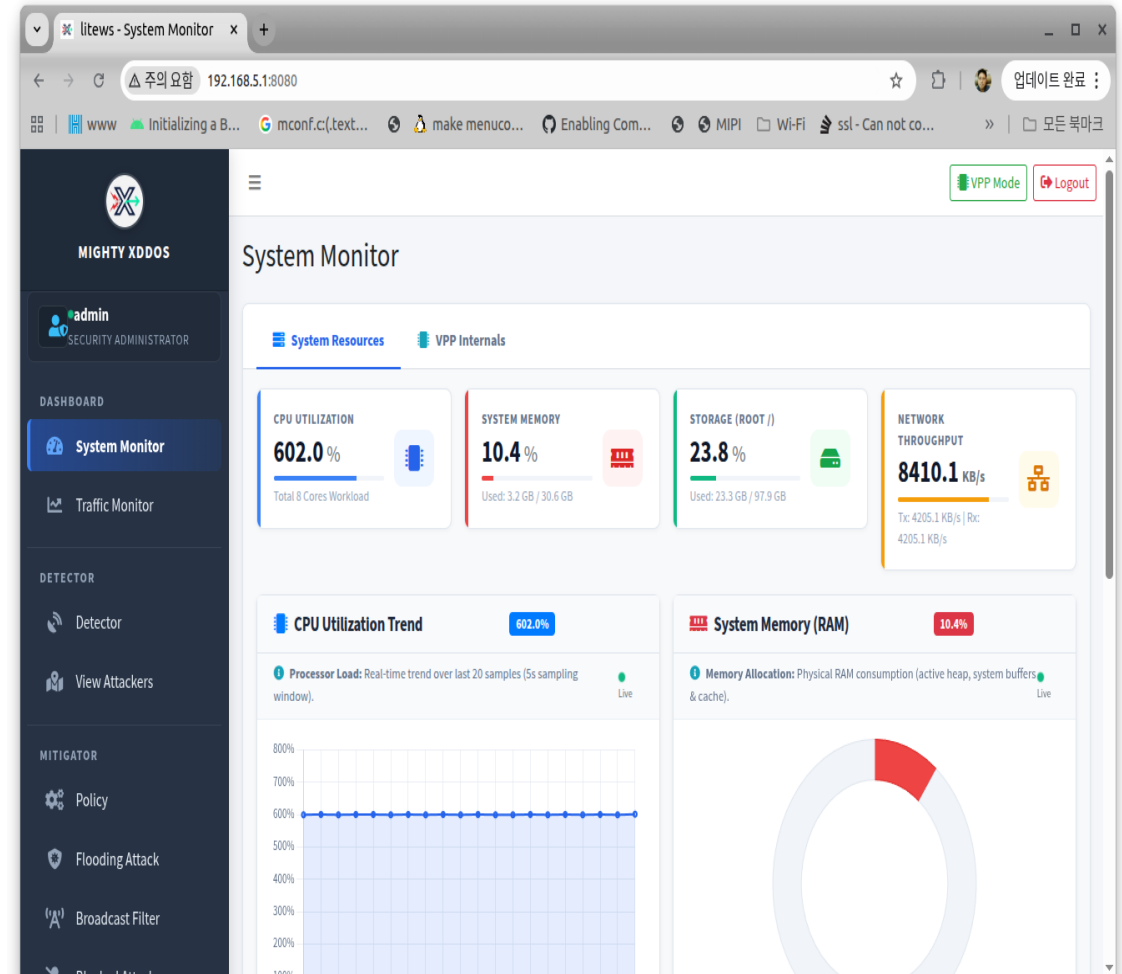


4. XDDoS User Interface(2) – WebUI(1)

Login page

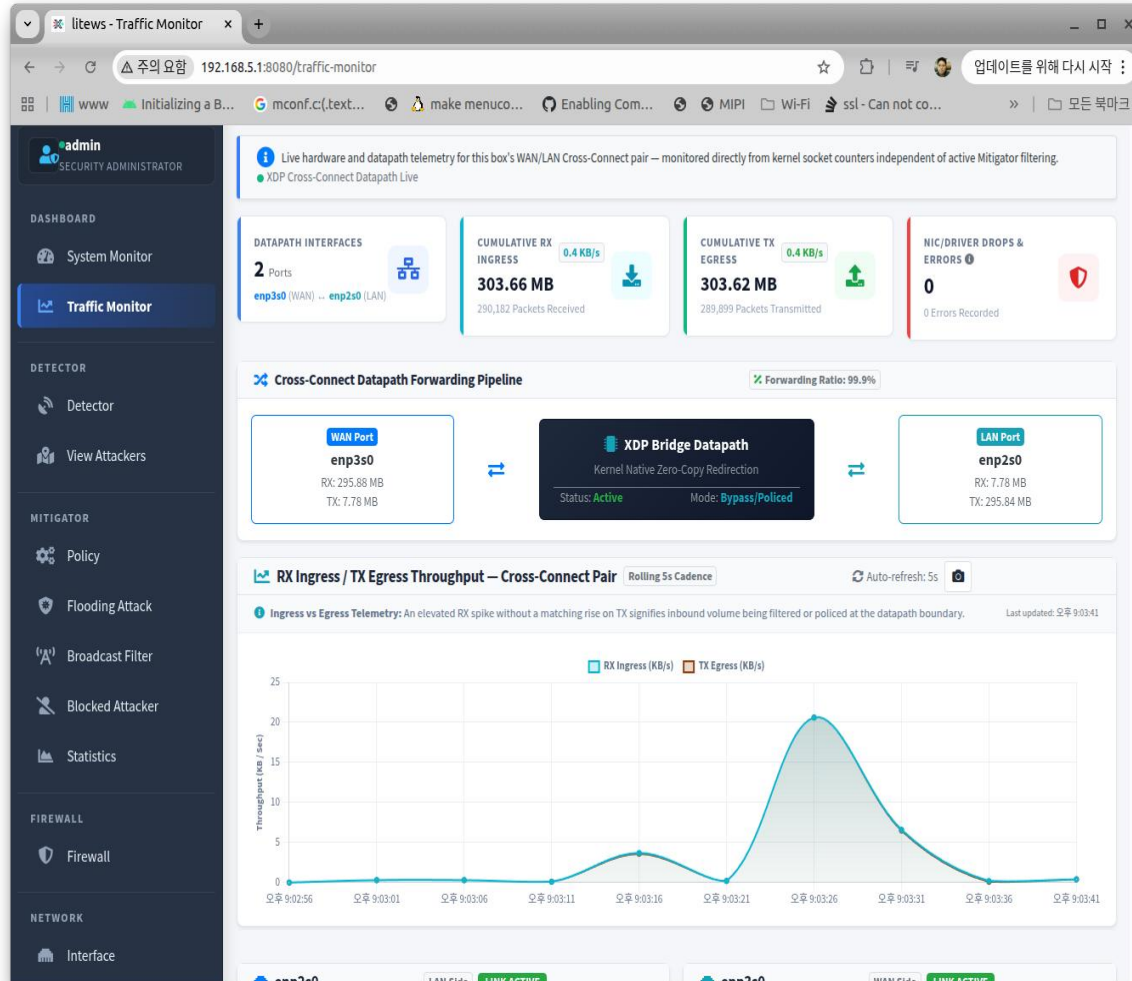


DashBoard – System Monitor Page

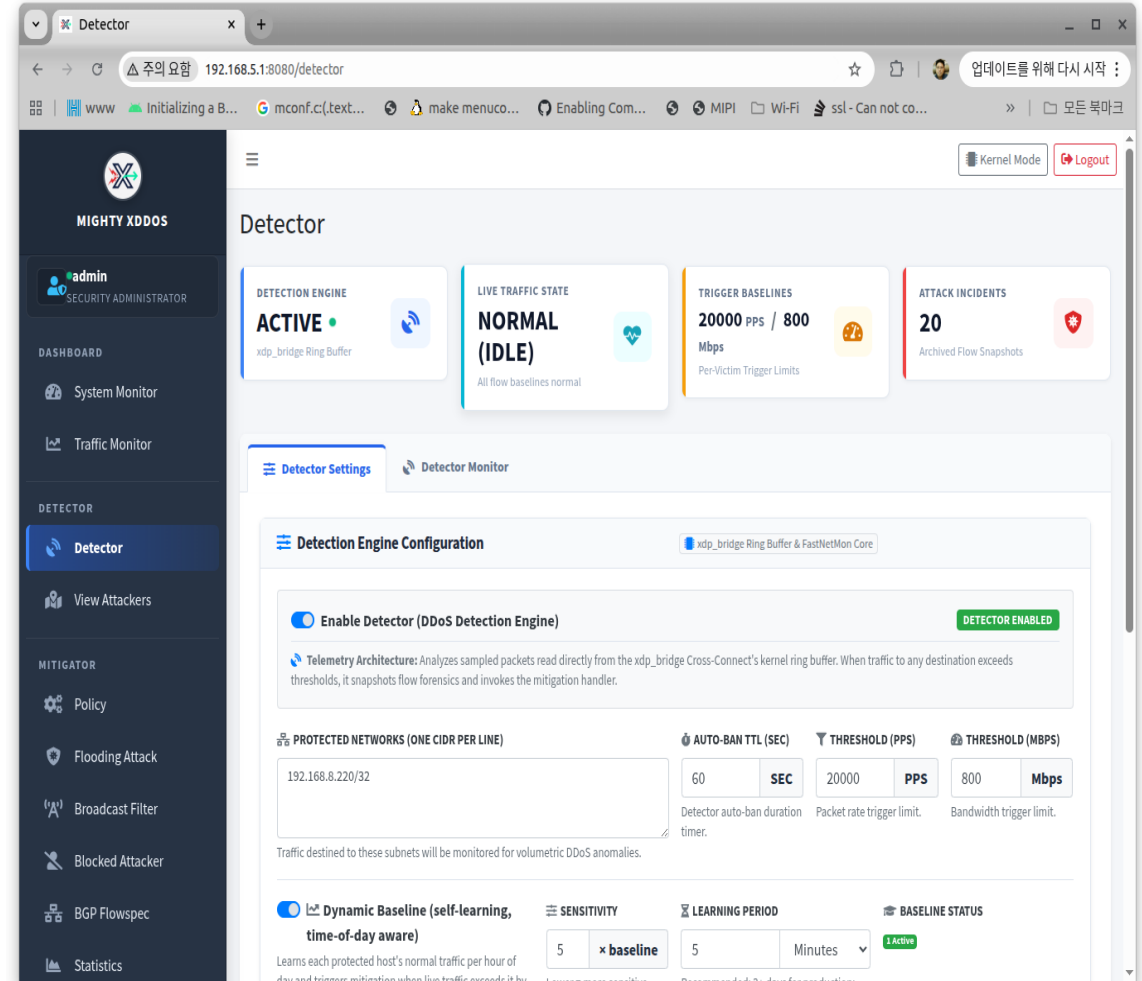


4. XDDoS User Interface(2) – WebUI(2)

DashBoard – Traffic Monitor Page



Detector Page#1



4. XDDoS User Interface(2) – WebUI(3)

Detector Page#2

The screenshot shows the 'Detector' page of the XDDoS web interface. The browser address bar shows '192.168.5.1:8080/detector'. The interface has a dark sidebar with a menu including 'admin', 'Dashboard', 'System Monitor', 'Traffic Monitor', 'Detector', 'View Attackers', 'Mitigator', 'Policy', 'Flooding Attack', 'Broadcast Filter', 'Blocked Attacker', 'BGP Flowspec', and 'Statistics'. The main content area is titled 'Detector' and contains four summary cards: 'DETECTION ENGINE ACTIVE' (xdp_bridge Ring Buffer), 'LIVE TRAFFIC STATE TRAFFIC DETECTED' (High flow anomaly detected!), 'TRIGGER BASELINES 20000 pps / 800 Mbps' (Per-Victim Trigger Limits), and 'ATTACK INCIDENTS 20' (Archived Flow Snapshots). Below these is a 'Detector Monitor (Live Telemetry)' section with a 'TRAFFIC ACTIVE' status and a 'Real-time Stream' button. It displays a 'FastNetMon Engine Stream' with live datapath telemetry. A terminal window shows the following output:

```
FastNetMon 1.2.10 Try Advanced edition: https://fastnetmon.com/product-overview/
IPs ordered by: bytes
Incoming traffic      74475 pps   23 mbps   1893 flows
192.168.8.220        74475 pps   23 mbps   989 flows  *banned*

Outgoing traffic      0 pps     0 mbps    0 flows

Internal traffic      0 pps     0 mbps

Other traffic         0 pps     0 mbps

Screen updated in:    5.3e-05 sec
Traffic calculated in: 0 sec 67 microseconds
Not processed packets: 0 pps
```

Detector - View Attackers page

The screenshot shows the 'View Attackers' page of the XDDoS web interface. The browser address bar shows '192.168.5.1:8080/detector/attackers-map'. The interface has a dark sidebar with a menu including 'admin', 'Dashboard', 'System Monitor', 'Traffic Monitor', 'Detector', 'View Attackers', 'Mitigator', 'Policy', 'Flooding Attack', 'Broadcast Filter', 'Blocked Attacker', 'BGP Flowspec', and 'Statistics'. The main content area is titled 'View Attackers' and contains a note: 'Only attacks fastnetmon traced to a single, distinct source IP are plotted here. Most floods (especially spoofed-source ones) involve many/random source addresses, where showing any one of them as "the attacker" would be misleading rather than useful – those are intentionally left off the map. An empty map most of the time is expected, not a sign something's broken.' Below this is an 'Attacker Locations' section with a map showing '1 plotted'. A tooltip for IP '23.34.172.165' is displayed, showing: 'Location: Unknown, United States', 'Target victim: 192.168.8.220', 'Attack type: unknown', and 'Time: 09 09 26 11:19:21'.

4. XDDoS User Interface(2) – WebUI(4)

Mitigator - Policy page

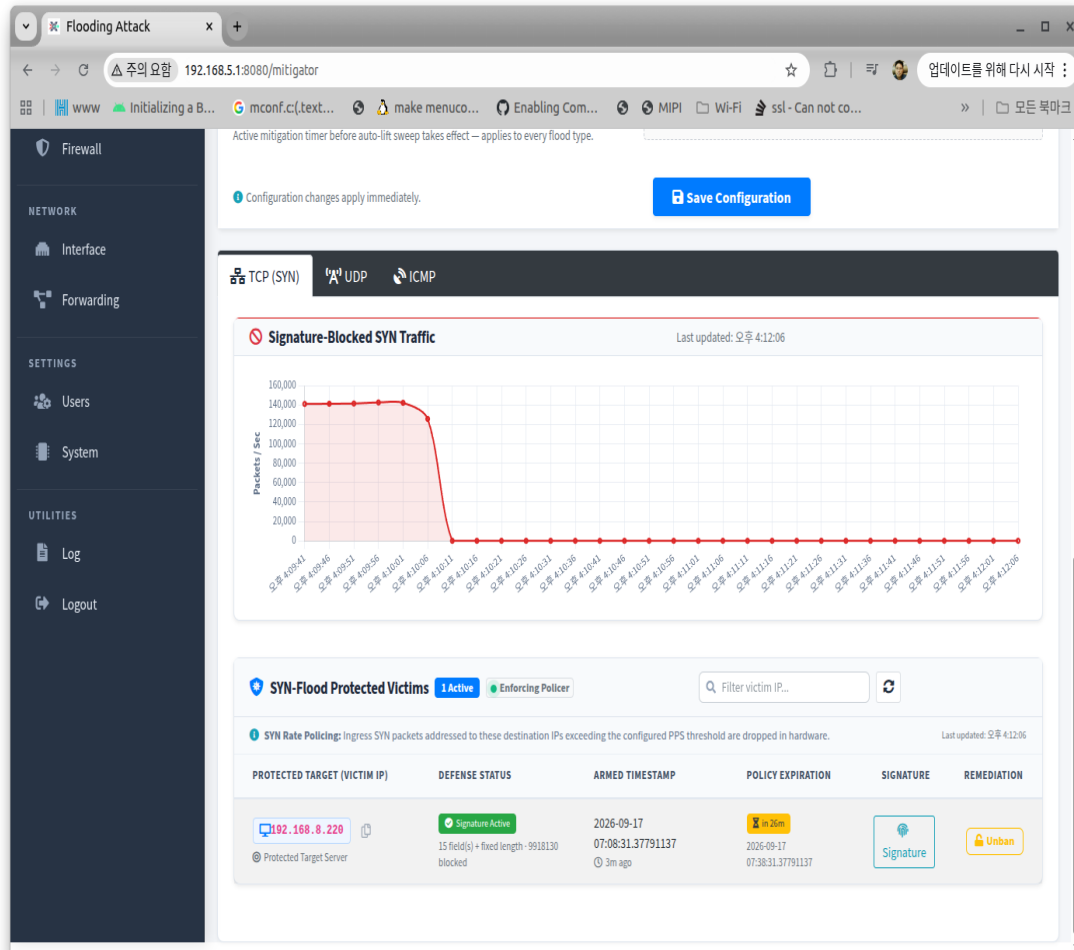
The screenshot shows the 'Policy' page of the Mitigator interface. The left sidebar contains navigation links: MIGHTY XDDoS, admin (SECURITY ADMINISTRATOR), DASHBOARD (System Monitor, Traffic Monitor), DETECTOR (Detector, View Attackers), and MITIGATOR (Policy, Flooding Attack, Broadcast Filter, Blocked Attacker, Statistics). The main content area is titled 'Policy' and features a 'VPP Mode' button and a 'Logout' button. The primary section is 'ACL / Mitigator Interface Binding Scope', which explains that it controls bridge member interfaces for enforcement. It offers two options: 'All (LAN & WAN)' (Default) and 'WAN Only'. Below this is a 'WAN Service Port Policy' section, marked as 'ZTNA-STYLE, AUTO-GENERATED', with an 'ENABLED' status and an 'Add Port Rule' button. A 'Save' button is also present.

Mitigator - Blocked Attacker page

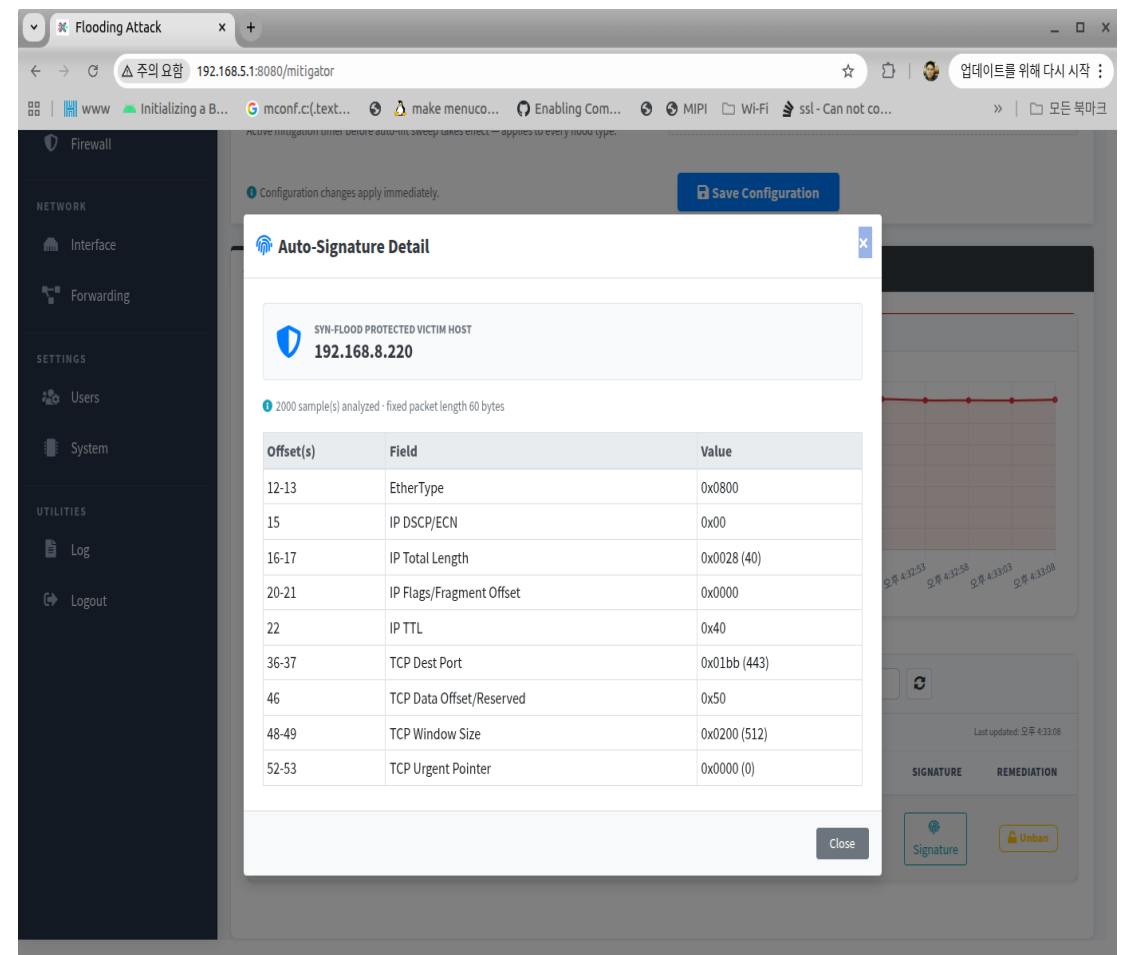
The screenshot shows the 'Blocked Attacker' page of the Mitigator interface. The left sidebar is identical to the Policy page. The main content area is titled 'Blocked Attacker' and includes a 'VPP Mode' button and a 'Logout' button. It displays four key metrics: 'MITIGATION ENGINE' (ACTIVE, L2 Bridge Hardware Policer), 'PROTECTED VICTIMS' (1, Flood-Type Rate Policed), 'BLOCKED ATTACKERS' (0, Hardware ACL Drop Enforced), and 'FLOOD DROP RATE' (—, Live Drop Share (Active Tab)). Below these is a 'Blocked Attacker Source IPs' section with a search bar and a table. The table has columns for 'ATTACKER SOURCE IP', 'TARGETED VICTIM', 'BLOCKED AT', 'EXPIRATION / TTL', and 'REMEDIATION'. A green checkmark icon and the text 'No Hostile Sources Blocked' are displayed, indicating that the hardware ACL filter table is clean.

4. XDDoS User Interface(2) – WebUI(5)

Mitigator - Flooding Attack/TCP(SYN) page

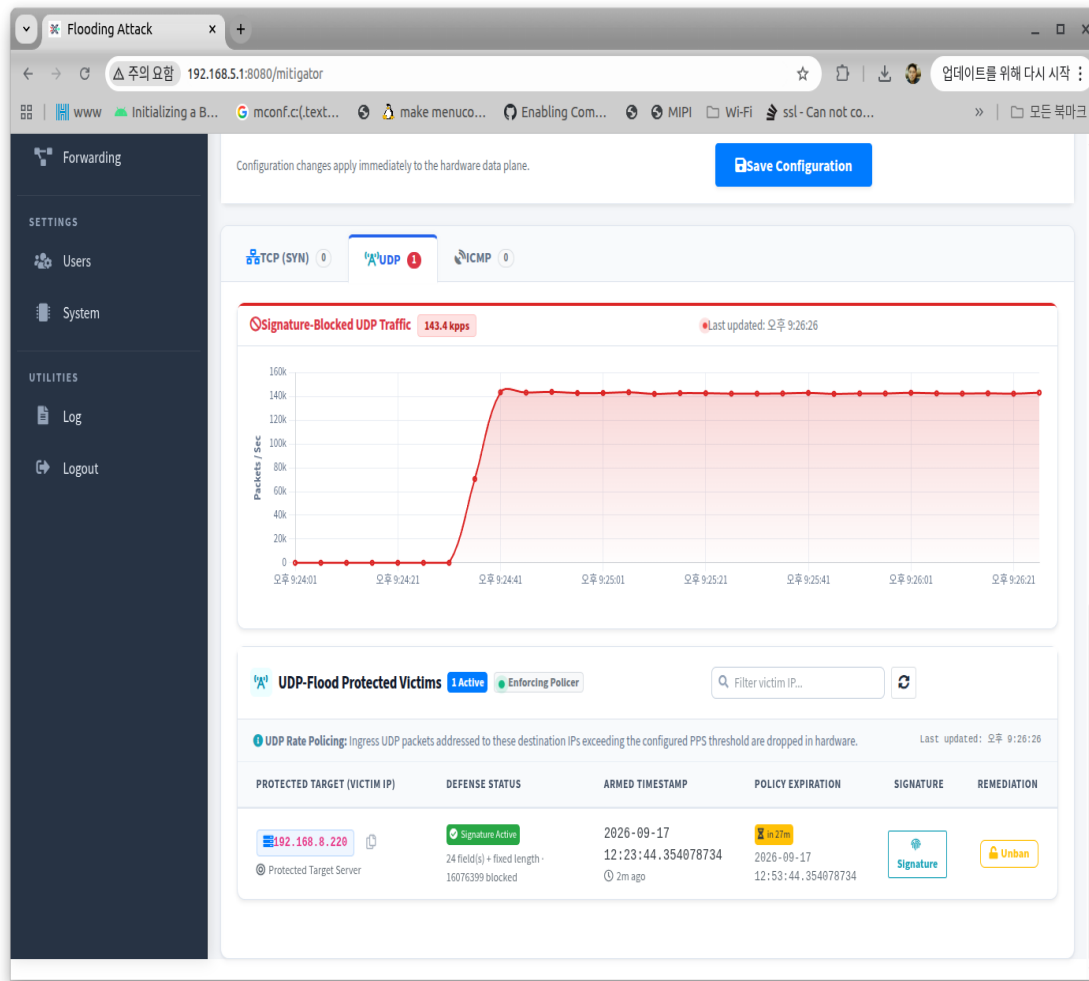


Mitigator - Flooding Attack/TCP(SYN) page - Signature

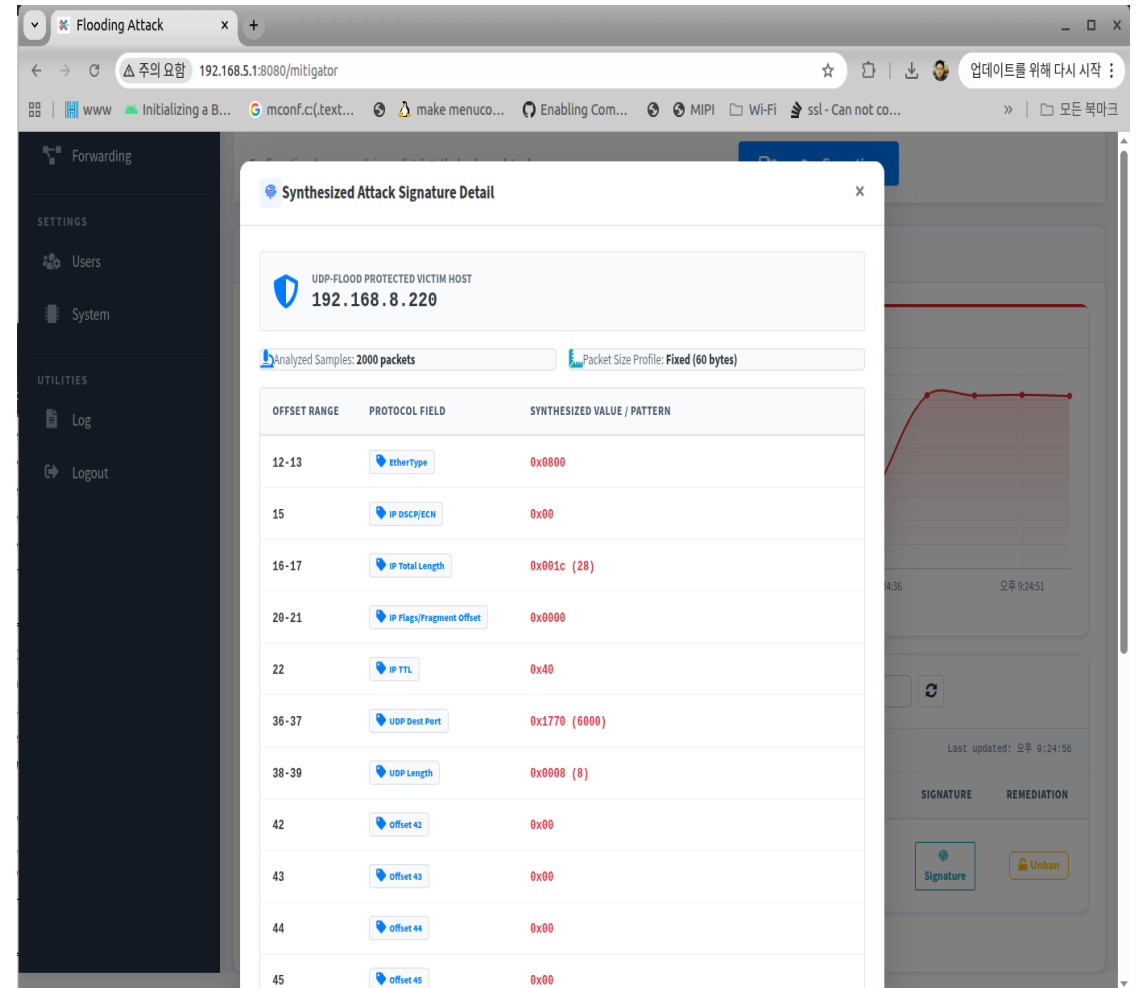


4. XDDoS User Interface(2) – WebUI(6)

Mitigator - Flooding Attack/UDP page

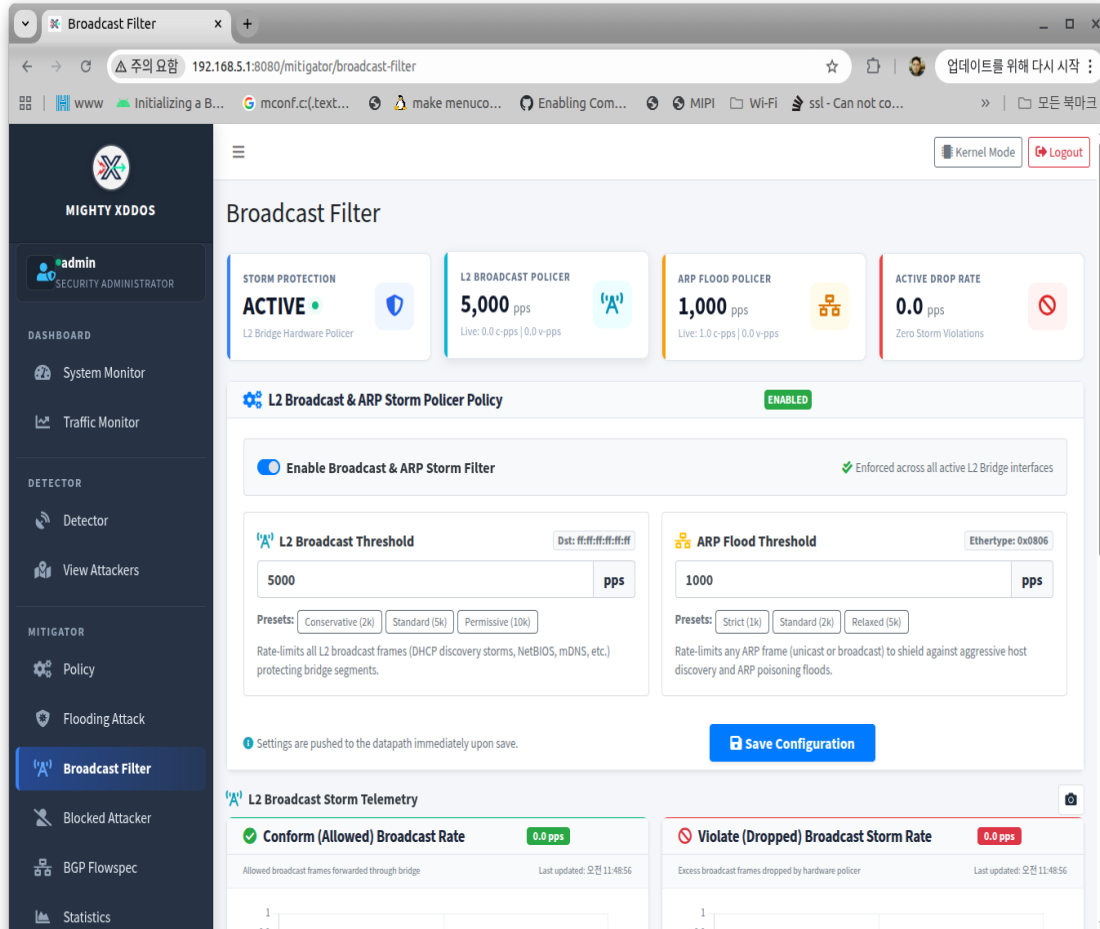


Mitigator - Flooding Attack/UDP - Signature page

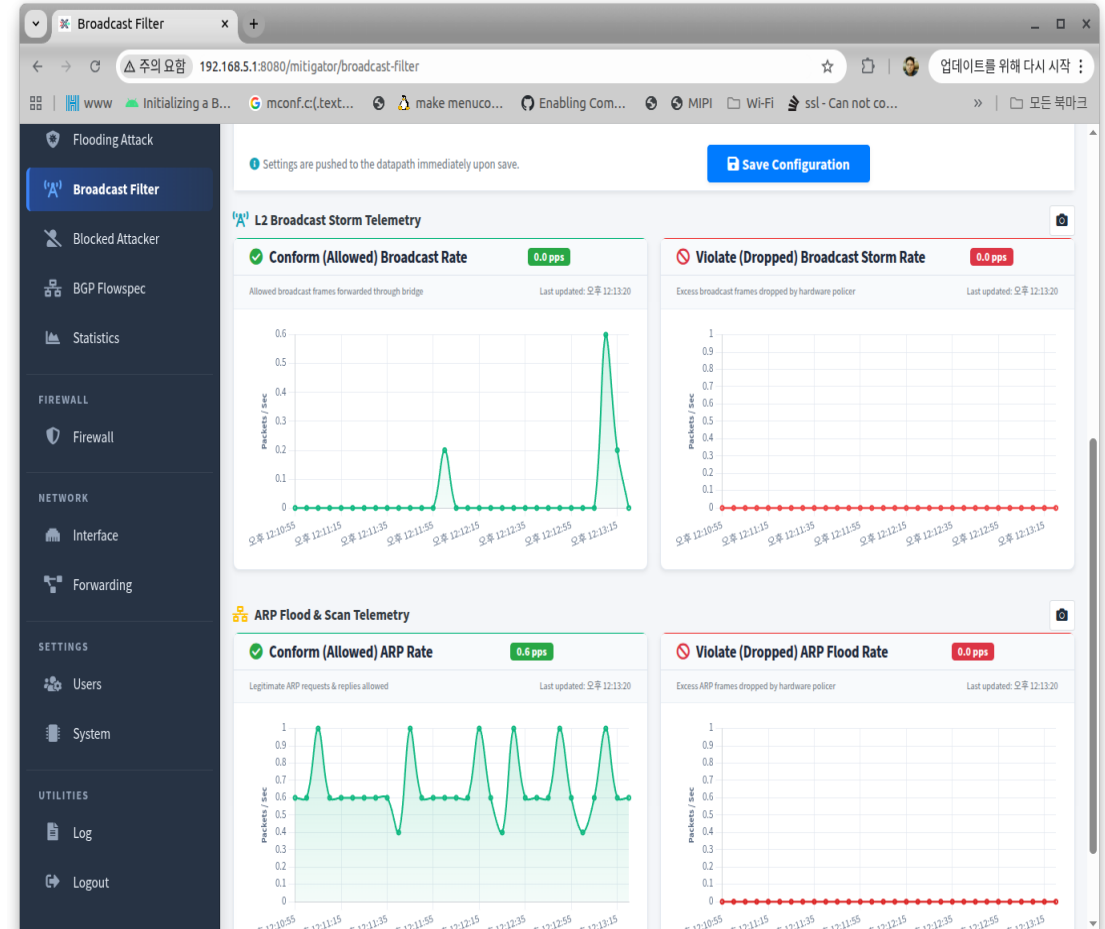


4. XDDoS User Interface(2) – WebUI(7)

Mitigator - Broadcast Filter Page#1



Mitigator - Broadcast Filter Page#2



4. XDDoS User Interface(2) – WebUI(8)

Mitigator - BGP Flowspec Page#1

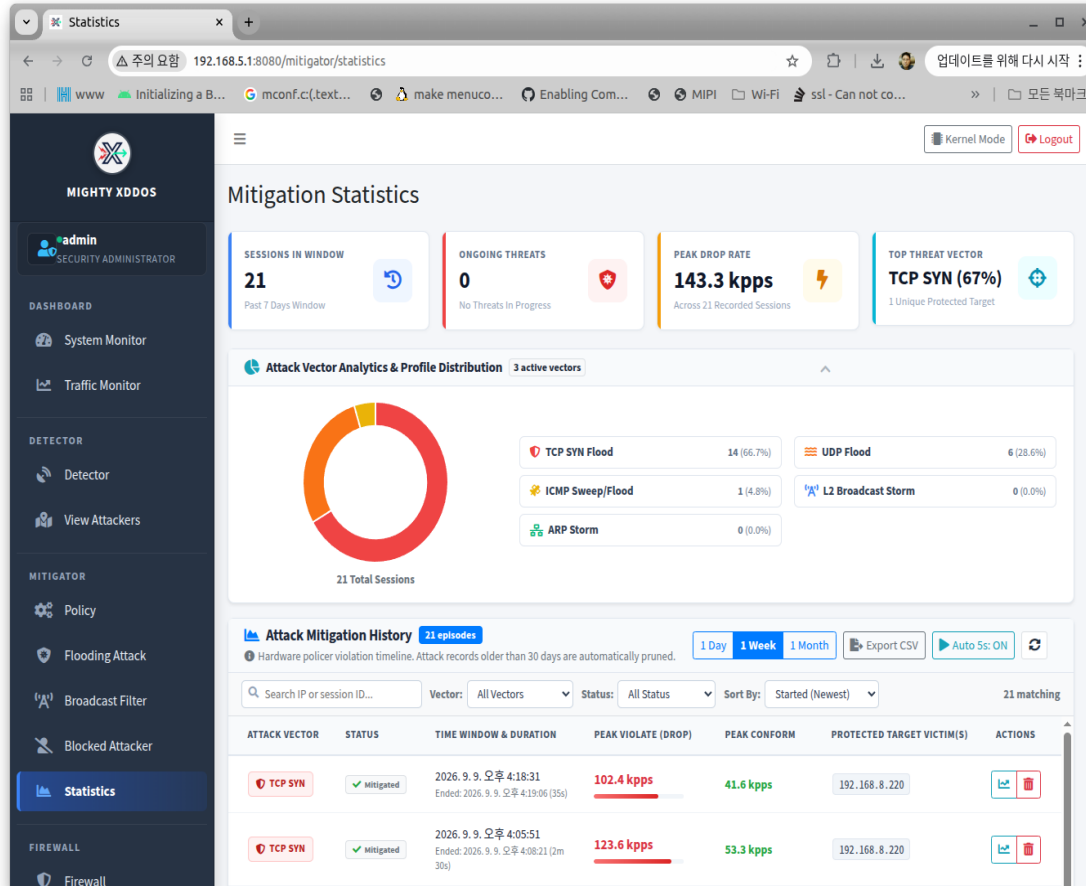
The screenshot shows the 'BGP Flowspec' configuration page in the Mitigator web interface. The left sidebar contains navigation links: admin, DASHBOARD, DETECTOR, MITIGATOR, and UTILITIES. The main content area displays the 'BGP Flowspec Engine Configuration' with a status bar showing 'ENGINE ACTIVE'. Below this, there are four configuration sections: 'Enable BGP Flowspec Engine (Upstream DDoS Offloading)', 'LOCAL AS NUMBER' (AS 65001), 'ROUTER-ID (BGP ID)' (192.168.8.114), and 'LINK SATURATION THRESHOLD' (5%). The 'Upstream BGP Neighbors (Peering Routers)' section shows a single peer configuration for 192.168.8.139 with Remote AS 65000 and a Hold Time of 90s.

Mitigator - BGP Flowspec Page#2

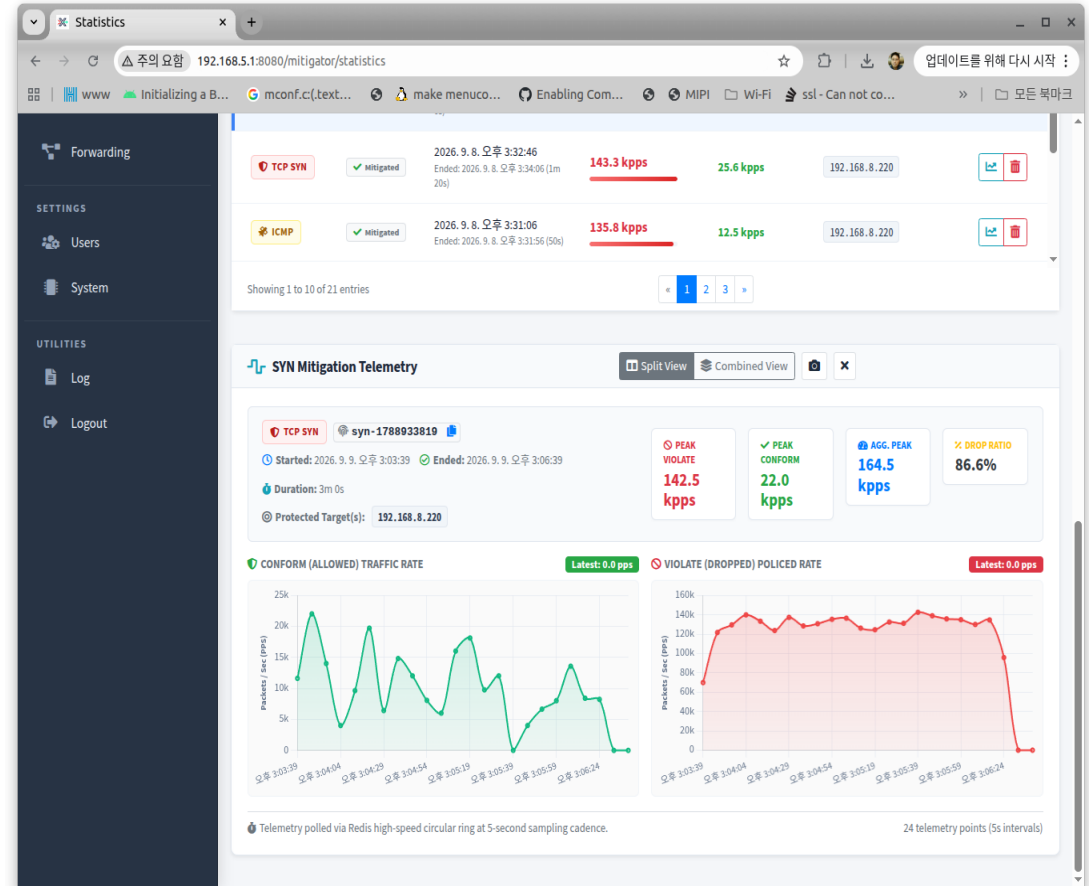
The screenshot shows the 'BGP Flowspec' configuration page in the Mitigator web interface, specifically the 'Upstream BGP Neighbors (Peering Routers)' section. It displays a table of upstream neighbors with columns for Upstream Neighbor IP, Remote AS, BGP Session State, Session Uptime, Prefixes / NLRI (RX / TX), and Connection Health. The table shows one neighbor with IP 192.168.8.139, Remote AS 65000, and a session state of 'ESTABLISHED'. Below the table, there is a section for 'Active Flowspec Rules (Upstream Advertised)' with a search bar and a table of rules. The table has columns for Flood Type, Protected Victim IP, Flowspec NLRI Match Filter (RFC 8955), Offloaded Action, and Announced At. One rule is shown for 'UDP Flood' targeting 192.168.8.220 with a match filter of 'dst=192.168.8.220 proto=UDP dst-port=6000 pktlen=28' and an action of 'discard'.

4. XDDoS User Interface(2) – WebUI(9)

Mitigator - Statistics Page#1



Mitigator - Statistics Page#2



4. XDDoS User Interface(2) – WebUI(10)

Network - Interfaces Page

Network Interfaces

DATAPATH ENGINE
VPP FASTPATH
Hardware DPDK poll-mode driver

ETHERNET INTERFACES
6
Configured Physical & Bond Ports

L2 BRIDGE GROUPS
1
Active L2 cross-connect & domains

VRRP REDUNDANCY
STOPPED
0 Virtual Router Instances

Ethernet & Bonding Interfaces

INTERFACE	STATUS	IP ADDRESS	NETMASK	ROLE	DPDK	LINK INFO	ACTION
enp2s0	↑ up	0.0.0.0	0.0.0.0	LAN	DPDK	✓ Link 1000Mb/s (Full)	Edit
enp3s0	↑ up	0.0.0.0	0.0.0.0	WAN	DPDK	✓ Link 1000Mb/s (Full)	Edit
enp1s0	↑ up	192.168.5.1	255.255.255.0	MGMT	—	✓ Link 2500Mb/s (Full) Port: TP Mac: 2500baseT/Full Auto-Neg: On	Edit
enp4s0	↓ down	—	—	—	—	No Link Port: TP Mac: 2500baseT/Full Auto-Neg: On	Edit

Network - Forwarding Page

FORWARDING ENGINE
eBPF XDP
eBPF/XDP High-Speed Linux Datapath

L2 BRIDGE GROUPS
1
Configured Cross-Connect Pair

DYNAMIC ARP ENTRIES
6
Live MAC Addresses Learned by the Bridge

ROUTING TABLE
3
Active IPv4 Forwarding Database Records

Cross-Connect Status
xdp_bridge Layer-2 Cross-Connect pair

NAME	TYPE	MEMBERS	XDP MODE
bridge0	↔ Cross-Connect	enp3s0 enp2s0	native

Learned MAC Table
Live MAC learning table for this Cross-Connect pair
Learned entries: 6
Last updated: 오후 12:14:57

MAC ADDRESS	BRIDGE DOMAIN	INTERFACE	STATIC	BVI
94:83:c4:0e:ac:78	↔ bridge0	enp3s0	no	no
98:83:89:9a:9b:c6	↔ bridge0	enp3s0	no	no
94:83:c4:0e:ac:77	↔ bridge0	enp3s0	no	no
f0:a7:31:a0:f4:d6	↔ bridge0	enp2s0	no	no
66:84:04:a9:79:f8	↔ bridge0	enp3s0	no	no

4. XDDoS User Interface(2) – WebUI(11)

Utilities - Log Page#1

System Logs

Log Monitor Archive Settings

System Logs

Filter log content... All Levels 100 Lines

Syslog Monitor

Sep 17 11:52:59 mightyxddos mighty-ddos[2727]: **WARN:** /usr/local/mightysg/etc/mightymail.conf not found, skipping email

Sep 17 11:52:59 mightyxddos mighty-ddos[2728]: BAN ip=192.168.8.220 direction=incoming pps=27875 attack_type=syn_flood source_ip_count=1 - mitigation applied, email alert sent

Sep 17 11:53:50 mightyxddos mighty-ddos[2751]: UNBAN (Detector-internal, no mitigation change) ip=192.168.8.220 direction=incoming pps=27875

Sep 17 12:00:24 mightyxddos redis-server[1064]: 1064:M 17 Sep 2026 12:00:24.100 * 1 changes in 900 seconds. Saving...

Sep 17 12:00:24 mightyxddos redis-server[1064]: 1064:M 17 Sep 2026 12:00:24.100 * Background saving started by pid 2861

Sep 17 12:00:24 mightyxddos redis-server[2861]: 2861:C 17 Sep 2026 12:00:24.107 * DB saved on disk

Sep 17 12:00:24 mightyxddos redis-server[2861]: 2861:C 17 Sep 2026 12:00:24.108 * Fork CoW for RDB: current 0 MB, peak 0 MB, average 0 MB

Sep 17 12:00:24 mightyxddos redis-server[1064]: 1064:M 17 Sep 2026 12:00:24.201 * Background saving terminated with success

Sep 17 12:00:39 mightyxddos mighty-ddos[2894]: **WARN:** /usr/local/mightysg/etc/mightymail.conf not found, skipping email

Sep 17 12:00:39 mightyxddos mighty-ddos[2896]: BAN ip=192.168.8.220 direction=incoming pps=26502 attack_type=syn_flood source_ip_count=51 - mitigation applied, email alert sent

Sep 17 12:02:34 mightyxddos mighty-ddos[2933]: UNBAN (Detector-internal, no mitigation change) ip=192.168.8.220 direction=incoming pps=26502

Sep 17 12:14:30 mightyxddos system[1]: Starting motd-news.service - Message of the Day...

Sep 17 12:14:30 mightyxddos system[1]: motd-news.service: Deactivated successfully.

Sep 17 12:14:30 mightyxddos system[1]: Finished motd-news.service - Message of the Day.

Sep 17 12:15:17 mightyxddos dbus-daemon[1050]: [system] Activating via systemd: service name='org.freedesktop.timedate1' unit='dbus-org.freedesktop.timedate1.service' requested by ':1.12' (uid=0 pid=3302 comm='timedactl status')

Utilities - Log Page#2

System Logs

Log Monitor Archive Settings

Log Disk Usage & Retention Policy

Total System Log Disk Usage (under /var/log)

1892.24 MB / 1,024.00 MB (100.0%)

This indicates the cumulative size of all system, service, and custom log files.

Rotation Interval: Daily

Max Retention Copies: 7

Max File Size Limit (MB): 0

Force rotate immediately if any log exceeds this size, regardless of interval. Set to 0 to disable size limit check.

Compress Archived Logs (.gz) ☒

HDD Warning Threshold (%): 85

Send warning email when disk space usage exceeds this percentage.

Current disk usage: 25.9%

Force Rotate Now

Save Settings

4. XDDoS User Interface(3) – CLI(1)

```
(mighty) 192.168.5.1 — Konsole
파일(F) 편집(E) 보기(V) 책갈피(B) 플러그인 설정(S) 도움말(H)
복사(C) 붙여넣기(P) 찾기(F)

mightyxddos# ?

Available commands:
show          Display information
enable        Enter privileged mode
help          Show available commands (same as ?)
exit          Close shell

mightyxddos# show ?

Possible completions:
interface     Show ethernet interfaces
vrrp          Show VRRP settings and daemon status
route         Show routing table
firewall      Show firewall rules
bridge        Show L2 bridge group(s)
detector      Show DDoS detection engine status
baseline      Show Dynamic Baseline learned per-host traffic profile
signature     Show Auto-Signature Synthesis progress for currently-armed victims
flooding-attack Show DDoS Mitigator flood defense policies and protected victims
blocked-attacker Show blocked attacker source IPs (from Detector's per-attack flow analysis)
broadcast-filter Show Broadcast/ARP Storm Filter settings and live policer counters
app-filter    Show Application Filter (Slow Connection Guard, Server Health) settings and live status
policy        Show Mitigator Policy settings
bgp           Show BGP Flowspec peering/rule status
mac-security  Show MAC Flooding Protection settings
system        Show system settings

mightyxddos# show
show  Display information
enable Enter privileged mode
help  Show available commands (same as ?)
exit  Close shell
```

```
(mighty) 192.168.5.1 — Konsole
파일(F) 편집(E) 보기(V) 책갈피(B) 플러그인 설정(S) 도움말(H)
복사(C) 붙여넣기(P) 찾기(F)

blocked-attacker Show blocked attacker source IPs (from Detector's per-attack flow analysis)
broadcast-filter Show Broadcast/ARP Storm Filter settings and live policer counters
app-filter       Show Application Filter (Slow Connection Guard, Server Health) settings and live status
policy           Show Mitigator Policy settings
bgp              Show BGP Flowspec peering/rule status
mac-security     Show MAC Flooding Protection settings
system           Show system settings

mightyxddos# enable
Password: *****
mightyxddos(en)# ?

Available commands:
show          Display information
interface     Network interface configuration
vrrp          VRRP cluster configuration
route         Routing table management
firewall      Firewall rule management
bridge        L2 Cross-Connect/Bridge Domain group management
arp-static    Static ARP Termination entry management
detector      DDoS detection engine control (start/stop/restart)
mitigate      Unban DDoS-mitigated victims/sources
system        System operations
write         Commit staged changes to the system
clear         Discard staged changes
exit          Return to view mode

mightyxddos(en)# show
show  Display information
interface Network interface configuration
vrrp  VRRP cluster configuration
route Routing table management
firewall Firewall rule management
bridge L2 Cross-Connect/Bridge Domain group management
```

4. XDDoS User Interface(3) – CLI(2)

```
(mighty) 192.168.5.1 — Konsole
파일(F) 편집(E) 보기(V) 책갈피(B) 플러그인 설정(S) 도움말(H)
복사(C) 붙여넣기(P) 찾기(F)

clear          Discard staged changes
exit           Return to view mode
mightyxddos(en)# show interface
Interface Status IP Address Link State
-----
enp1s0 UP 192.168.5.1/24 up (Speed: 2500Mb/s, Duplex: Full)
enp2s0 UP - up (Speed: 1000Mb/s, Duplex: Full)
enp3s0 UP 192.168.8.114/24 up (Speed: 1000Mb/s, Duplex: Full)
enp4s0 DOWN - down
enp5s0 DOWN - down
enp6s0 DOWN - down

Interface enp1s0:
Link Detection : up (Speed: 2500Mb/s, Duplex: Full)
Supported Ports : TP
Max Link Mode : 2500baseT/Full
Auto-Negotiation : supported (currently on)

Interface enp2s0:
Link Detection : up (Speed: 1000Mb/s, Duplex: Full)
Supported Ports : TP
Max Link Mode : 2500baseT/Full
Auto-Negotiation : supported (currently on)

Interface enp3s0:
Link Detection : up (Speed: 1000Mb/s, Duplex: Full)
Supported Ports : TP
Max Link Mode : 2500baseT/Full
Auto-Negotiation : supported (currently on)

Interface enp4s0:
Link Detection : down
Supported Ports : TP
```

```
(mighty) 192.168.5.1 — Konsole
파일(F) 편집(E) 보기(V) 책갈피(B) 플러그인 설정(S) 도움말(H)
복사(C) 붙여넣기(P) 찾기(F)

Max Link Mode : 2500baseT/Full
Auto-Negotiation : supported (currently on)

Interface enp6s0:
Link Detection : down
Supported Ports : TP
Max Link Mode : 2500baseT/Full
Auto-Negotiation : supported (currently on)

mightyxddos(en)# show bridge
Name : bridge0
Type : xconnect
Members : enp3s0, enp2s0
Running : on
Mode : native
Learn : on
UU-Flood : on
Local IP : 192.168.8.114/255.255.255.0
Detector Sample Rate: 1-in-75

mightyxddos(en)# show route
Codes: C - connected, S - static, D - default
Type Destination Gateway Interface Protocol
-----
[D] 0.0.0.0/0 192.168.8.1 enp3s0 -
[C] 192.168.5.0/24 0.0.0.0 enp1s0 kernel
[C] 192.168.8.0/24 0.0.0.0 enp3s0 kernel

mightyxddos(en)#
show          Display information
interface     Network interface configuration
vrrp          VRRP cluster configuration
route         Routing table management
firewall      Firewall rule management
bridge        L2 Cross-Connect/Bridge Domain group management
```

4. XDDoS User Interface(3) – CLI(3)

```
(mighty) 192.168.5.1 — Konsole
파일(F) 편집(E) 보기(V) 책갈피(B) 플러그인 설정(S) 도움말(H)
복사(C) 붙여넣기(P) 찾기(F)

ACL/Mitigator interface binding scope: all

WAN Service Port Policy: enabled
Proto Port Description
-----
tcp    443   https
tcp    8080   https
udp    6000   test udp
icmp    icmp test
tcp    80     http
tcp    22    ssh to nanopi

VPP raw output (show acl-plugin acl):
acl-index 0 count 1 tag {mightyxddos-acl-builtin-phys}
  0: ipv4 permit+reflect src 0.0.0.0/0 dst 0.0.0.0/0 proto 0 sport 0-65535 dport 0-65535
    applied inbound on sw_if_index: 1, 3
    applied outbound on sw_if_index:
    used in lookup context index: 1, 5
acl-index 1 count 1 tag {mightyxddos-acl-egress-wan}
  0: ipv4 permit+reflect src 0.0.0.0/0 dst 0.0.0.0/0 proto 0 sport 0-65535 dport 0-65535
    applied inbound on sw_if_index:
    applied outbound on sw_if_index: 2
    used in lookup context index: 2
acl-index 2 count 1 tag {mightyxddos-acl-egress-lan}
  0: ipv4 permit src 0.0.0.0/0 dst 0.0.0.0/0 proto 0 sport 0-65535 dport 0-65535
    applied inbound on sw_if_index:
    applied outbound on sw_if_index: 1, 3
    used in lookup context index: 0, 4
acl-index 3 count 10 tag {mightyxddos-acl-wan-svc-ports}
  0: ipv4 permit src 0.0.0.0/0 dst 0.0.0.0/0 proto 17 sport 0-65535 dport 68
  1: ipv4 permit src 192.168.8.139/32 dst 192.168.8.114/32 proto 6 sport 0-65535 dport 179
--More--
```

```
(mighty) 192.168.5.1 — Konsole
파일(F) 편집(E) 보기(V) 책갈피(B) 플러그인 설정(S) 도움말(H)
복사(C) 붙여넣기(P) 찾기(F)

mightyxddos(en)#
mightyxddos(en)# show baseline
Dynamic Baseline   : on
Sensitivity        : 5.0x learned baseline
Learning Period    : 5 minute(s)
                    (shorter than the 3 day(s) recommended for production —
                    for real deployments, restore this to at least that before relying
                    on Dynamic Baseline for automatic bans.)
Protected Networks : 192.168.8.220/32

Host      Status First Seen      SYN pps  UDP pps  ICMP pps
-----
192.168.8.220 Active 2026-09-15 15:06:05 0.0      0.1      0.0
mightyxddos(en)#
mightyxddos(en)#
mightyxddos(en)#
mightyxddos(en)# show signature
Flood Type Victim      Samples Status
-----
SYN          192.168.8.220 2000 Active

Signature detail: SYN / 192.168.8.220
Fixed packet length: 60 bytes
Offset(s) Field      Value
-----
12-13      EtherType      0x0800
15          IP DSCP/ECN      0x00
16-17      IP Total Length  0x0028 (40)
20-21      IP Flags/Fragment Offset 0x0000
22          IP TTL          0x40
36-37      TCP Dest Port  0x01bb (443)
46          TCP Data Offset/Reserved 0x50
```

4. XDDoS User Interface(3) – CLI(4)

```
(mighty) 192.168.5.1 — Konsole
파일(F) 편집(E) 보기(V) 책갈피(B) 플러그인 설정(S) 도움말(H)
복사(C) 붙여넣기(P) 찾기(F)

mightyxddos(en)# show bridge
Name      Type      State BD-ID Learn UU-Flood ARP-Term LAN-Isolation Learn-Limit Mac-Age BVI IP
BVI Status Members
-----
bridge0   bridge-domain UP      1    on   on    off    off          2000      5 min  192.168.8.114/24 (bvi0)
UP        enp2s0, enp3s0, enp4s0

L2 FIB (MAC address table, all bridge domains):
Mac-Address  BD-Idx If-Idx Age(min) Static Filter BVI Interface
-----
98:83:89:9a:9b:c6 1      2      0      off  off  off  enp3s0
8a:0a:45:f2:e8:21 1      1      0      off  off  off  enp2s0
50:3e:aa:3f:99:e2 1      2      0      off  off  off  enp3s0
10:e7:7a:e3:d2:e5 1      3      0      off  off  off  enp4s0
b0:b0:00:00:00:00 1      4      no     on   off  on   bvi0
9e:13:60:8f:6d:b6 1      2      2      off  off  off  enp3s0
94:83:c4:0e:ac:78 1      2      0      off  off  off  enp3s0

L2FIB total/learned entries: 7/6

Static ARP Entries:
Bridge IP Address MAC Address
-----
mightyxddos(en)#
```

show

Display information

interface

Network interface configuration

vrrp

VRRP cluster configuration

route

Routing table management

firewall

Firewall rule management

bridge

L2 Cross-Connect/Bridge Domain group management

```
(mighty) 192.168.5.1 — Konsole
파일(F) 편집(E) 보기(V) 책갈피(B) 플러그인 설정(S) 도움말(H)
복사(C) 붙여넣기(P) 찾기(F)

Enabled      : Yes
Ban Duration  : 30 min
Auto-Unban   : idle 60 sec or longer (per flood type) triggers an early unban, ahead of Ban Duration

Protocol Flood Defense Policies:
Protocol  Defense Method                                     Threshold
-----
SYN        Auto-Signature Synthesis (interim: Reset Challenge) 20000 pps
UDP        Auto-Signature Synthesis (interim: Rate Limit)      1000 pps
ICMP       Auto-Signature Synthesis (interim: Rate Limit)      1000 pps

SYN Reset Challenge (Interim, until a signature is active):
Challenged (SYN-ACK reflected): 157641
Verified (RST matched):        1
Permitted (already verified):   0
Whitelist TTL:                  300s

SYN Auto-Signature Synthesis:
No victim currently armed.

UDP Auto-Signature Synthesis:
No victim currently armed.

ICMP Auto-Signature Synthesis:
No victim currently armed.

(see 'show signature' for byte-level signature field detail)

Flooding Attack Protected Victims:
No victims currently protected.

--More--
```

4. XDDoS User Interface(3) – CLI(5)

```
(mighty) 192.168.5.1 — Konsole
파일(F) 편집(E) 보기(V) 책갈피(B) 플러그인 설정(S) 도움말(H)
복사(C) 붙여넣기(P) 찾기(F)

Enabled : Yes
L2 Broadcast Threshold : 5000 pps (per port)
ARP Threshold : 1000 pps (per port)
L2 Multicast Threshold : 5000 pps (per port)

Interface Policer Conform Pkts Violate Pkts Classify Table Sessions
-----
enp2s0 ARP 54 0 8 1
enp2s0 L2-Broadcast 0 0 7 1
enp2s0 L2-Multicast 3 0 5 1
enp2s0 L2-Control (exempt) - - 6 1
enp3s0 ARP 19 0 12 1
enp3s0 L2-Broadcast 0 0 11 1
enp3s0 L2-Multicast 0 0 9 1
enp3s0 L2-Control (exempt) - - 10 1
enp4s0 ARP 4 0 16 1
enp4s0 L2-Broadcast 6 0 15 1
enp4s0 L2-Multicast 2 0 13 1
enp4s0 L2-Control (exempt) - - 14 1

VPP raw output (show policer name mitigate-l2-arp-flood-enp2s0):
Name "mitigate-l2-arp-flood-enp2s0" type 1r2c cir 480 eir 0 cb 60000 eb 0
rate type kbps, round type closest
conform action transmit, exceed action drop, violate action drop

Policer at index 1: single rate, not color-aware
cir 213044 tok/period, pir 1 tok/period, scale 16
cur lim 3932160000, cur bkt 3928227840, ext lim 0, ext bkt 4291035136
last update 21356058
conform 54 packets, 3240 bytes
exceed 0 packets, 0 bytes
--More--
```

```
(mighty) 192.168.5.1 — Konsole
파일(F) 편집(E) 보기(V) 책갈피(B) 플러그인 설정(S) 도움말(H)
복사(C) 붙여넣기(P) 찾기(F)

Protected Services:
192.168.8.218 ports 80,443 nanopi

Slow Connection Guard:
Enabled : Yes
Mode : reset
Slow Request : 10 s, below 64 bytes/s
Silent Hold : 30 s
Slow Read (zero window) : 30 s
Per-source Limits : 256 concurrent, 100 new/s
Block : 5 resets in 60 s, for 10 min

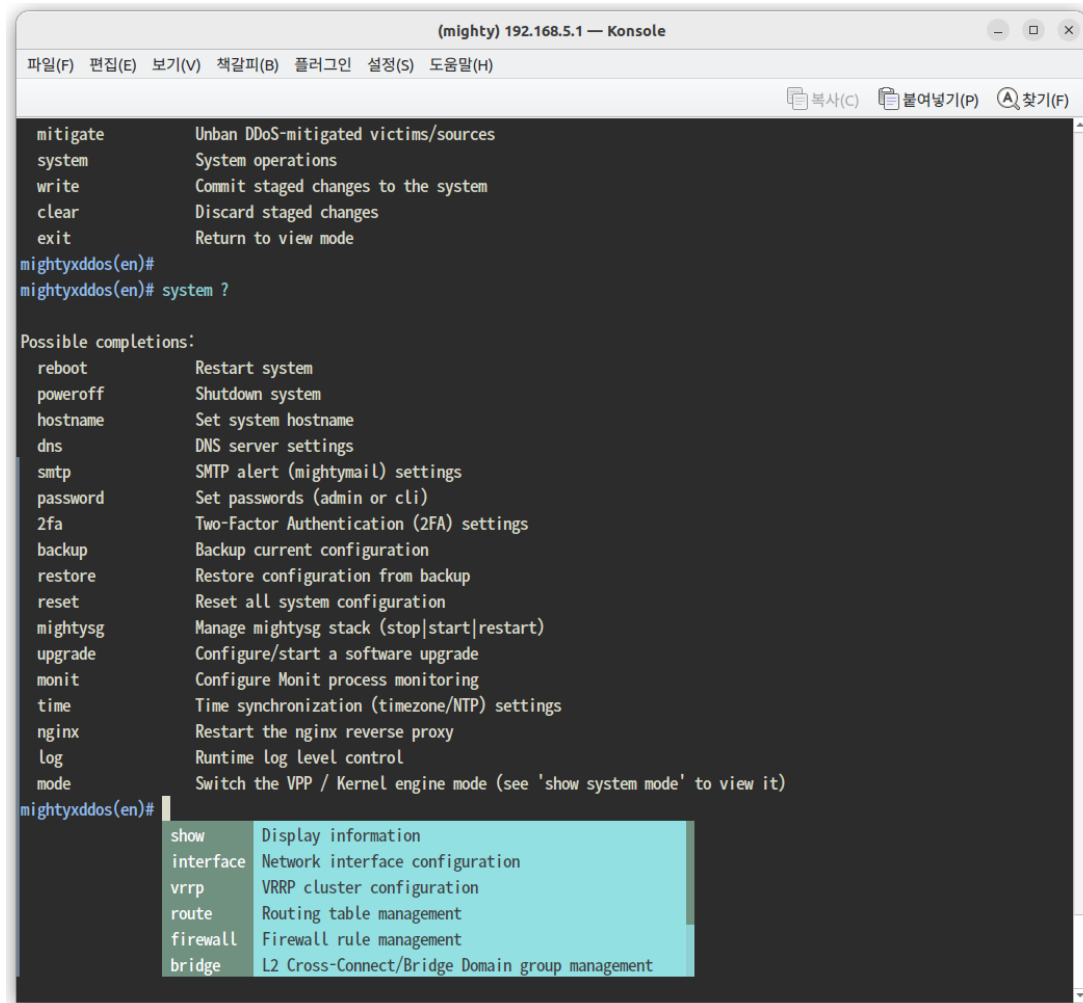
Server Health Detection:
Enabled : Yes
Action : detect
Degraded When : SYN-ACK response < 50% or server RST > 30%, 3 s in a row (from 50 SYN/s)
Offender : >= 5% of new connections and >= 20 new/s

Live Counters:
Tracked Connections : 0
Slow Request / Silent Hold / Slow Read : 789 / 0 / 0
Connection Limit / Rate : 0 / 0
Resets Sent (failed) : 789 (0)
Packets Dropped : 6376
Sources Blocked : 0 (since virtserver start)

Server State SYN/s Resp % RST % Active
192.168.8.218 idle 0 0 0 0

Recent Detections (grouped by source, service and type):
Last Seen Source Service Detection Count Result
--More--
```

4. XDDoS User Interface(3) – CLI(6)



The screenshot shows a terminal window titled "(mighty) 192.168.5.1 — Konsole". The terminal displays a list of commands and their descriptions, followed by a prompt for the user to enter a command. The commands listed are:

- mitigate: Unban DDoS-mitigated victims/sources
- system: System operations
- write: Commit staged changes to the system
- clear: Discard staged changes
- exit: Return to view mode

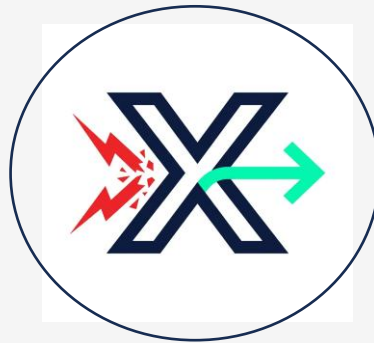
The user enters the command "mightyxddos(en)# system ?". The terminal then displays a list of possible completions:

- reboot: Restart system
- poweroff: Shutdown system
- hostname: Set system hostname
- dns: DNS server settings
- smtp: SMTP alert (mightyemail) settings
- password: Set passwords (admin or cli)
- 2fa: Two-Factor Authentication (2FA) settings
- backup: Backup current configuration
- restore: Restore configuration from backup
- reset: Reset all system configuration
- mightysg: Manage mightysg stack (stop|start|restart)
- upgrade: Configure/start a software upgrade
- monit: Configure Monit process monitoring
- time: Time synchronization (timezone/NTP) settings
- nginx: Restart the nginx reverse proxy
- log: Runtime log level control
- mode: Switch the VPP / Kernel engine mode (see 'show system mode' to view it)

The user enters the command "mightyxddos(en)# show". The terminal then displays a list of possible completions:

- show: Display information
- interface: Network interface configuration
- vrrp: VRRP cluster configuration
- route: Routing table management
- firewall: Firewall rule management
- bridge: L2 Cross-Connect/Bridge Domain group management

Appendix



A. Testbed(1)

```
<TCP/UDP/ICMP Flooding attacks – random source IPs>  
$ sudo hping3 -S -p 443 --flood --rand-source 192.168.8.218  
$ sudo hping3 -udp --flood -p 6000 --rand-source 192.168.8.218  
$ sudo hping3 --icmp --flood --rand-source 192.168.8.218
```

```
<L2 broadcast attack>  
$ sudo hping3 --udp -d 100 -p 6000 --flood 192.168.8.255
```

```
<unicast + multicast MAC Flooding attack>  
$ sudo macof -d 192.168.8.218 -i enp2s0
```

```
<unicast MAC Flooding attack>  
$ sudo macof -d 192.168.8.218 -i enp2s0 -e 8a:0a:45:f2:e8:21
```

```
<ARP Flooding attack>  
$ sudo nping --arp-type ARP-request --dest-mac ff:ff:ff:ff:ff:ff --source-ip rand \  
--rate 10000 -c 100000 --interface enp2s0 192.168.8.218
```

- Web Server
- Mail Server
- DNS Server
- Game Server
- LLM Serving
- ...



192.168.8.218/24



enp2s0(LAN)

enp3s0(WAN)

192.168.8.114/24
(Bridge Interface IP)

XDDoS Mitigator

Attacker



BGP Router

192.168.8.139/24

DDoS Attack



DDoS Attack

1GbE Switching Hub

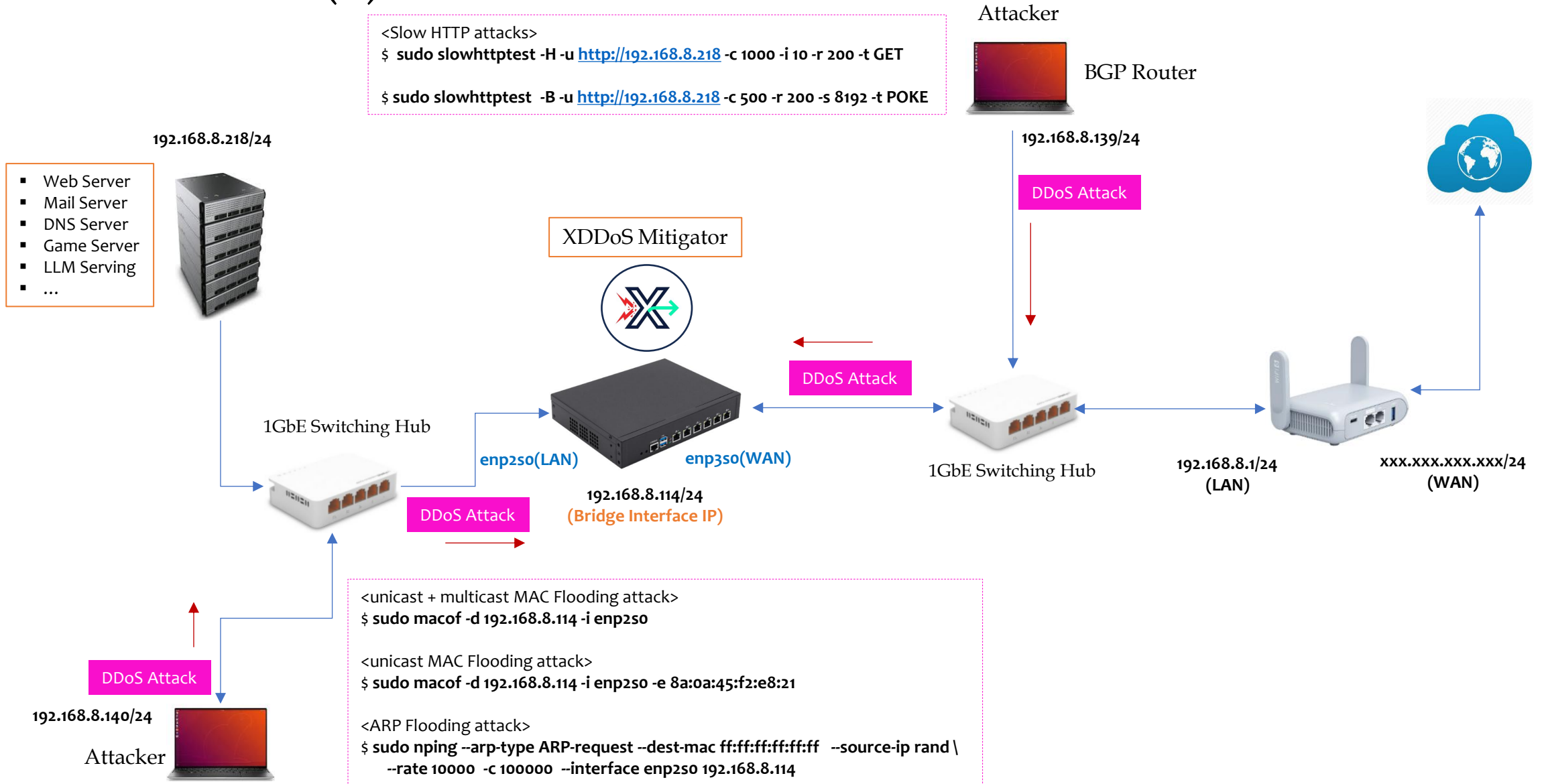
192.168.8.1/24
(LAN)



xxx.xxx.xxx.xxx/24
(WAN)



A. Testbed(2)



B. References

- [1] <https://www.dpdk.org/>
- [2] <https://fd.io/>
- [3] <https://ebpf.io/>
- [4] <https://fastnetmon.com/>
- [5] <https://prometheus.io/>
- [6] <https://grafana.com/>
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- [9] Impressive Packet Processing Performance Enables Greater Workload Consolidation, Intel
- [10] <https://medium.com/google-cloud/forwarding-over-100-mpps-with-fd-io-vpp-on-x86-62b9447da554>
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- [12] <https://www.netgate.com/resources/articles-vector-packet-processing>
- [13] <https://www.datadoghq.com/blog/xdp-intro/>
- [14] https://www.tutorialspoint.com/data_communication_computer_network/token_bucket_algorithm.htm
- [15] vLLM : <https://vllm.ai/>
- [16] SGLang: <https://www.sglang.io/>
- [17] Ollama: <https://ollama.com/>
- [18] MAX: <https://www.modular.com/>
- [19] And Google and Gemini~

Faster Connectivity, Flawless Security

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